SD-19

Parts, Materials, and Processes Management Guide

A Guidebook of Best Practices for Oversight of Parts Selection in Defense Systems



DEFENSE STANDARDIZATION PROGRAM OFFICE

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STDZ

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Foreword

The December 2023 version of the SD-19 introduced a new paradigm for parts, materials, and processes (PM&P) management. This approach increases discipline in program office oversight by applying best practices from MIL-STD-11991, *General Standard for Parts, Materials, and Processes*, and the SD-26, *Diminishing Manufacturing Sources and Material Sources (DMSMS) and PM&P Management Contracting Guide*. The new paradigm encompasses review and approval of a PM&P selection organization's plan to meet MIL-STD-11991 requirements, verifying that the processes in the Plan are being followed, and validating that those processes are effective. The new paradigm also includes collecting and analyzing records on PM&P selection oversight activities and their effects on cost, schedule, and performance; obtaining part selection data and using tools to analyze the data in support of best practices; and revisiting PM&P selections when a risk change is triggered.

This document reinforces that paradigm by providing best practices for a program office to implement a PM&P Management Program and an associated PM&P Management Plan. While the material covered is similar to the December 2023 version, this document is substantially reorganized based on lessons learned from multiple training events.

This publication should be used by Department of War (DoW) acquisition activities, defense contractors, and government design organizations that make PM&P selections. The document provides guidance for government and industry managers to achieve a more disciplined and integrated PM&P management approach. This approach ensures that all PM&P selections meet their allocated and derived requirements and properly balance the costs and risk of engineering design considerations for defense systems, such as performance specifications, obsolescence resilience, supply chain risks, and system security weaknesses and vulnerabilities.

Recommended changes to this publication should be sent to the Defense Standardization Program Office, 8725 John J. Kingman Road, Stop 5100, Fort Belvoir, VA 22060-6220, or via email to DSPO@dla.mil.



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Section 1. Introduction

1.1 Organization of This Document

The principal audience for this document is the program office responsible for Department of War (DoW) systems and equipment at any point in their life cycle, from early development through design, production, sustainment, modification, and eventually to disposal. This document provides guidance on conducting risk-based program office oversight of the selection of parts, materials, and processes (PM&P) for these systems.

This document refines and reorganizes the content from the December 2023 version. That previous version established a new paradigm by shifting the focus from “parts management” to the more comprehensive “PM&P management oversight.” This new edition leaves that paradigm intact while making the following improvements:

- Consistent terminology: Replaces all remaining references to “parts management” with “PM&P management.”
- Clearer explanation: Reorganizes the sections to better explain the details of the new paradigm.
- Streamlined use: Aims to make the implementation of the paradigm more efficient.

The following paragraphs describe the content of each section and highlight the changes made to each.

Section 1

Section 1 defines key terms, describes the importance of effective PM&P management, summarizes DoW policy and guidance, and portrays the high-level contents of a program office PM&P Management Program and Plan. Primary changes to Section 1 include:

- Refined the definitions of PM&P management and parts;
- Expanded the relative roles of government and contractors in PM&P management;
- Expanded the consequences of inadequate PM&P management;
- Incorporated changes to the policy and guidance associated with PM&P management;
- Provided a clearer depiction of the steps in a program office PM&P Program along with a new graphic that depicts those steps; and
- Revised the organization of a program office PM&P Management Plan.

Section 2

Section 2 explains the use of a risk assessment and the establishment of a risk profile to drive the level and extent of PM&P management oversight. Primary changes to Section 2 include:

- Updated references;
- Increased emphasis on the risk assessment being conducted in conjunction with other risk assessments and not being conducted by PM&P practitioners;
- Moved most of the material on a formal risk assessment to a new Appendix C; and
- Included a more detailed explanation and illustration of a risk profile.

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Section 3

Section 3 lists the questions to be answered to determine the appropriate level and type of program office oversight and discusses the determination of skills and number of people needed to conduct that oversight. Primary changes to Section 3 include:

- Moved more detailed material on determining the specifics of what oversight should entail to later sections of the document;
- Incorporated all the material from Section 4 of the December 2023 version of the document on engaging and obtaining the necessary subject matter experts; and
- Moved material in December 2023 version of Section 3 dealing with the Program Office PM&P Management Plan to Section 11, which consolidates information on that Plan.

Section 4

Section 4 discusses all aspects of the oversight functions associated with the review and approval of a contractor or government design organization (GDO) PM&P Management Plan. Primary changes to Section 4 include:

- Moved all former Section 4 content to Section 3;
- Captured and reorganized material on the contractor or GDO PM&P Management Plan, previously in Sections 3 and 5, to form the new Section 4;
- Expanded the section on tailoring MIL-STD-11991 requirements for developing a contractor or GDO PM&P Management Plan; and
- Relocated material in the previous version of Section 5 dealing with the Program Office PM&P Management Plan to Section 11, which consolidates information on that Plan.

Section 5

Section 5 explains considerations for deciding the level of oversight associated with verifying that the contractor or GDO is following the processes in its PM&P Management Plan. Primary changes to Section 5 include:

- Integrated, reorganized, and revised material on verification in the previous versions of Sections 3 and 5 to form the new Section 5; and
- Relocated material in the previous version of Section 5 dealing with the Program Office PM&P Management Plan to Section 11, which consolidates information on that Plan.

Section 6

Section 6 provides details on oversight considerations for validating that the contractor or GDO PM&P selection processes are effective. Primary changes to Section 6 include:

- Moved material from the prior version of Section 6 on record keeping to the new Sections 7 and 9;
- Reworked and updated the material on validation and monitoring in the previous versions of Sections 3 and 5 to form the new Section 6; and
- Relocated material in the previous version of Section 5 dealing with the Program Office PM&P Management Plan to Section 11, which consolidates information on that Plan.

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Section 7

Section 7 proposes a framework for PM&P management record keeping. It provides examples of useful metrics to collect. Primary changes to Section 7 include:

- Made minor revisions to incorporate changes to emphasize that record keeping also applies to situations where a GDO selects parts; and
- Split off material from the prior record keeping section on ways to use metrics to improve the efficiency and effectiveness of PM&P management into Section 9.

Section 8

Section 8 identifies not previously covered special considerations for PM&P selection by government organizations. Primary changes to Section 8 include:

- Added material and created a new section.

Section 9

Section 9 describes ways to use metrics to improve the efficiency and effectiveness of PM&P management. Primary changes to Section 9 include:

- Established a new section on this subject using content previously included under record keeping.

Section 10

Section 10 covers detecting risk changes after initial PM&P selection, assessing those changes to determine any risk increase, and making PM&P changes if the additional risk is too high. Primary changes to Section 10 include:

- Consolidated material into its own section.

Section 11

Section 11 describes the preferred contents of a program office PM&P Management Plan. Primary changes to Section 11 include:

- Consolidated material previously found throughout the entire guidebook into a single section.

Section 12

Section 12 discusses PM&P management support tools, services, and training. Primary changes to Section 12 include:

- Deleted references to archived Warfighting Acquisition University (WAU) material.

Appendices

Appendix A contains details on pertinent policy and guidance. Questions for common types of program reviews are listed in Appendix B. Appendix C describes the elements of a formal PM&P initial risk

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assessment. Appendix D provides some considerations for specifying PM&P requirements on contract. Primary changes to the appendices include:

- Updated Appendix A to reflect the latest issuances.
- Appendix C was extracted from text formerly included in section 2.

1.2 Definitions and Responsibilities

1.2.1 What Is Parts, Materials, and Processes Management?

PM&P management is a systems engineering discipline for *selecting and assuring the performance* of parts and assemblies of parts, while accounting for the materials and processes used to manufacture them, throughout all phases of a system's (or equipment's) life cycle from initial design through disposal. All PM&P used in the design, development, fabrication, procurement, and sustainment of DoW systems and equipment are included in PM&P management. Furthermore, it applies to prototypes, engineering models, and all production versions of those systems. Consequently, both the government and contractor systems engineering communities are involved.

> ***A "part"*** *is any single unassembled hardware item of a major or minor component, accessory, or attachment that is not normally subject to disassembly without the destruction or impairment of its design use. Examples include but are not limited to elements inside custom devices, resistors, integrated circuits, and connectors.*[1]
>
> *The source of this definition defines subcategories as follows.*
>
> - *Electrical, electronic, and electromechanical parts carry, store, or transmit electrical current, and/or electromagnetic or optical waves in the intended application.*
> - *Mechanical parts have a single, non-electrical function (other than electrical grounding) and contain one or more necessary material items.*
>
> ***A "material"*** *is a metallic or nonmetallic element, alloy, mixture, or compound used in a manufacturing operation that becomes a permanent portion of the manufactured item, or which can leave a remnant, residue, coating, or other substance that becomes or affects a permanent portion of a manufactured item. Environmental materials (moisture, oxygen in the air, etc.) such as those used in tooling or equipment not intended to modify or leave residues are not meant to be covered by this definition.*[2]
>
> ***A "process"*** *when used in the definition of PM&P management is an operation, treatment, or procedure used during fabrication of materials, parts, subassemblies, and/or assemblies that modifies an existing configuration, creates a new configuration that alters the form/fit/function, and/or changes the chemical, physical, and/or the mechanical properties of the parent item.*[3]

---

[1] Military Standard (MIL-STD)-11991B, *General Standard for Parts, Materials, and Processes*, October 3, 2023, https://www.dau.edu/sites/default/files/2024-01/MIL-STD%2011991B.pdf. Also note that many companies use the word "component" in lieu of "part" and the use of the word "item" in the definition will be found in forthcoming versions of the MIL-STD.

[2] Ibid.

[3] Ibid.

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PM&P Selection Considerations

PM&P selection decisions should result in achieving application requirements throughout a system's life cycle. During design, PM&P selection decisions are based on the thoughtful assessment and balancing of numerous, overlapping engineering design considerations (many of which are not the responsibility of the PM&P management practitioner, whether in government or industry).

Engineering design considerations include performance, cost, quality, qualification, reliability, maintainability, supportability, standardization, technology features and life cycle stage, manufacturing processes and producibility, Diminishing Manufacturing Sources and Material Shortages (DMSMS) risk, system security, cyber weaknesses and vulnerabilities, hardware and software assurance, supply chain risk, susceptibility to counterfeiting, unauthorized tampering, and use of hazardous materials.[a]

The selection decision also varies as a function of criticality,[b] the application of the part within the design, program duration, risk that the program office is willing to accept, and other factors.

[a] Performance and cost are included as design consideration even though they are often optimized based on constraints from the other design considerations. If the optimized values are unacceptable, then trades will be made.

[b] This document uses the term critical to encompass either mission essential or safety critical parts. A mission essential part is an item whose failure implies that the mission likely will not be completed. A safety critical part is an item that if missing or not conforming to the design data, quality requirements, or overhaul and maintenance documentation, would result in an unsafe condition. The part itself is not inherently critical. Instead, it's the use of the part in the design. The program office, with input from the contractor, determines criticality.

PM&P selection is associated with all initial designs, redesigns, engineering changes, configuration changes, PM&P replacements, and PM&P used to maintain a system during sustainment. PM&P management should not be confused with the materiel management logistics function defined as "the phase of military logistics that includes managing, cataloging, demand and supply planning, requirements determinations, procurement, distribution, overhaul, and disposal of materiel."[4]

1.2.2 Government and Contractor PM&P Management Roles

The government program office's role in PM&P management is primarily contractor oversight, which may encompass PM&P approval. Contractors select the majority of PM&P for most DoW systems, especially during design or redesign. Contractors can also perform a substantial amount of PM&P selection activities during sustainment (e.g., performing maintenance, changing manufacturing processes, resolving obsolescence issues, avoiding lead time disruptions, rectifying performance degradations, developing post-production engineering changes and redesigns, adding and meeting new requirements); however, the degree of contractor involvement depends on the system. Contractors may also replace unavailable parts during production and sustainment. When selecting or replacing parts, contractors, along with their subcontractors, also establish the supply chain for those parts.

As indicated in the above part selection considerations, many factors affect a decision to choose or approve the use of a part for a DoW system. All those considerations can be integrated into a single question associated with the words "assuring the performance of" found in the PM&P management definition: "Does the part meet all of its allocated and derived requirements?" Answering this question is a key role of the PM&P management practitioners choosing the part and overseeing the PM&P selection process and its results.

[4] Department of Defense Instruction (DoDI) 4140.01, *DoD Supply Chain Materiel Management Policy*, March 6, 2019; Change 1 effective September 2, 2025, https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/414001p.pdf.

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The question is answered through the systems engineering process' decomposition of system-level requirements down to the subsystem, assembly, and part level. This requirement decomposition process is not limited to performance or physical requirements. Every consideration should lead to allocated and/or derived requirements, though some requirements associated with the supply chain or system security are not always decomposed to the part level through the systems engineering process. Other requirements, such as reliability, can sometimes be addressed through redundancy rather than by forcing all parts to meet a reliability target.

The PM&P management practitioner, whether in industry or the program office, needs to ensure that the correct subject matter experts (SMEs) are involved in assessing the PM&P selection considerations listed above. While PM&P management practitioners should have some level of knowledge in *all* the disciplines associated with the considerations, the PM&P management practitioner should engage other disciplines to solicit expert assessments.[5] Using these external inputs and their own knowledge of the system and its requirements, the PM&P management practitioner should integrate all information to decide (1) which PM&P to select and (2) how to oversee the selection process.

Although industry largely selects the PM&P for defense systems, there are three general situations where the government role sometimes includes PM&P selection:

1. Direct PM&P selection associated with a (re)design or engineering change;
2. PM&P replacement due to unavailability during maintenance/sustainment; or
3. PM&P replacement due to new, unacceptable risks resulting from changes to the part itself or changes to its supply chain.[6]

Oversight of government PM&P selection activities is also a government role performed by the program office PM&P management practitioners whose responsibilities are nearly identical to the oversight of contractor PM&P selection.[7] Examples of government PM&P selection follow:

Direct PM&P selection

- Initial design or redesign. Occasionally, a GDO may conduct initial design or major redesign of a system without using a contractor. The government may also develop (and implement) minor redesigns or engineering change proposals (ECPs) during production or sustainment. For example, minor system modifications may be made to reduce operating and support cost or make a small capability change.
- Requiring government furnished property (GFP) be used in the design.

PM&P replacement due to unavailability

- Replacing a part due to an obsolescence or another availability issue during sustainment. For example, the government may face such an availability issue when performing depot maintenance in an organic facility.

[5] Some of these considerations (e.g., performance and quality) are assessed by PM&P management practitioners; some of these considerations (e.g., supply chain risk management (SCRM) and hardware and software assurance) are usually assessed by SMEs in other functions and these experts do not consider themselves to be PM&P management practitioners; and some of these considerations (reliability and maintainability) are assessed by both.

[6] This situation is analogous to the need for requalification due to changes in design or manufacturing.

[7] One key difference is that the program office controls contractor design and PM&P selection activities through legally binding contract requirements. There is no contractual document when the government is performing those same activities; consequently a different enforcement mechanism would be needed. Another potential difference is in the program office's ability to make PM&P changes. It may be easier for a program office to influence design changes when the designer is another government organization.

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PM&P replacement due to new, unacceptable risks

- After a supplier issues a product change notice (PCN) during sustainment. For example, manufacturers issue PCNs regularly. If the impact on operations is unacceptable, the government may select a different part.
- Change to the supply chain or support infrastructure. Events can occur after initial PM&P selection and approval that could invalidate the part selection decision by introducing a greater risk to a DoW system. For example, there may be a buyout of a domestic supplier by a potential foreign entity owned supplier or the introduction of representatives of a potential foreign entity on a company's board of directors. Such changes may be unacceptable and lead to a part replacement.

In this latter situation, where PM&P may be replaced because of new risks, the government role also includes monitoring for the triggering events which are occurrences that affect the risk profile of a part or system and assessing the acceptability of the associated risk changes. The PM&P management practitioner is not responsible for monitoring, detecting, or assessing PM&P changes. That is a materiel management responsibility. The role of the PM&P management practitioner is to encourage the performance of these functions.

1.3 Importance and Applicability

PM&P management is applicable throughout a product's life cycle from prototyping to engineering models to advanced development to production and to sustainment. Therefore, PM&P management applies to several of the acquisition pathways wherever PM&P are selected.[8] While the extent of PM&P selection may vary, Urgent Capability Acquisition (UCA), Middle Tier Acquisition (MTA), Major Capability Acquisition (MCA), and even Defense Business Systems pathways all require PM&P management. For those acquisition pathways where timelines are accelerated, PM&P management becomes more critical.[9]

While the following uses MCA terminology, equivalent concepts apply to the other pathways. Due to the long-lasting impact of the selection of key PM&P, PM&P management processes should be in place by the preliminary design review (PDR). Some PM&P selections will have been made by PDR. High level part specifications should generally be understood at this phase of acquisition in instances where PM&P have not yet been selected and plans should be in place to address any knowledge gaps. Necessary test and analysis assets should also be identified.

At critical design review (CDR), although design work may not be complete, parts qualification methodologies and plans for additional tests and analyses should be in place. Parts reliability and quality should be assured for selections that have been made. Risk mitigations for restricted PM&P, parts being used outside of their specifications, and manufacturing processes should also be in place.

PM&P management continues to be important during production, operation, and support. For example:

- PM&P changes may happen because of obsolescence or other supply chain issues;
- PCNs may be issued;
- Production lot variations may occur; or
- Engineering changes of various scopes may be implemented.

[8] Appendix A.1 contains pertinent guidance related to the acquisition pathways.

[9] Concepts in this introductory material were extracted from Locker, David, "Parts Management with MIL-STD-11991," U.S. Army Capabilities Development Command Aviation and Missile Center, December 2023, https://www.dau.edu/sites/default/files/2024-02/SD-19%20Parts%20Management%20Guide%2C%20December%202023_0.pdf.

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It is imperative that PM&P management planning and activities be reviewed and adjusted to accommodate such changes.

1.3.1 PM&P Management Benefits

Disciplined PM&P management has generally been recognized as a contributor to the long-term success of any system. Selecting the right PM&P is fundamental to achieving many engineering and manufacturing objectives, and influencing cost, schedule, and performance. Disciplined PM&P management processes improve operational readiness and reduce life-cycle costs by promoting the use of common, widely available, reliable parts.[10] For example, through disciplined PM&P management, the Javelin missile successfully transitioned into low rate and full rate production with limited issues.

Javelin Success Story

Javelin is a portable anti-tank missile that can be used against a variety of targets by selecting the proper attack mode. It is autonomously guided to the target after launch, leaving the gunner free to reposition or reload immediately after launch. The Javelin PM&P Management Program included:

- Assessment of performance for each part, along with detailed tolerance analysis and derating criteria and
- Assembly level testing to provide assurance of part performance.

As a result, parts issues were identified early, and adjustments were made to support qualification, user testing, and the efficient transition to production. Special test protocols were devised and implemented to assess the long term reliability of the electronics and energetic devices. These protocols identified a systemic construction flaw in an optical sensor that would have caused significant reductions in storage life. Corrective actions were developed and implemented in time for production and supported other program offices use of the same sensor.

Beyond these traditional benefits, reasons for conducting more disciplined PM&P management have been reinforced in today's environment.

- PM&P management reduces the likelihood of system compromise resulting from supply chains for system components including hardware, material, software, data, algorithms, and humans. Compromise of a system component means that its content, function, quality, and/or reliability has been modified in some way to be something other than what the system expected. The supply chains for system components can be compromised by natural and geopolitical events or the active introduction of manipulated parts, materials, software, or information, as well as interferences with networks and processes. Similarly, the information that these systems depend on to operate can be compromised, impacting the system's support and/or manufacturing. As a result, the operators of critical systems impose a myriad of restrictions on how components can be sourced to minimize the risk of compromised components finding their way into critical systems.[11]
- Disciplined PM&P management also reduces the likelihood that cyber attackers exploit cyber physical system vulnerabilities in software, hardware, firmware, adjacent systems in the network, energy power supplies, and user interface vulnerabilities and to reduce the likelihood of impact to

[10] National Air Standard (NAS) 1524, Standardization Savings, Identification, and Calculation, supplies formulas and examples to calculate the cost of adding a part for a specific system or local organization. While some of these calculations' given variables and examples may be specific to the aerospace field, the formulas translate across all fields and can offer a framework for understanding the benefits of part standardization.

[11] Extracted from University of Maryland, *Safeguarding Critical System Supply Chains Against Compromise Workshop, July 14, 2022*, December 2022.

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the system's operational mission.[12] Cyber physical systems utilize technologies that combine the cyber and physical worlds in a way that can respond in real time to their environments.[13] Examples of cyber physical systems include the smart grid, autonomous vehicles, vehicle charging systems, satellite communication systems, and embedded systems. PM&P management is part of a holistic approach to cyber physical systems security from a system engineering perspective. In particular, PM&P selection is combined with designing in required hardware and software assurance, cybersecurity attack countermeasures, SCRM, and anti-tamper in a way that evaluates threats.

1.3.2 Consequences of Inadequate PM&P Management

Similarly, there are negative consequences from inadequate PM&P management.

- Poor performance. The most important result of effective PM&P management is that parts will meet their allocated and derived requirements defined through the systems engineering process. If PM&P management is not performed properly, parts may not meet their requirements. For example, the system may not meet its performance requirements when actual performance specifications are aggregated to the system level.
- Improper parts utilization. Poor PM&P management increases the likelihood that parts will be used outside of their rated specifications. Similarly, known problematic PM&P (e.g., the use of hazardous materials, life limiting designs or construction features, or known problematic suppliers) are more likely to be selected if insufficient attention is given to PM&P approval. Both situations contribute to poor performance, poor reliability, insufficient readiness, security vulnerabilities, safety concerns, and increased cost.
- Poor reliability. Parts have both performance and reliability requirements. When a part does not meet its reliability requirements, it is likely that the system will also not meet its reliability requirements when overall reliability is calculated. Furthermore, poor reliability can lead to decreased readiness.
- Increased DMSMS issues. Poor PM&P management could lead to the selection of parts that are at or near the end of their production life. That implies that parts may become obsolete sooner than parts that are at the earlier stages of their life cycles. In addition to affecting production schedules, there are costs associated with resolving these problems and these costs will be incurred sooner than for parts in the earlier stages of their life cycles. If the obsolescence is not identified proactively, resolution costs tend to increase.
- Increased cost—unplanned rework, greater footprint. Many of these situations lead to cost increases as noted in other bullets. Cost increases, due to ineffective PM&P management, are largely from the need for a resolution, which results in unplanned expenses. In addition, these unplanned expenses are almost always higher than the costs that would have been incurred if the right PM&P had been selected initially.
- Overreliance on sole source. Multi-source is a desirable part characteristic that is more likely to be obtained when PM&P management is performed effectively. Sole source situations may contribute to declining readiness and schedule slippage. Multiple suppliers decrease the likelihood that parts will become unavailable.
- Higher likelihood of cyber exploitation and system compromise. Ineffective PM&P management could lead to the selection of PM&P with known vulnerabilities in situations where cyber and other threat attacks are likely to be employed by adversaries. System compromise occurs when these attacks are successful.

[12] DiMase, Daniel, Zachary A. Collier, John Chandy, Brian S. Cohen, Gloria D'Anna, Holly Dunlap, John Hallman, Jay Mandelbaum, Judith Ritchie, and Ly Vessels, "A Holistic Approach to Cyber Physical Systems Security and Resilience," *2020 IEEE Systems Security Symposium (SSS)*, Crystal City, VA, 2020, 1–8.

[13] National Institute of Standards and Technology, *Framework for Cyber-Physical Systems: Volume 1, Overview*, NIST Special Publication 1500-201, version 1.0 (June 2017), https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.1500-201.pdf.

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- Increased likelihood of supply chain disruptions. For example, counterfeit and/or maliciously tampered items could penetrate the DoW supply system especially associated with obsolete parts that are a lucrative target for counterfeiters. The cost to eliminate counterfeit parts from the supply system is high. Malicious tampering might make it easier for systems to be compromised. Another example is foreign ownership, control, or influence (FOCI). FOCI is an undesirable part characteristic. It creates opportunities for malicious tampering and the loss of intellectual property or sensitive and classified information which could lead to compromise as well.
- Increased likelihood of an occurrence triggering any of the aforementioned consequences. Occurrences such as changes in ownership or control of a subcontractor, the parts themselves, parts manufacturers, parts supply chains, etc. may occur at any point in a system's life cycle. Such occurrences may "trigger" increased risk. Insufficient oversight associated with the identification of such triggering events and assessment of whether something needs to be done to address the changed risks may lead to any of the consequences of inadequate government oversight of PM&P selection.

Many program offices have suffered such negative consequences as described above by taking a laissez-faire approach to PM&P management oversight. In some cases, program offices treat problems only when they materialize and this lack of insight into PM&P has led to increased failures, delays in fielding, greater performance risk, additional testing, and higher cost to resolve the problem. Some actual electronic examples include:

- Use of pure tin. Failures led to short circuits due to pure tin whisker growth which prevented assets from returning to the field.
- Supplier switch from gold wire bonds to copper wire bonds to reduce the cost of a plastic encapsulated microcircuit. Improper testing resulted in increase to field failures.
- Parts with untreated gold leads increase risk of solder joint integrity due to gold embrittlement. Delayed hardware fielding.
- Parts with improper derating. Tantalum capacitor failures led to delays in delivery of hardware.
- Purchase of parts from unauthorized distributors. This situation leads to increased testing because of increased counterfeit risk.

A non-electronic example in the area of adhesive bonding follows. During the early years, bond-line failures disrupted a system's production line and flight tests. Problems included inadequate design, inadequate surface preparation, inadequate testing, and inadequately defined manufacturing processes. In addition, there was lack of attention to both operator and environmental variables, significantly affecting bond line integrity and causing covers or stand-off de-bonding, bracket bond failures, and de-bonding of the thermal protection system.

One program office experienced production issues due to excessively viscous silicone adhesive. The chosen adhesive for the process met all design requirements, except for the potential viscosity range. The manufacturer supplied a lot that, while it met the data sheet requirements for viscosity, was too viscous for the spray nozzles. The contractor applying it had to add solvent by trial and error. This delayed production and increased the risk that the adhesion system would fail. The PM&P selection fix, in this case, was to have more rigid receipt inspections of incoming product, test to internal/design specifications rather than manufacturer specifications/data sheets, and to identify one or more alternatives if the standard adhesive could not be procured to design specifications for a particular batch/lot.

Another program office elected to eliminate PM&P management for the development phase of a new capability to be incorporated into an existing system under the assumption that PM&P management increased cost and slowed the development process. During the system qualification test of the new capability, a new assembly failed, requiring root cause analysis that identified several issues with the development effort. The developer had introduced ball grid array devices with lead-free solder balls while

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using tin-lead solder and a tin-lead solder reflow profile. The tin-lead reflow profile did not completely reflow the lead-free balls, resulting in solder joints with lower reliability. Costly rework became necessary because another part had performance issues over the required operating temperature range that was being screened with assembly level testing. In addition, one critical part in the design was obsolete. All these issues resulted in expensive and time-consuming efforts to rebuild hardware to complete system qualification testing.

1.4 Policy and Guidance

PM&P management reflects two broad types of policy and guidance,[14] overarching and that which is pertinent to engineering design considerations across multiple disciplines. The result is a complex PM&P management landscape that requires an understanding of DoW policy and guidance that spans numerous topics.

1.4.1 Overarching Policy and Guidance

The first type of policy and guidance focuses on the general procedures associated with the PM&P management process itself. It does not include all the individual considerations that are considered and balanced in selecting a PM&P during design, whether a new design or a redesign.

Engineering for defense systems has always included PM&P selection, even if it has not always been explicitly stated as such. Engineers identify parts that will meet the allocated and derived requirements for system elements by balancing various design considerations. PM&P selection is an input to the configuration management process when establishing a product baseline from the functional baseline for all elements and enabling elements of a system's design.

The DoW's overarching PM&P management policy is explicitly found in DoDI 5000.88. DoDI 5000.88 establishes policy, assigns responsibilities, and provides procedures for applying engineering to defense systems. In section 3.6.f. "Parts Management," this policy document specifically states the following:

> *The PM [program manager] will ensure that a parts management process is used for the selection of parts during design to consider the life cycle application stresses, standardization, technology (e.g., new and aging), reliability, maintainability, supportability, life cycle cost, and diminishing manufacturing sources and material shortages. As applicable, parts management requirements should be specified in the RFP's statement of work for the Technology Maturation and Risk Reduction (TMRR), Engineering and Manufacturing Development (EMD), and production acquisition phases.*[15]

Section 3.7.f of DoDI 5000.88 is about standardization and references DoDI 4120.24,[16] which in turn references DoDM 4120.24, where Enclosure 4.2.c states:

> *Parts Management. Program offices must apply standardization processes to improve parts commonality, which may include cross program technical requirements and a business case analysis. Program offices should ensure that a parts management process is used to reduce the proliferation of parts and associated documentation and promote*

[14] The Defense Standardization Program PM&P management website, accessible at https://www.dsp.dla.mil/Programs/Parts-Management/, contains some policy and guidance documents. That website also contains information and proceedings of past PM&P management-related conferences.

[15] DoDI 5000.88, *Engineering of Defense Systems,* November 18, 2020, https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500088p.PDF. Note that this document is yet to be updated to replace "parts management" with "PM&P management."

[16] DoDI 4120.24, *Defense Standardization Program*, March 31, 2022, https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/412024p.pdf.

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the use of parts with acceptable performance, quality, and reliability, as specified in Military Standard (MIL-STD)-3018 (Reference (p)). Total ownership cost analysis must also be applied in the selection of parts for the program. [17]

As is the case for PM&P management policy, PM&P management guidance is included in engineering guidance, most importantly in the following guidance documents associated with DoDI 5000.88:

- Engineering of Defense Systems Guidebook.[18] This guidebook describes the activities, processes and practices for program managers, systems engineers, and other acquisition personnel during systems development, which includes pre-material development decisions and each of the acquisition pathways. For additional information on PM&P management content corresponding to the acquisition pathways, see Appendix A.1 of this document.
- Systems Engineering Guidebook.[19] This guidebook describes systems engineering guidance and best practices for program managers and systems engineers with DoW acquisition program responsibilities. For additional information on where PM&P management content is located, see Appendix A.2 of this document.
- Department of Defense Systems Engineering Plan (SEP) Outline.[20] This guidance document describes the structure and content of an SEP. PM&P management is included in a mapping of design considerations into contracts (in Table 2.5-1 of the SEP). For additional information on examples of PM&P management-related questions that can be used to prepare for systems engineering technical reviews and integrated logistics assessments, see Appendix B of this document.

1.4.2 Policy and Guidance Pertinent to Engineering Design Considerations

The second type of policy and guidance is that which is pertinent to engineering design considerations. The selection, procurement, and use of PM&P for DoW systems must examine and balance numerous engineering design considerations. During parts selection, an engineer determines how each design consideration affects the allocated and derived requirements (e.g., performance specifications, physical characteristics, and risk reduction) for the part to be selected.

Design considerations fall into three categories. One category, systems engineering, includes the application of traditional design considerations, such as reliability, maintainability, supportability, manufacturing, and quality, as part of the design process for a DoW system. The other two categories—system security engineering and SCRM—represent relatively new areas of growing importance, due to their potential impact on DoW system designs. The content in Appendix A.3 through A.5 contains additional information on the policy and guidance documents pertinent to the relevant design considerations across the three categories.

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[17] Department of Defense Manual (DoDM) 4120.24, *Defense Standardization Program (DSP) Procedures*, October 15, 2018, https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodm/412024m.pdf.

[18] Office of the Under Secretary for Research and Engineering, Office of the Deputy Director for Engineering, *Engineering of Defense Systems Guidebook*, February 2022, https://www.cto.mil/wp-content/uploads/2023/06/Eng-Def-Sys-2022.pdf. Note that this document is yet to be updated to replace "parts management" with "PM&P management."

[19] Ibid.

[20] Office of the Deputy Director for Engineering, *Department of Defense Systems Engineering Plan (SEP) Outline*, Version 4.1, May 2023, https://www.cto.mil/wp-content/uploads/2023/06/SEP-Outline-4.1.pdf.

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1.5 Necessary Actions to Achieve the Vision and Paradigm for PM&P Management

Given the benefits of disciplined PM&P management and the potential consequences of laissez faire PM&P management oversight, this document conveys guidance on a government PM&P management oversight paradigm with the following overarching vision.

Overarching PM&P Management Vision

The more disciplined selection, procurement, and usage of PM&P on DoW systems to meet system requirements while balancing the costs and risks associated with the engineering design considerations used in PM&P selection considerations.

The vision can be achieved by emphasizing and applying the practitioner-oriented best practices found in this document and the Standardization-related Document 26 (SD-26), *DMSMS and PM&P Management Contracting Guide* to conduct PM&P management oversight throughout the life cycle of DoW systems and equipment.[21] These best practices are encompassed through the following:

- Establishment of a program office PM&P Management Program
- Development and implementation of a program office PM&P Management Plan

The program office PM&P Management Plan differs from the contractor or GDO PM&P Management Plan. The program office Plan explains how the program office will oversee (verify and validate) the processes found in the GDO or contractor PM&P Management Plan. Section 11 of this document describes the contents of a program office PM&P Management Plan in more detail. Section 4 of this document contains additional information about a GDO or contractor PM&P Management Plan.

The steps in a program office PM&P Management Program are listed below and pictorially depicted in Figure 1. These steps represent the actions that a program office should take to oversee PM&P management.

1. Assess risk to create a risk profile used to determine the level and type of oversight to be performed by the program office.
2. Set requirements for a contractor PM&P Management Plan that describes processes to assure that PM&P selections meet all allocated and derived requirements.
3. Verify that the contractor is following the processes in its Plan.
4. Validate that the contractor's PM&P selection is effective.
5. Keep records and develop metrics designed to improve PM&P selection.
6. Perform steps 2–5 when PM&P selection is performed by GDOs.
7. Evaluate metrics and changing oversight processes where needed.
8. Detect risk changes generated by triggering events and adjust PM&P selections when warranted.

PM&P management practitioners generally do not assess risk in step 1 and detect and assess risk changes in step 8. The PM&P community should try to ensure that these activities occur in conjunction with normally performed risk management activities such as for SCRM, industrial base assessments, or

[21] SD-26, *DMSMS and PM&P Management Contracting Guide*, Defense Standardization Program Office (DSPO), June 1, 2023, https://www.dau.edu/sites/default/files/2024-02/SD-26%20DMSMS%20and%20Parts%20Management%20Contracting%20Guide%2C%20June%202023.pdf.

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mission engineering. The outcome of step 1 is a risk profile that should show a combination of the severity and likelihood of risks (e.g., security, availability, performance) on groupings of parts (usually subsystems). The level and extent of oversight may vary for the different aspects of the system shown on the risk profile.

The risk profile drives the content of a program office PM&P Management Plan, which lays out the level and type of PM&P oversight.[22] The Plan begins with establishing PM&P selection requirements often based on an appropriately tailored MIL-STD-11991 that meets the needs of the program office.[23,24] The contractor or GDO PM&P Management Plan will be designed to meet these requirements, and it encompasses the processes to select PM&P.

If a program office manages the acquisition and/or support for multiple systems, the program office should apply each of the steps of its PM&P Management Program separately for each system. In addition, a program office should apply each step separately to every contractor or GDO selecting PM&P for the system. This is important because whether a system is being developed, modified, manufactured, or sustained can imply different PM&P selection risks, varying needs for subject matter expertise, and diverse levels of rigor for a program office PM&P Management Plan. It is not uncommon for a program office to simultaneously have contractors for development, production, and/or sustainment.[25]

As indicated in Figure 1, the sections of the program office PM&P Management Plan align with the steps in the program office PM&P Management Program. Although oversight of design/development activity is different than oversight of maintenance, all contractors and GDOs could be encompassed in a single program office PM&P Management Plan as long as any differences among them are highlighted.

Sections 2–10 of this document contain detailed discussions of all aspects of a program office PM&P Management Program as shown in Figure 1. Section 11 separately describes the corresponding program office PM&P Management Plan(s).

[22] Section 11 of this document provides a more detailed description of a program office PM&P Management Plan.

[23] No mention of MIL-STD-3018 is made in this document because has been cancelled. Future contracts may reference the general requirements of MIL-STD-11991 instead of MIL-STD-3018 when only more limited requirements are needed.

[24] In certain highly critical, sensitive, and technically complex situations, more rigorous PM&P selection requirements should be imposed. Lesser requirements may be needed when PM&P selection is minimal and the program office has confidence in the PM&P selection processes.

[25] Multiple contracts for similar functions on a system may have identical program office PM&P Management Plans.

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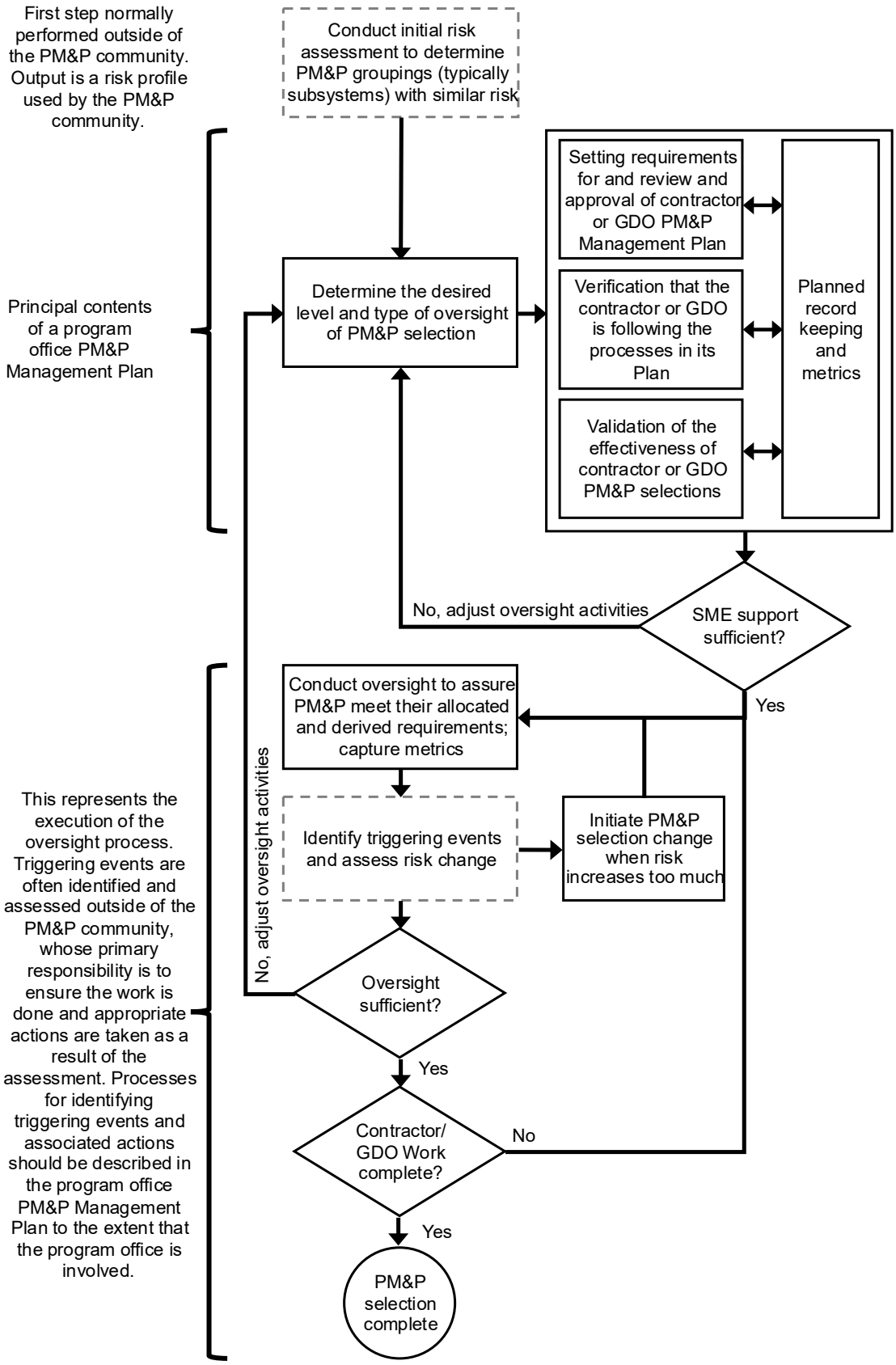


Figure 1. Program Office PM&P Management Program

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Section 2. Conducting an Initial Risk Assessment

The initial risk assessment is the first step in a program office PM&P Management Program. As shown in Figure 1, it results in a risk profile to be used in determining the level and type of a program office's PM&P management oversight, which primarily encompasses:

- Setting requirements for and reviewing and approving a contractor and/or GDO PM&P Management Plan;
- Verifying that the contractor and/or GDO making PM&P selections is following the processes in its Plan; and
- Validating that the contractor and/or GDO processes are effective in selecting PM&P.

This risk assessment should not be conducted in isolation. It should be performed in conjunction with other risk assessments. There should be a collaboration with SCRM and other common risk assessments such as those for safety, mission, security, and programmatic risks. The principal role of the PM&P community is not to conduct the risk assessment itself; rather the community should ensure that the assessment takes place and that the resulting risk profile becomes the basis for establishing the level and type of PM&P selection oversight.

The initial risk assessment gives program managers a means to prioritize resource requirements. While risk may change with the threat environment, suppliers, and regulations, the *initial* assessment reflects a baseline need for oversight resources to mitigate the risks to an acceptable level. If this level of resources is not provided, a program office should conduct additional risk assessments to understand the level and type of oversight that can be applied and highlight to decision makers the additional risk resulting from inadequate resourcing.

Because rigorous oversight over all aspects of the system may not be needed, the risk profile should show risk-based aggregations of parts (typically by subsystem which is the term this document will use) that may need special attention and different levels of oversight. Risk determination and aggregation is at the discretion of the program office based on the complexity of the system, system requirements, the extent of quantitative risk information available, part criticality, and mission criticality. Too high a level of aggregation could lead to inadequate attention being focused on critical parts within that aggregation.

Even though the risk assessors are not in the PM&P community, the community can support the process. Therefore, Subsection 2.1 and subsection 2.2 along with Appendix C respectively describe a less complex and a more complex risk determination and aggregation approach that may be used and tailored to the unique parts within the system. Because the risk assessment should not be stand-alone, the subsection 2.1 approach is more practical than the formal processes referenced in subsection 2.2. Subsection 2.3 suggests considerations for refining those approaches further to help finalize the risk profile.

2.1 Risk Determination and Aggregation via Subject Matter Experts

Since risk determination and aggregation are highly complex and dependent upon a system's design, missions, and operating environments, this first approach relies heavily on SMEs. The experts should first identify the critical system functions and the subsystems used to perform those functions. For many systems, these critical functions are those involved with

- Information and communication technology (ICT),
- Propulsion,

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- Fire control,
- Sensors, or
- Command and control.

To generate a list of subsystems where PM&P selection risks are highest, these lists of critical functions and subsystems should be further refined (through both additions and subtractions) by the SMEs because of their knowledge of

- The system itself,
- Missions,
- Operating environments, and
- Threats.

The concept of different risk tolerances as a function of mission may be an overriding consideration by the SMEs. The space community uses four risk tolerance levels when determining the level and extent of a program office's PM&P selection review and approval:[26]

- Minimum practical,
- Low,
- Moderate, and
- High.

To supplement subject matter expertise principally for ICT, the program protection plan (PPP) identifies critical ICT component supply chain risks and recommended mitigation actions. Program protection planning identifies critical program information (CPI), threats to CPI, and the associated risk and then determines those critical subsystems/components where protection measures are needed.

Program Protection Definitions

Program protection is: "The integrating process for managing risks to DoW warfighting capability from foreign intelligence collection, from hardware, software, and cyber vulnerability or supply chain exploitation, and from battlefield loss throughout the system life cycle."[a] Program protection planning identifies CPI, threats to compromise CPI, and the associated risk. It is DoW policy that: "CPI will be identified early and reassessed throughout the RDT&E program so that CPI protections requirements and countermeasures may be identified and applied as the CPI is developed and modified throughout the life cycle as needed." In addition, "CPI protection measures will be integrated and synchronized, then documented within the PPP in accordance with [DoDI 5000.02[b]].[c]"

CPI are: "U.S. capability elements that contribute to the warfighters' technical advantage, which if compromised, undermines U.S. military preeminence. U.S. capability elements may include, but are not limited to, software algorithms and specific hardware residing on the system, its training equipment, or maintenance support equipment."[d]

[a] Warfighting Acquisition University, DAU Glossary, "program protection."

[b] DoDI 5000.02, *Operation of the Adaptive Acquisition Framework*, January 23, 2020 incorporating change 1, effective June 8, 2022, https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500002p.PDF.

[c] DoDI 5200.39, *Critical Program Information (CPI) Identification and Protection Within Research, Development, Test, and Evaluation (RDT&E)*, May 28, 2015, incorporating change 3, effective October 1, 2020, https://www.esd.whs.mil/portals/54/documents/dd/issuances/dodi/520039p.pdf.

[d] Ibid.

[26] Aerospace Report No. TOR-2011(8591)-21, Mission Assurance guidelines for A-D Mission Classes, June 3, 2011, https://s3vi.ndc.nasa.gov/ssri-kb/static/resources/TOR-2011(8591)-21.pdf.

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In practice, there often are some additional critical areas, not explicit in the PPP, that are similarly vulnerable to exploitation in a way that degrades the level of assurance. For example, every subsystem using logic-bearing microelectronics devices will typically not be explicit in a PPP. Reasons for such omissions include:

- Identifying subsystems associated with CPI is partially subjective;
- The PPP may not be current;[27]
- Program offices often attempt to minimize CPI because of constraints on funding and people; and
- The scope of program protection activities could be limited because no standardized CPI and critical functions and components identification process is consistently applied across DoW.

Therefore, a program office should supplement PPP-derived subsystems with other subsystems identified by SMEs from all the system security engineering disciplines as discussed above.[28]

2.2 Risk Determination and Aggregation via Formal Analysis

A more formal risk determination and aggregation is usually not needed since the PM&P risk assessment should be determined as part of another required risk assessment. If a program office determines that a formal analysis should be conducted, the steps in the risk assessment could be as follows.

1. Identify and organize PM&P selection risks and risk tolerances to be considered in a program office PM&P Management Program;
2. Classify the identified risks by likelihood and severity of consequence; and
3. Aggregate risks by subsystem (or possibly some other grouping of parts).

These steps are pictured in Figure 2.

[27] The PPP should be aligned with the Life Cycle Sustainment Plan (LCSP) and the Systems Engineering Plan (SEP) during all phases of the acquisition life cycle to ensure any changes in the status of CPI and critical components are effectively managed.

[28] While it would be good if this were a standard system security engineering process, it may be necessary to seek information about the need for additional protection on a case-by-case basis.

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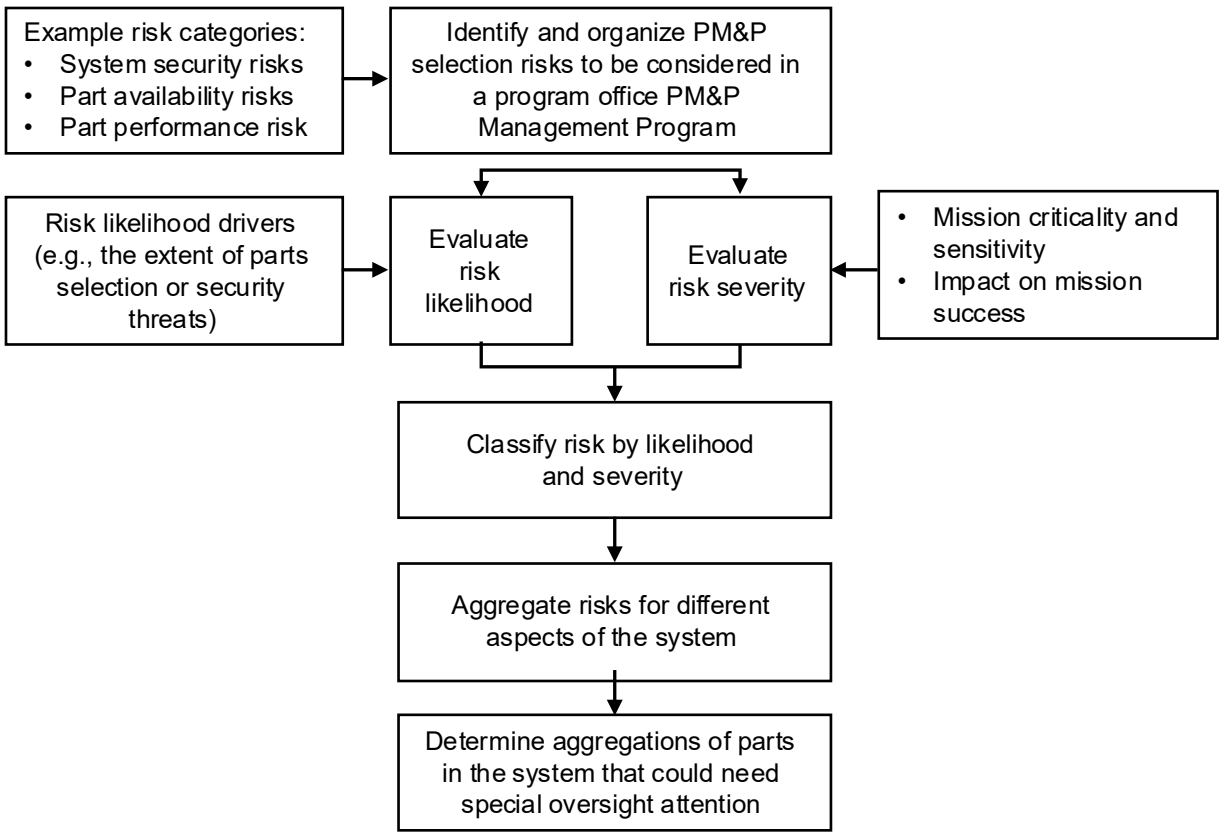


Figure 2. Potential Risk Assessment Steps

Appendix C contains details of a formal risk assessment.

2.3 Finalizing the Risk Profile

Simply having a list of high-risk subsystems may not be sufficient to determine level and type of oversight since there are variations in how oversight is performed. The same level and type of oversight may not be appropriate for every high-risk subsystem and for every part within those subsystems. This section provides the program office with considerations for further prioritizing these subsystems and parts to enable better oversight differentiation.

The first consideration should be the Military Department determination of part criticality. It may be appropriate to apply less rigorous oversight to non-critical parts. Part criticality may be further refined by separately categorizing parts associated with Technology Area Protection Plans, Science and Technology Protection Plans, and PPPs.[29]

Another consideration is part complexity. Complex microelectronic parts such as application specific integrated circuits or microcontrollers may be given more rigorous oversight. The same applies to programmable devices such as field programmable gate arrays.

Mission criticality is a third differentiation factor. Parts associated with subsystems involved in national defense missions with the highest criticality should receive greater levels of oversight. Similarly, parts

[29] DoDI 5000.83, *Technology and Program Protection to Maintain Technological Advantage*, May 21, 2021, https://www.esd.whs.mil/Portals/54/Documents/DD/issuances/dodi/500083p.pdf.

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identified with the most important missions for the system itself (but not at the national defense level) may be given greater oversight than other critical parts.

Parts within a subsystem may also be prioritized through the level of supply chain risks they represent. The following questions can help differentiate identify some of the more prevalent risks:

- Is this part from a sole-source or single-source supplier?
- Does the part have a manufacturing lead time greater than 12 months?
- Is there a known DMSMS issue?
- Does the part depend on rare earth elements or other materials with volatile supply chains?
- Is the underlying technology for this part more than 10 years old?
- Is this a new/unproven technology with no significant field or test history?
- Is this part type (e.g., high-density field programmable gate arrays, connectors) a known target for counterfeiting?
- Is there a history of GIDEP (Government-Industry Data Exchange Program) alerts for this part or its manufacturer?
- Is the part identified as CPI within the Program Protection Plan?
- Has the part exhibited high failure rates during developmental testing or in previous applications?
- Does the part have a high individual unit cost (e.g., >$50,000 per unit)?

After all these considerations are taken into account, the parts associated with high-risk subsystems will be divided into prioritized groups for which different levels of oversight may be required as illustrated in Figure 3.

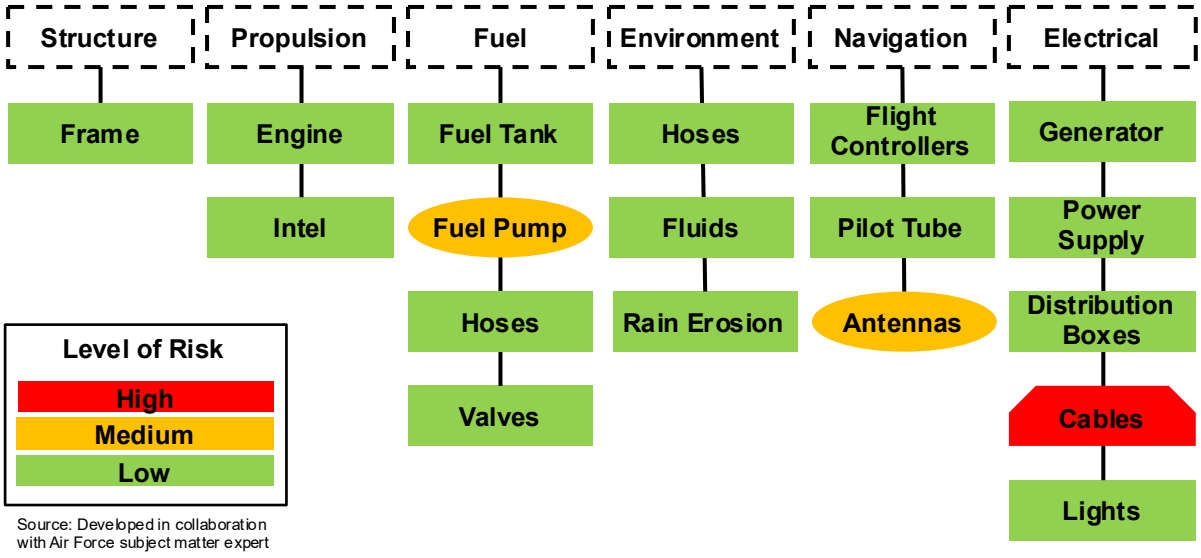


Figure 3. Illustrative Risk Profile

This illustrative profile integrates risk severity, risk likelihood, and other considerations deemed important into a single red-yellow-green indicator. As such, the profile implies that additional oversight may be needed for PM&P selection for fuel pumps and navigation antennas and even more intense oversight may be needed for electrical cables. A more complex profile might not aggregate all these factors to enable a more refined variation and determination of the level and type of oversight.

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Section 3. Overview of Considerations in Determining the Level and Type of Program Office Oversight

Section 3 discusses high-level *considerations* for deciding the specifics of the PM&P management oversight to be performed by the program office. Figure 4, extracted from Figure 1, highlights the pertinent material.

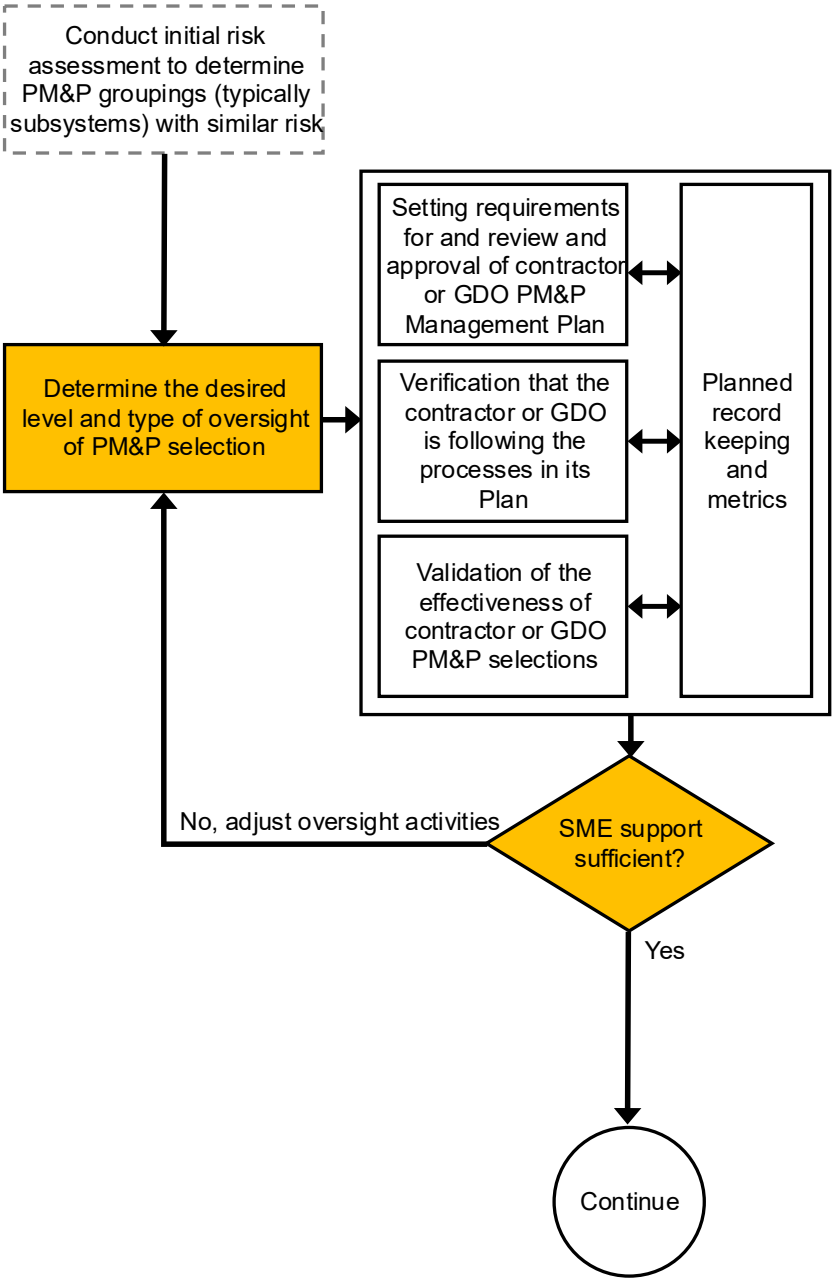


Figure 4. Overview of Determining Level and Type of Oversight

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Based on the risk profile, the program office should establish the level and type of oversight to achieve the desired level of assurance that parts meet their allocated and derived requirements. Program offices should recognize that although the level and type of oversight impacts cost, acquisition cost increases for PM&P management are historically minimal while delivering substantial benefits in system life-cycle cost. Acquisition cost increases should not drive reduced priority to sustainment concerns.

The program office's PM&P management oversight will be performed on contractors and GDOs if they are involved. This oversight must ultimately be enabled and based on contract requirements or the equivalent to perform specific PM&P management functions and to obtain the data necessary to conduct the oversight.

Figure 4 implies that the determination of the level and type of oversight is a two-step process. The first step develops an initial determination based on the material in sections 4–8, which contain greater detail on the level and type of program office oversight associated with a contractor or GDO PM&P Management Plan, verification that the processes in that Plan are being followed, validation of parts selection effectiveness, and planned record keeping. The second step revises that initial determination based on the availability of the necessary subject matter expertise needed to perform the oversight.

Figure 5 portrays another way of visualizing this two-step process. Answers to key oversight related questions shown in section 3.1 should be developed from the risk profile. Those answers imply the need for a specific set of knowledge, skills, and abilities (KSAs) as shown in section 3.2. Next, section 3.3 covers estimating the number of full-time equivalents (FTEs) of people with those KSAs. Finally, section 3.3 discusses obtaining access to those FTEs and, if there are constraints placed on securing those FTEs, how the level and type of oversight should be adjusted.

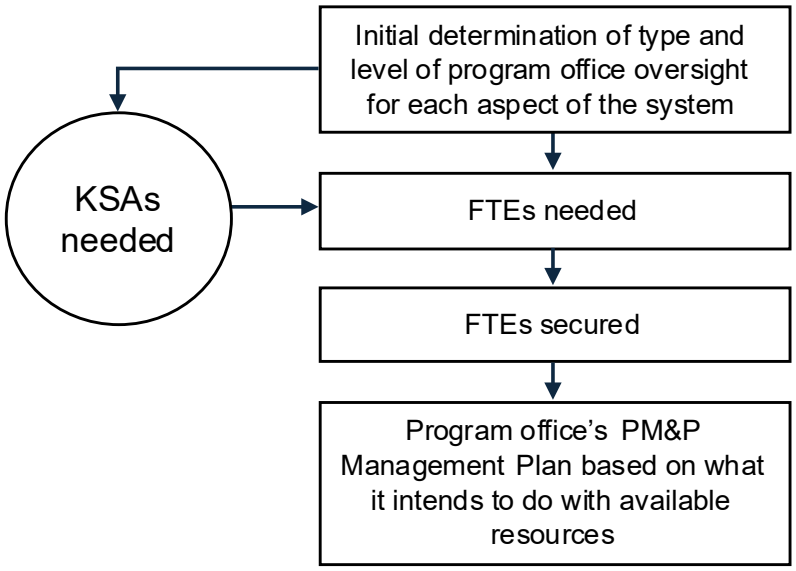


Figure 5. Securing Personnel Resources to Support Level and Type of Oversight

3.1 Questions Defining Oversight Needs

The risk profile informs an initial determination of the level of program office oversight necessary for the contractor's or GDO's PM&P management activities.[30] Therefore, the risk profile assists a program office

[30] The final determination of oversight activities will be articulated in the program office PM&P Management Plan. Changes from the initial determination may be the result of unavailability of the requisite government subject matter expertise or a decision to accept greater risk in the PM&P selection process. Greater oversight intensity implies lower risk that the PM&P do not meet all the program office's objectives or, equivalently, less rigorous PM&P management objectives.

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in answering the following types of questions.[31] The questions are organized around the types and levels of oversight activities the program office may perform and are used to determine the desired level of effort associated with those activities.

Questions on establishing the requirements for a contractor (or GDO) PM&P Management Plan are as follows:

- Does the program office need to receive and approve a contractor (or GDO) PM&P Management Plan?
- What document does the program office need to use to specify broad requirements for PM&P?
- Does the program office need to impose further process requirements?
- To what extent does the program office need to specify PM&P technical requirements and prohibitions?[32]

Questions on determining the level and type of a program office's verification efforts include:

- Does the program office need to verify that a contractor (or GDO) is following the processes in its Plan?
- What data should be reported to the program office?
- What reviews and inspections should the program office conduct to enable it to verify that the contractor (or GDO) is following the processes in the contractor (or GDO) PM&P Management Plan?
- What level of collaboration and coordination between the program office and the contractor (or GDO) on its PM&P Management Plan is required for efficient and effective implementation, such as working groups or integrated product teams?

Questions on determining the level and type of a program office's validation efforts include:

- Does the program office need to validate the effectiveness of the contractor's (or GDO's) PM&P management processes?
- Should the validation efforts vary? Same for all PM&P? Focus on critical PM&P? Tailored for different aspects of the system?
- What data should be reported to the program office, and what reviews and inspections should the program office conduct to enable it to validate the effectiveness of the processes in the contractor (or GDO's) PM&P Management Plan?
- What parts does the program office need to approve before the PM&P selected are used in the system?
- What information should be included in the request for PM&P approval?

Using the answers to the above questions, program offices should:

- Specify the knowledge, skills, and abilities (KSAs) needed,
- Determine the associated level of effort, and
- Secure access to the personnel who will provide the needed level of effort by KSA.

The following three subsections discuss each of the above bullets.

[31] The order of these questions is generally aligned to the illustrative PM&P management requirements in the SD-26.

[32] Note that how the contractor or GDO should select PM&P is generally not a contract requirement. That function is inherent in the systems engineering process by which PM&P are selected to meet a design in a way that satisfies all system requirements.

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3.2 Specifying the Knowledge, Skills, and Abilities Needed

Traditionally, PM&P management KSAs have focused on the multiple disciplines of engineering. The following list is representative of potential, traditional KSAs for overseeing contractor PM&P selection and approval:

- Knowledge of mechanical, electrical, and chemical engineering
  - Abilities in circuit, structural, thermal, and stress/strength analysis
  - Abilities in corrosion and degradation analysis
- Knowledge of material properties, including surface properties
  - Abilities in construction and physical analysis
- Knowledge of material characterization techniques
- Knowledge of part construction and manufacturing processes[33]
- Knowledge of assembly processes
- Knowledge of welding, soldering, brazing, mechanical joining, and adhesive joining
- Knowledge of additive manufacturing
- Knowledge of part manufacturer business practices and critical customers
- Knowledge of reliability, quality, and design engineering
  - Skill in fault tree, failure, and root cause analysis
  - Skill in statistics and failure distributions
- Knowledge of historical failure mechanisms
- Knowledge of accelerated testing techniques
- Knowledge of inherent design problems that prevent parts from meeting allocated and derived requirements.

Such traditional KSAs are necessary but not sufficient in today's PM&P selection environment. Supply chain issues, highlighted by the pandemic, and the risks associated with system compromise by adversaries, have demonstrated the need for additional KSAs associated with system security engineering and SCRM:

- Knowledge of SCRM
  - Ability to analyze suppliers' capacity to meet demand based on broad economic conditions and supplier financials
  - Ability to analyze suppliers' capacity to meet demand based on capacity, infrastructure, human capital, and transportation/distribution limitations
  - Ability to analyze suppliers' capacity to meet demand based on political and regulatory, environmental, and other compliance factors
  - Ability to analyze DoW wholesale suppliers' capability to keep stock on hand, based on factors such as commonality of parts with other DoW systems, expected demand rates, and costs
  - Ability to assess risks associated with FOCI
  - Ability to assess risks associated with counterfeit and malicious tampering
- Knowledge of system security engineering
- Knowledge of critical hardware/software vulnerabilities
- Knowledge of cyber-attack opportunities and mitigation of risk
- Knowledge of techniques to secure networks and systems
- Knowledge of program protection planning and CPI and critical ICT component identification
- Knowledge of criteria for achieving desired levels of hardware and software assurance.

[33] Usually developed one part type at a time to build knowledge base.

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The risk assessment may not specifically highlight all the exact skills needed. Some KSAs may be a very small part of the government's oversight and approval function. Therefore, most program offices should obtain access to key personnel possessing skills and experience representative of the comprehensive list of KSAs on an as needed basis. Because few individuals have the broad levels of experience necessary to encompass all or even a majority of this long list of KSAs, access to such a set of KSAs will normally be obtained from a wide variety of people, not a small set of experts.

3.3 Determining the Necessary Level of Effort

There may not be a formal way to calculate a level of effort required as a function of the risk assessment. Program offices may ultimately rely on the judgement of experienced government SMEs in combination with a workload analysis tool. Highly critical missile defense program offices with intense PM&P management oversight suggest 4 FTEs. Some Acquisition Category (ACAT) I aviation and missile program offices with rigorous PM&P management oversight use 1–3 FTEs.

The level of effort should be a function of:

- Oversight activities,
- Number of (critical) PM&P in the system, and
- Life cycle stage of the system.

If the risk assessment concluded that only program office approval of a contractor PM&P Management Plan is a sufficient level of oversight, then a minimum of ~0.25 FTE may be adequate.[34] If additional oversight is needed to address risk (e.g., PM&P approval, monitoring contractor PM&P selection activities, …), then the level of oversight effort will vary as a function of the number of (critical) PM&P. During design and development, the minimum level of effort should be 1 FTE for ACAT I program offices. For other ACAT levels, and non-ACAT program offices, a single FTE could be shared among several program offices.[35] Generally, a lower level of support is needed during production and sustainment because PM&P selection activities will be much lower. The final determination of PM&P management staffing levels should be reflected in the program office's technical staffing plan included in the SEP.[36]

3.4 Securing Access to the Necessary Level of Effort

The KSAs may be provided by a combination of government and contractor personnel. In practice, however, contractors will often be responsible for providing the majority of the KSAs. Some of the KSAs may not be common in the government, and if found in government, such personnel may not be available when needed because they are in high demand.

Usually, program offices have Systems Engineering and Technical Assistance (SETA) contracts to provide on-call subject matter expertise for many of the various KSAs needed to support PM&P management. Therefore, program offices should define the statement of work/capabilities for those SETA contracts to include the necessary level of effort and response time for the PM&P management KSAs. Associated risk increases should be documented in the program office PM&P Management Plan.

If resource or other constraints lead to a decision not to secure the necessary level of effort, the program office PM&P Management Plan should document both the additional risk tolerated and the actual level and type of oversight to be conducted.

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[34] A thorough contractor PM&P Management Plan should take about 1–3 person weeks to review.

[35] Larger non-ACAT program offices may require one FTE.

[36] Office of the Deputy Director for Engineering, *Department of Defense Systems Engineering Plan (SEP) Outline*, Version 4.0, May 2023, https://www.cto.mil/wp-content/uploads/2023/06/SEP-Outline-4.1.pdf.

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Section 4. Review and Approval of a Contractor or GDO PM&P Management Plan

Figure 6 highlights the elements of Figure 1 that relate to the review and approval of a contractor or GDO PM&P Management Plan.

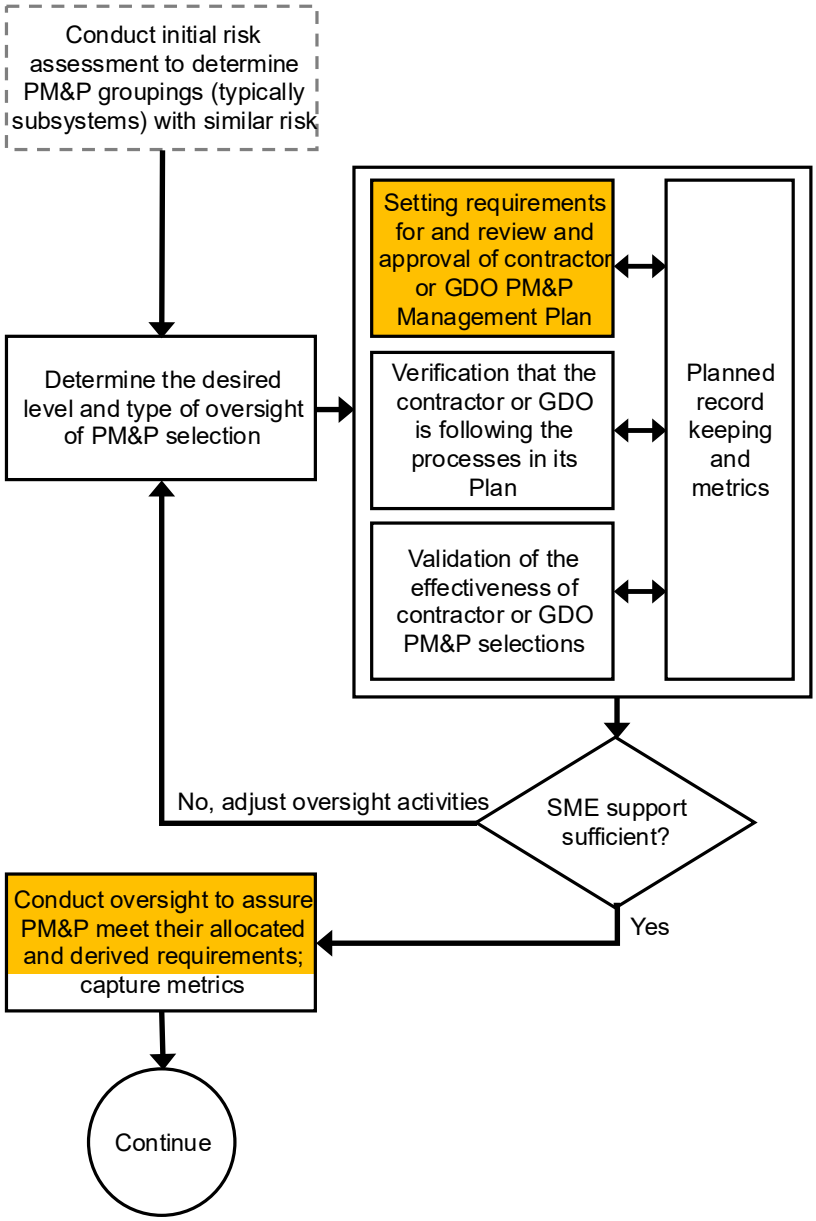


Figure 6. Review and Approval of Contractor or GDO PM&P Management Plan

Section 4.1 covers the establishment of PM&P selection requirements based on MIL-STD-11991. Options include:

- No contractor PM&P Management Plan required

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- Require a contractor PM&P Management Plan in accordance with the general requirements of MIL-STD-11991 tailored to the specific situation
- Require a contractor PM&P Management Plan in accordance with the general and detailed requirements of MIL-STD-11991 tailored to the specific situation
- Supplement the manufacturing process requirements of MIL-STD-11991
- Supplement PM&P requirements and prohibitions of MIL-STD-11991
- Replace MIL-STD-11991 with an alternative requirements document under circumstances where more rigorous requirements are necessary.

Section 4.2 identifies the most important oversight considerations for reviewing and approving a prime contractor, GDO, or subcontractor PM&P Management Plan.

4.1 Establishing PM&P Selection Requirements

4.1.1 Determining Whether a Contractor or GDO PM&P Management Plan Is Needed

Nearly all system development, production, modification, and support contracts should have a requirement for a contractor PM&P Management Plan that the program office must approve. The same applies in the rare situation where a GDO is involved, although the arrangement with a GDO would generally not be through a formal contract. A contractor or GDO Plan may not be needed in the following cases:

- There is no possibility of new or different PM&P being selected or used
- PM&P selection activity is minimal
  - No PM&P will be selected for critical/sensitive applications
  - No PM&P will be selected where there is an especially high consequence of failure
  - The program office has a high degree of confidence that the contractor's or GDO's internal PM&P selection processes are adequate
  - The program office has a high degree of confidence that the contractor or GDO will follow its documented PM&P selection processes.

Such conditions are unusual. Depending on the results of its risk assessment, the program office should determine whether the contractor PM&P Management Plan describes the processes for meeting either the general requirements or all (general and detailed) requirements of MIL-STD-11991.

4.1.2 Tailoring MIL-STD-11991 Requirements

Some considerations for limiting the contractor or GDO PM&P Management Plan to the general requirements of MIL-STD-11991 follow:

- The system is expected to perform minimal or no critical/sensitive functions and program office approval is required for PM&P associated with those critical/sensitive applications.
- There are minimal or no system applications with an especially high consequence of failure and program office approval is required for PM&P associated with those applications where there would be a high consequence of failure.
- The program office wants to reduce the burden on the contractor or GDO and therefore the government accepts any implied risk.
- The program office was not able to secure the required level of effort for SMEs to oversee the detailed requirements of MIL-STD-11991 and therefore the government accepts any implied risk.

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The program office should recognize that production and support contracts may involve PM&P selection if approved PM&P become unavailable. Therefore, a tailored contractor PM&P Management Plan may be applicable, and the program office may limit the Plan to the general requirements of MIL-STD-11991. If the contract contains a requirement that the program office must approve the use of all PM&P that have not been previously approved in drawings or similar documents, then the Plan could be limited to the program office approval process itself. Another contract option, in lieu of a formal approval process, is to require a program office determination of what to do when a part is not available.

Subcontractors selecting PM&P for the system (often in the first two tiers below the prime contractor) should also be required to prepare a PM&P Management Plan as part of the flow down of contract requirements. In some cases, subcontractors accept the prime contractor PM&P Management Plan and therefore no independent document is produced. In other situations, subcontractors develop separate Plans. When a GDO is involved, this discussion applies to the GDO's supply chain.

Regardless of whether only the general requirements or all requirements of MIL-STD-11991 are invoked, the contractor or GDO Plan should describe how the contractor or GDO intends to address those requirements along with:

- Requirements stated elsewhere in MIL-STD-11991 (see Figure 7) when referenced by either the general or detailed requirements (e.g., when the general requirements indicate that the requirements in an appendix apply). This is important when the program office either adds or deletes requirements which may refer to other requirements that should not be deleted accidentally.
- Other PM&P management requirements explicitly included in the contract (or analogous document when a GDO is involved) that add to, modify, or replace MIL-STD-11991 (e.g., inclusion of other commercial standards, requirements associated with problematic additive manufacturing processes, exception reporting, finishes and coatings, soldering).[37]

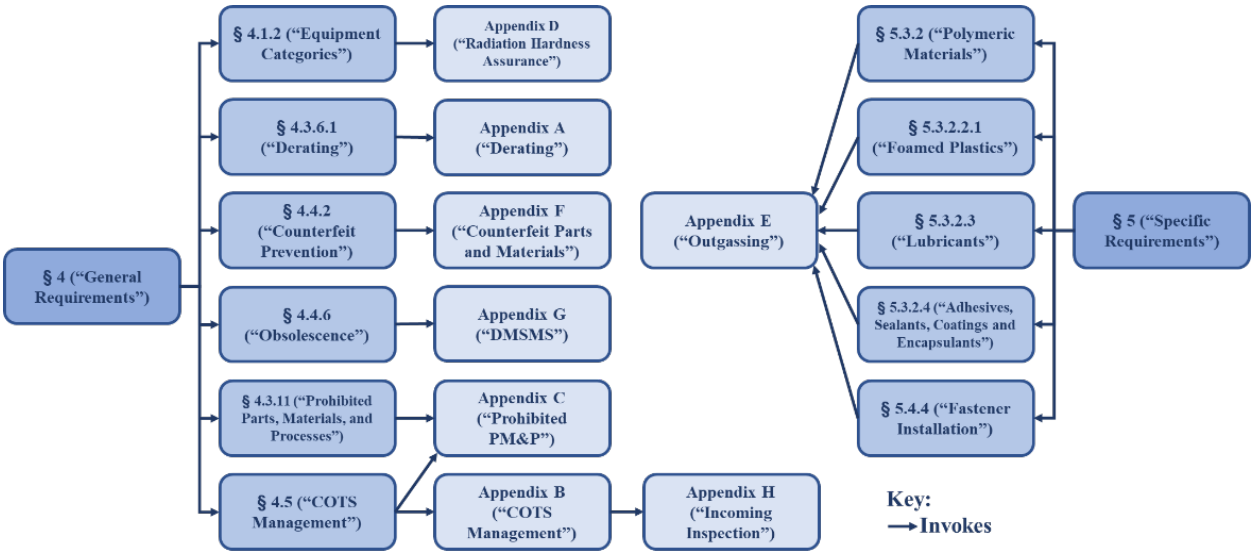


Figure 7. Mapping of Requirements in MIL-STD-11991

[37] The program office may want to clarify the definition of a part for additive manufacturing. For example, part specification should include process control for key variables and materials, feedstock, quality, and meeting application requirements.

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The contractor or GDO Plan should exclude any MIL-STD-11991 requirements that the program office eliminates to remove unnecessary, overly costly, and/or burdensome requirements based on the results of the risk assessment.

This alignment with MIL-STD-11991 does not imply that the level of detail found in the contractor or GDO PM&P Management Plan is similar across program offices. The program office should tailor the Plan requirements for the specific situation. The program office may explicitly include requirements that add to, modify, or replace those in MIL-STD-11991. Often, these changes are made to add emphasis in a way that improves the likelihood of avoiding situations that experience has shown may lead to problems when the system is fielded. Sometimes these changes eliminate requirements that do not apply to the system.

Finally, tailoring of the Plan may be proposed by the contractor or GDO. The contractor or GDO should propose a PM&P Management Plan to meet the program requirements, which may include some exceptions or modifications to MIL-STD-11991 as suitable to meet all system performance requirements. MIL-STD-11991 requirements represent extensive lessons learned on PM&P selection for military applications, but do not substitute for the critical thinking needed to implement an effective PM&P program. Many additional factors (e.g., criticality, operating environment, quality, reliability, cost, redundancy) should be carefully considered when applying these requirements and the processes to meet those requirements to each unique situation. The program office acceptance of the proposed alternative will be indicated by an approval of the Plan.

To better enable such tailoring, the program office should work with the contractor or GDO early and often to explain the intent of each requirement and describe what it wants to see in the contractor or GDO PM&P Management Plan, which will become the basis of acceptance or rejection of that Plan. This basis will be a function of knowledge of contractor processes, system requirements and system specifications, and prior experience for similar situations. Once the contractor or GDO has submitted a proposed Plan, the program office will review the Plan and either request updates to the Plan if portions are unacceptable or explicitly explain the changes needed. If changes are made to the proposed Plan, the program office will review changes and accept the Plan once it is satisfied with the revisions. The following lists more detailed steps to enable contractor or GDO tailoring:

Before the contract is finalized

- Program office determines what it needs and includes those needs in the statement of work, which will eventually become part of the contract.
- Program office ensures that system requirements and system specifications contain sufficient information for the contractor or GDO to write a PM&P Management Plan.

These two “before the contract is finalized” bullets represent good request for proposal development practices. A program office cannot have discussions with individual, competing contractors about specific expectations for a Plan outside of general industry meetings or open question and answer sessions. The objective of these bullets is to provide potential contractors with sufficient information about what the program office wants to see in a Plan so contractors can properly consider that information in its proposal.

After the contract is signed

- Program office meets with the contractor or GDO to explain 1) any added requirements and deleted MIL-STD-11991 requirements and (2) what it wants to see in the contractor or GDO PM&P Management Plan (i.e., the basis of acceptance or rejection of the Plan). What the program office wants to see will be a function of:
    - *Knowledge of contractor processes.* The contractor may have established processes that help assure meeting PM&P requirements that the proposed PM&P Plan should leverage. Examples may include preferred parts and supplier lists based on prior history in similar

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applications, or disciplined drawing review requirements that include component engineering and materials and processes engineering sign-off.

- *Other requirements in the system.* The PM&P Plan should efficiently collaborate and leverage related program requirements such as supply chain risk and obsolescence management.
- *System requirements and system specifications.* The system specification should include detailed life cycle environmental and operational profiles along with system reliability, availability, and maintainability requirements to allow derivation of part reliability requirements.
- *What the contractor knows.* The contractor may have established processes based on prior experience and successful performance history in applications similar to the program requirements that the PM&P Plan should leverage.

- Program office reviews the contractor or GDO PM&P Management Plan.
- Program office requests updates to the Plan if portions unacceptable; the program office must explain the changes it wants.
- Program office reviews the Plan changes.
- If the Plan changes are unacceptable, the program office should work with contractor or GDO to rewrite the Plan, perhaps providing the verbatim changes it wants.

Oversight may be limited to review and approval of the contractor or GDO PM&P Management Plan. If the program office does intend to limit its oversight in that way, the contractor or GDO Plan could be based on either the general or both general and detailed MIL-STD-11991 requirements per the criteria discussed above. However, even when the general requirements may be sufficient, the detailed requirements could be included to add rigor to the contractor's or GDO's processes in the limited oversight situation.

4.1.3 Information on Associated Formal Contract Requirements

The SD-26 illustrative requirements applicable to this section of the program office Plan are concerned with whether a contractor Plan should be required, and if required, whether the contractor Plan should be based on only the general requirements of MIL-STD-11991 or both the general and detailed requirements of MIL-STD-11991.[38]

SD-26 Description

The SD-26 is primarily composed of a series of tables and appendixes that provide illustrative language for DMSMS and PM&P management contract requirements along with implementation notes about their applicability as a function of different life cycle circumstances. The illustrative contract requirement wording should be tailored to complement the program office's Acquisition Strategy and its corresponding product support concepts, competition strategy, and intellectual property strategy. Not every requirement is applicable in all circumstances. Inclusion of any of the referenced contract requirements should be based on assessments of the specific risk to the program office and the availability of suitable, government subject matter expertise to review contract proposals and contractor submitted deliverables to ensure they meet the government's requirements. Other material in the SD-26 includes:

- Other sections of a contract where PM&P and DMSMS management concepts should be included and suggestions of what should be conveyed;
- Requirement applicability during each phase of the MCA pathway and as a function of the level of government involvement in PM&P and DMSMS management; and
- Associated contract data requirements lists and data item descriptions.

[38] See SD-26 illustrative requirements 24, 25a, and 25b.

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The SD-26 contains two other illustrative contract requirements to tailor MIL-STD-11991 that should be reflected in the contractor PM&P Management Plan. One requirement is associated with manufacturing processes that should be included in the contractor Plan. The second adds additional PM&P specification requirements or PM&P prohibitions; additional guidance on these requirements is shown in Appendix C of MIL-STD-11991.

The contract should specify that the program office has the right to review and inspect material needed to review and approve the contractor Plan and documents referenced by that Plan. The material to be inspected or reviewed should be specified along with the period of performance of the contract when this review and inspection can occur.

4.1.4 Establishing More Rigorous PM&P Selection Requirements

Based on lessons learned, some program offices have determined that minimum PM&P selection requirements should exceed those of MIL-STD-11991 because of the criticality and/or demanding nature of the system's missions. In those situations, the contractor or GDO PM&P Management Plan should be based on documents most appropriate to the specific situation. In the case of the Missile Defense Agency (MDA), for example, an internal document is used instead.[39]

4.2 Oversight Considerations

A program office should concentrate its review of the contractor or GDO PM&P Management Plan on the processes associated with the principal PM&P management technical requirement areas of highest concern. At a high level, the program office review of the contractor's or GDO's Plan should ensure that the Plan addresses the general requirements in MIL-STD-11991, informed by detailed requirements where applicable and managing risks for areas such as obsolescence, counterfeits, lead-free electronics, plastic encapsulated microcircuits, and additive manufacturing in part through the submission of BOMs or parts lists.

Ultimately, the contractor's or GDO's Plan should provide the program office with confidence that sufficient PM&P selection rigor will be applied. The following lists technical requirement areas that are usually among the most important regardless of whether the general requirements or all requirements of MIL-STD-11991 are invoked. Government practitioners should be familiar with MIL-STD-11991 prior to reviewing the contractor or GDO Plan. The contractor or GDO Plan should:

- Define contractor- or GDO-customer interfaces, such as IPTs or working groups, to address issues, review PM&P selections, and review qualification tests by both the prime and its sub-tiers. Include configuration control and part disposition documents, focal points, change concurrence processes, and process verification.
- Describe how part application requirements are derived from system requirements and detailed PM&P and vendor selection processes that consider life-cycle cost, supportability, and availability. (See MIL-STD-11991 sections 4.2 and 4.2.1.)
- Identify restricted or prohibited PM&P based in part on lessons learned and controlled part categories; review any proposed use with customer where applicable. (See MIL-STD-11991 section 4.3.11 and Appendix C.)
- Explain how PM&P qualification processes will assure meeting application requirements (e.g., baselining requirements using industry and military standards as a starting point). Identify PM&P qualification methodologies and test assets along with vendor qualification processes. Provide an order of precedence that favors the selection of standard parts. (See MIL-STD-11991 sections 4.3.4 and 4.3.5.)

[39] MDA-QS-003-PMAP-Rev C, *Parts, Materials, and Processes Mission Assurance Plan*, n.d.

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- Incorporate suitable industry standard processes with suitable performance levels defined, such as definition of any part grades, classes, reliability levels assumed to meet application requirements (e.g., Class 3 for Initial Production Check [IPC] J-STD-001 soldering) to assure sufficiency and provide for efficient PM&P selection. (See MIL-STD-11991 section 4.3.3.)
- Prioritize part classes based on considerations of quality, reliability, and obsolescence timelines.
- Encourage part standardization. (See MIL-STD-11991 section 4.3.3.)

Part Standardization Benefits

- **Reduced acquisition lead-time and costs.** Government and industry avoid the expenses and delays of designing and developing items by using common parts.
- **Delayed or mitigated DMSMS and part obsolescence.** Standardization of parts increases demand and the likelihood of sustained sources.
- **Increased reliability.** Increased item standardization leads to a more extensive performance history and user experience that can improve the likelihood of reliable equipment performance.
- **Enhanced logistics readiness and interoperability.** Common components simplify logistics support, enhance substitutability and interoperability, and translate to savings in procuring, testing, warehousing, and transporting parts.
- **Increased safety of systems and equipment.** Preferred PM&P reduce risk and reduce mission failure and loss of life.

- Define stress derating levels and indicate how customer communications will demonstrate how part applications are within stress limits. (See MIL-STD-11991 Appendix A.)
- Explain processes for assuring that COTS assemblies will meet application requirements and for accounting for GFP performance. (See MIL-STD-11991 section 4.5 and Appendix B.)
- Include processes to notify customers when parts are not procured from authorized sources along with counterfeit mitigation efforts. (See MIL-STD-11991 section 4.4.2 and Appendix F.)
- Document processes to monitor failures to identify whether they are caused by PM&P issues and to coordinate with customer on resolutions and corrective action when that is the case. (See MIL-STD-11991 sections 4.3.6 and 4.4.3.)
- Document effective processes to flow down requirements to subcontractors and to require separate subcontractor Plans (when the prime contractor or GDO is unable to flow down requirements, then the prime contractor is responsible). Address subcontractor data and parts lists. (See MIL-STD-11991 section 4.6.)
- Establish a configuration control process, including change notification, approval, and alternate or substitute parts and materials. Include a PM&P documentation process. (See MIL-STD-11991 sections 4.4.5 and 4.4.9.)
- Include processes to access all necessary reference documents.
- Provide terms and definitions.

When only the general requirements of MIL-STD-11991 are invoked, there may be more cycles of review to ensure all the requirements are documented, because of the absence of detail. A program office may create its own checklist associated with the above list of requirement areas (or modifications thereof) if determined to be useful.

There are other considerations, beyond processes for meeting PM&P management technical requirements, associated with reviewing and approving a contractor or GDO PM&P Management Plan.

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These considerations generally provide the program office with additional leverage to enforce the implementation of changes that it deems to be most important. The contractor or GDO Plan should:

- Inform the program office of the contractor's lines of responsibility for the processes to meet the above technical requirements;
- Provide the program office with the right to review and inspect data necessary to perform the functions established in the program office PM&P Management Plan (e.g., a level 3 technical data package [TDP] when reviewing PM&P selections);
- Ensure the right to review and inspect all referenced documents;
- Require program office approval on changes; and
- Detail the program office's process to institute part changes when the government determines that a contractor- or GDO-selected part is not suitable.

The last bullet is very important. If there is a problem with a part when a product is delivered, whatever the cause, the program office normally carries all the risk and pays for corrective action depending on the terms and conditions in the contract. The contractor or GDO PM&P Management Plan can establish processes to replace PM&P during design. However, even when PM&P selection occurs later in the life cycle, the program office should seek a mechanism to reject unacceptable PM&P before they are installed on a system. The life-cycle cost of a system increases when corrective actions are taken after a system is fielded as compared to when the PM&P is initially selected. Life-cycle costs also increase when spares are procured to replace failed parts.

Typically, the review and approval process to meet each (or key) MIL-STD-11991 requirement should be assessed as compliant, something minor to fix, or non-compliant. If the contractor or GDO tailors a requirement, the program office judges the acceptability of that tailoring. The program office should only accept the contractor or GDO or subcontractor PM&P Management Plan when the adequacy of every process necessary to meet MIL-STD-11991 general or general and detailed requirements has been determined. Work should not be allowed to start unless all processes are compliant. A thorough review of a contractor Plan usually takes two to four weeks.

As a function of risk, the program office may review and even approve subcontractor PM&P Management Plans in conjunction with the prime contractor. The level of program office involvement in the review of subcontractor Plans varies:

- The program office may perform its own independent review of the Plan;[40]
- The program office may be a participant in the prime contractor or GDO review of the Plan;
- The program office may review the prime contractor's or GDO's processes for reviewing the Plan;
- The program office may review the Plan only when problems arise; or
- The program office may not review the Plan.

For the fourth bullet, if a problem arises, the program office should learn about it as a result of its participation in technical, production, or other reviews.[41] When that occurs, the program office may review subcontractor PM&P Management Plans to determine the cause of the problem (e.g., the prime contractor may not have established the right requirements for the subcontractor) and to help formulate mitigations.

[40] This is a common practice for MDA for important subsystems.

[41] Periodic auditing of subcontractors is a more rigorous way to determine whether there are potential issues with subcontractors as compared to total reliance on scheduled reviews. Such audits may be unannounced.

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Section 5. Verification that the Contractor or GDO Is Following the Processes in Its Plan

Figure 8 highlights the portion of Figure 1 associated with the verification that the contractor or GDO is following the processes in its PM&P Management Plan.

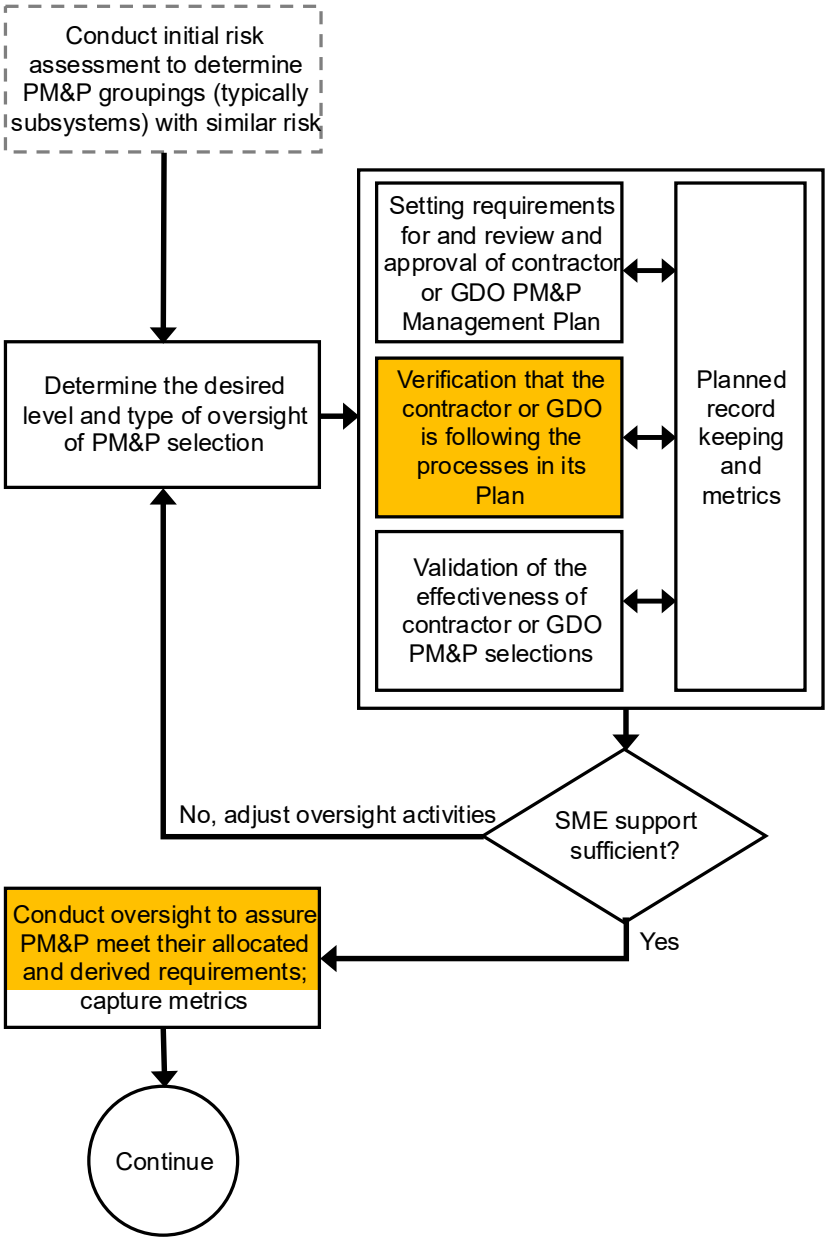


Figure 8. Verification that the Contractor or GDO Is Following the Processes in Its PM&P Management Plan

In theory, problems associated with not following processes may be uncovered during oversight of contractor or GDO PM&P selection effectiveness. However, the advantage of separating verification oversight is that risks may be discovered earlier and addressed before they impact PM&P selection.

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5.1 Information on Associated Contractual Requirements

Most verification activities are not contractual because verification is a government process. The SD-26 illustrative requirement in this area (illustrative requirement 28) is concerned with the availability of documents and data on the processes in the contractor Plan.

There is another contractual possibility not explicit in the SD-26. SD-26 illustrative requirement 15, entitled "monitor, manage, and report subcontractor DMSMS capability," could be tailored to verify subcontractors are following the processes in their PM&P Management Plans. It is also possible for the program office to team with its prime contractor in these verification activities.

5.2 Oversight Considerations

Verifying whether a contractor or GDO is following the processes in its PM&P Management Plan is important because not following the agreed upon processes requires corrective action. The verification approach should differ by PM&P grouping as determined by the risk profile. However, verifying that a prime contractor or GDO *is* following the processes in its Plan may not reduce risk sufficiently, depending upon the prime contractor's or GDO's functions. For example, the prime contractor PM&P selection may be very small—all PM&P selection may occur at lower tiers of the supply chain. Thus, a minimum level of verification (to be used as the total scope of verification efforts only in special and limited circumstances) examines the prime contractor's or GDO's processes to:

- Flow down PM&P management requirements and
- Approve the PM&P Management Plan for the most critical subcontractors to ensure that those subcontractors are flowing down the appropriate PM&P management requirements throughout their supply chains.

There are different levels of verification rigor to be conducted in addition to this minimum level. Collaboration and consistent interaction through working groups and integrated product teams provides the most effective and efficient form of verification of PM&P Plan implementation. In addition, a combination of the five following approaches (ordered from most rigorous to least rigorous) will provide confidence in effective Plan implementation. All these approaches include program office (or program office representative) presence at contractor or GDO facilities. Although process documentation can be reviewed remotely, independent quality assurance requires on-site presence.

- ***Official program office audit.*** In the planning stage of this approach, the program office establishes criteria to verify whether the contractor or GDO follows the processes in its Plan. Next, program office SMEs perform official site visits to observe and inspect the contractor's or GDO's processes. Site visits form the basis for an evaluation of whether the criteria are being met. The results of the evaluation become the documented findings.
- ***Request Defense Contract Management Agency (DCMA) surveillance.*** DCMA surveillance is intended to verify that companies are fulfilling the provisions of their contracts with the DoW. Surveillance encompasses substantiation of costs, confirmation of product performance and service delivery, inspection of processes and procedures (such as testing and qualification), and other contracted activities.[42] When DCMA is the administrative contract officer, it may perform PM&P management surveillance in cases where risks associated with PM&P selection processes are deemed to be high and surveillance personnel/ resources are available. If a program office wanted to ensure that PM&P management surveillance occurs, it would likely prepare a Quality Assurance Letter of Instruction, a memorandum of agreement, and/or a memorandum of understanding that specifically requests that DCMA perform those functions. The request would also provide instructions on how and how often DCMA should perform those functions.

[42] Paraphrased from DCMA Manual 2303-01, *Surveillance*, June 9, 2025, https://www.dcma.mil/Portals/31/Documents/Policy/DCMA_MAN_2303-01_VOL3.pdf.

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Depending on the level of technical sophistication required, supplemental program office expertise may be essential, thereby making this approach very similar to the official program office audit.

- ***Embed engineers with the contractor or GDO.*** A program office may embed engineers (either full time or part time) at a contractor or GDO location. Usually this is done for technically demanding or especially critical/sensitive applications to ensure quality, to determine that all requirements have been met, and to observe testing. These engineers could also verify that the processes in the contractor or GDO PM&P Management Plan are being followed and periodically report their observations to the program office.
- ***Address PM&P management during formal program office reviews.*** A program routinely performs several different types of review. They are often called product assurance, management, or quality reviews. The scope of these reviews extends beyond PM&P selection processes and decisions, but all technical risks (e.g., obsolescence, performance, schedule, and cost) are included.
- ***Informal site visits and other interactions.*** This is the least formal verification approach. Day-to-day interactions between the contractor/GDO and the program office can be leveraged to form an opinion about the extent to which the contractor or GDO is following the process in its PM&P Management Plan. Interactions can take multiple forms. They may be system and/or subsystem IPT meetings or there could even be PM&P management working group meetings where PM&P approval is discussed and questionable PM&P selection decisions can be identified and resolved (e.g., selecting a better alternate part that is form/fit/function drop-in compatible and have better quality/reliability grade than the proposed part).

The first three approaches are more appropriate for the highest risk and mission criticality scenarios and PM&P groupings. The last two approaches apply to lower risk situations.

The following lists important PM&P selection practices that should be followed by the contractor or GDO and verified by the program office:

- Document and understand the performance requirements, including functions, reliability, quality, life-cycle cost, life-cycle stresses (higher level manufacturing processes, environmental, operating, time, corrosion/degradation, etc.).
- Document prohibited parts, materials, and processes; required performance requirements (such as military or automotive specifications); country of origin limitations; and similar criteria.
- Identify suitable specifications for defining PM&P capability requirements.
- Identify manufacturers with suitable product assurance programs and PM&P that meet suitable specifications.[43]
- Identify parts with maximum stress ratings that will meet program and/or application stress derating requirements.[44]
- Identify parts with suitable functional and tolerance requirements for the application (including assessment with circuit tolerance analysis for electronics).
- Document part manufacturer's qualification, quality, and reliability data that meets the application requirements and/or identify data gaps that may require additional testing or mitigation.
- Document the part construction (parts, materials, and processes) to confirm suitable product assurance (e.g., qualification tests suitable for the construction).
- Document manufacturing and test supply chain.
- Evaluate and document obsolescence and availability risks and mitigations.

[43] Most military applications will require PM&P that meet military, aerospace, or automotive specifications.

[44] MIL-STD-11991 defines stress derating levels suitable for most military applications.

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- Evaluate compatibility with higher level manufacturing processes and materials (e.g., solder and curing processes, assembly stress screening).
- Evaluate compatibility with assembly parts and materials (e.g., galvanic couples, outgassing effects).

Verification of these practices applies throughout the supply chain, especially when the prime contractor or GDO is the system integrator because that implies significant PM&P selection occurs at sub-tier vendors. Any of the five above verification approaches should include a flow down of requirements into the sub-tiers. If a program office wants DCMA PM&P management surveillance to extend to lower tiers of the supply chain, letters of delegation could be required. Because of the complexities and costs of such sub-tier surveillance, it is not generally pursued. Rather, the prime contractor's or GDO's processes for ensuring sub-tier PM&P management compliance would be reviewed.

The program office may repeat some of these verification approaches. While the use of embedded engineers and informal interactions are continuous activities, the others are not. Program office audits, program office technical reviews, DCMA surveillance, and evaluations of sub-tier vendors may be repeated periodically or at certain key points in the system's life cycle as a function of risk and past performance.

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Section 6. Validation of the Effectiveness of the Contractor or GDO PM&P Selections

Figure 9, extracted from Figure 1, highlights the portion of the overall PM&P Program flowchart associated with validation.

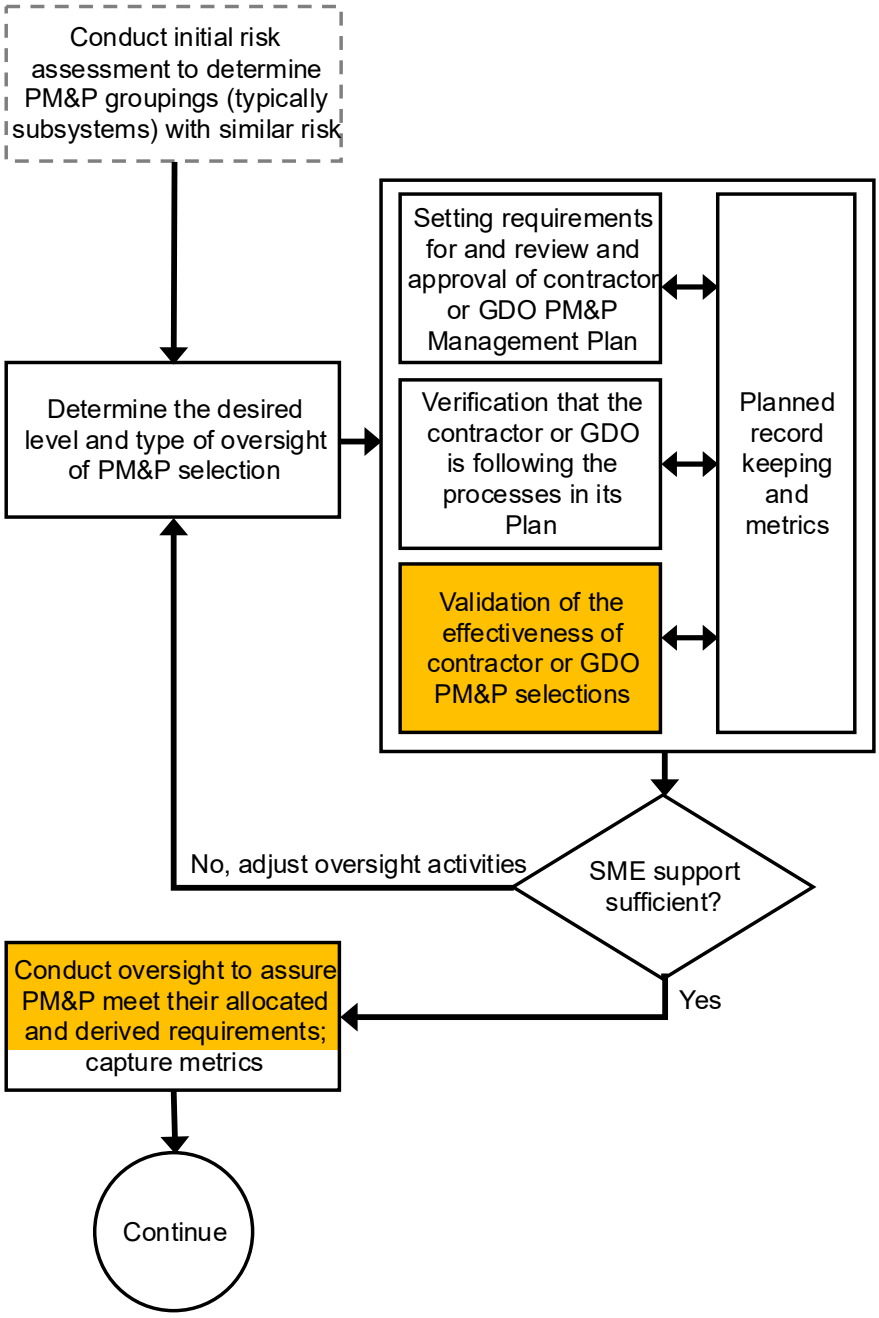


Figure 9. Validation that PM&P Selection Processes Are Effective

Validating the effectiveness of PM&P selections could be an important aspect of oversight to reduce PM&P selection risk. This section discusses oversight activities associated with its PM&P management

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practitioners obtaining data on the PM&P selected and then determining and acting on acceptability for some or all PM&P. Potential actions include approving or rejecting some or all PM&P or only suggesting PM&P changes to the contractor or GDO. The following specific validation approaches (ordered from least rigorous to most rigorous) are discussed in the next three subsections:

1. Program office monitoring of contractor or GDO PM&P selections to facilitate suggested PM&P changes or processes for PM&P rejection or approval
2. Assessing the viability of PM&P selected that do not meet program office preferences leading to PM&P rejection, suggested PM&P changes, or expedited PM&P approval processes through exception reporting
3. Specific program office approval or rejection of PM&P.

Any combination of these three approaches may be pursued. The risk profile should be used to determine the level and type of validation by PM&P grouping.

6.1 Monitoring

6.1.1 Monitoring Benefits

Under nearly all circumstances where oversight is deemed important, it should include the monitoring of PM&P selections through a review of parts lists and specifications as early as possible. Timeliness is key; unless the review is very soon after the time of PM&P selection, the development process may be impeded, and costs could increase when PM&P are rejected. Monitoring PM&P selections is one of the most important elements of program office oversight, especially when there is limited or no government PM&P approval. Under those circumstances, monitoring may be the program office's only opportunity to examine parts lists. Benefits of monitoring include:

- The quality (grade) of parts along with reliability can be easily determined and inquiries should be made on whether the parts meet allocated and derived requirements.
- Confidence in the PM&P that are used in DoW systems. The program office may have the ability to discuss and potentially influence the contractor's or GDO's processes. The program office will also be able to maintain a degree of awareness over what the contractor or GDO is doing.
- Identification of a contractor-chosen part from a preferred parts list (PPL) not preferable to the program office.[45] Contractors will generally select parts from their PPL. Because the contractor-chosen PPL is created to reflect the contractor's supplier relationships, price, and previous experience, a contractor-chosen PPL part selection may not necessarily be the best choice from a government perspective. For example, the military's operating environments may be different than the contractor's experience or there may be a better part being sold by a competitor to the selecting contractor.
- Negotiated PM&P changes before design lock. Monitoring enables the identification of potential PM&P selection problems early, which creates an opportunity to make changes that will not impact cost or schedule.
- Enhanced program office influence. Making changes during a formal PM&P approval process may be difficult because of impacts on redesign cost and schedule. The likelihood of those changes occurring early in the process is significantly higher even in situations where the program office does not have approval authority.
- Improved contractor or GDO PM&P selection. Government engineers monitoring contractor or GDO PM&P selection activities may have a much broader experience base than the engineers in a contractor program office or GDO. A government engineer may have had opportunities to monitor dozens of systems while a contractor or GDO engineer may only have worked on one or

[45] Appendix D.3 describes the difference between a program office–approved and a contractor-chosen PPL.

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two systems. Therefore, the government engineer may be able to suggest ways of improving PM&P selection.

- Improved planning efficiency and future resource availability when changes cannot be negotiated. Monitoring prepares the program office to program/budget for future mitigation costs. At a minimum, there will be improved efficiency and planning, and funding will be in place to resolve issues. While there is no guarantee that resolution costs will be lower, the improved efficiency is likely to result in a better use of personnel resources and avoid cost increases.
- Improved PM&P approval process. Knowing about issues in advance will improve the efficiency of the PM&P approval process. Review time will be reduced. A determination of possible alternatives may have occurred.
- Discovery of problematic PM&P. Some issues may escape notice in a PM&P approval process but can be detected when the program office is closely observing the selection process. The selection of known problematic PM&P can be identified.

Monitoring may be carried out remotely through conscientious review of BOMs throughout various stages of development. However, monitoring is more effective when the program office is also present at contractor/GDO and subcontractor facilities to observe the processes that were used to select parts for the BOMs.[46] SAE AS9100 may be an effective mechanism to enable site visits to lower-level suppliers.[47] Monitoring of BOMs is affected by the risk determination and aggregation. Focus may be placed on BOMs and items within the BOMs having the greater level of risk. However, monitoring is so important to oversight that no BOMs should be ignored. The program office should not wait until a BOM is nearly complete to monitor PM&P selection, as changes may be more difficult.

6.1.2 Information on Associated Contractual Requirements

There are no explicit contract requirements other than those related to the ability of the program office to review and inspect material needed to conduct monitoring. The material to be inspected or reviewed should be specified along with the period of performance of the contract (or similar arrangement with a GDO) when this review and inspection can occur.

6.1.3 Oversight Considerations

A combination of two monitoring methods should be used to examine all PM&P selection with the prime contractor or GDO and in the sub-tiers. In the first method, the program office interacts with the contractor or GDO during design and PM&P selection. Contractor or GDO processes are observed directly, discussions are held, and the preliminary outputs of those processes can be reviewed before being finalized. This approach gives the program office the earliest possible insight into the PM&P selection processes and preliminary decisions. The second method involves reviewing preliminary parts lists/BOMs and their updates. Such parts list reviews would be slightly later than the interactions in the first method, but still relatively early in the PM&P selection process.

The purpose of the reviews is to decide whether the selection is acceptable. The SD-18, *Part Requirement & Application Guide*, may have helpful background information to support that determination.[48] However, the SD-18 should not be the sole basis for decision-making because it is a legacy document that is not updated.

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[46] Monitoring processes need not be the sole purpose of the facility visit; monitoring can take place in conjunction with other activities to minimize any disruptions created by a government visit itself.

[47] SAE AS9100: Quality Management Systems—Requirements for Aviation, Space, and Defense Organizations.

[48] SD-18, *Part Requirement & Application Guide*, DSPO, n.d. https://www.navsea.navy.mil/Home/WarfareCenters/NSWCCrane/Resources/SD18.aspx.

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In either monitoring method, the reviews would take all the PM&P selection considerations into account. Fundamentally, there would be a determination of whether the parts meet all their requirements. For example:

- Have the proper grade parts been selected? Is there an overuse of commercial grade parts? Should more military or other higher performance (e.g., automotive or aerospace/defense) grade parts be used?
- Have prohibited PM&P been selected?
- Have all applications and operating environments been considered?
- Is reliability sufficient?
- Is the part shelf life adequate? Is the part nearing the end of its life cycle?
- Have cybersecurity vulnerabilities been mitigated?
- Are there risks for tampering or counterfeiting in the supply chain? Are lower tier vendors secure? Are there any unauthorized sources?
- Have parts with more common or standard packaging been selected to reduce chances of obsolescence?

The reviews should take a slightly different form for COTS assembles. Because there are no standard procedures used by the PM&P management community (both industry and government) to select COTS assemblies, a government-industry team, sponsored by DoW's DMSMS and PM&P Management Working Groups (WGs), examined the problems associated with choosing and using COTS assemblies throughout a system's life cycle. The team developed a COTS assembly checklist to help establish a structured and sound evaluation environment for program office personnel to compare various electronic COTS assemblies for use in safety critical or mission critical aerospace, defense, high performance, and space applications.[49] The questions found in the checklist provide a communications mechanism (from industry or GDOs to program office) for monitoring and assessing COTS assembly selections. Program offices should encourage industry and GDOs to use the checklist for selecting COTS assemblies.

A parts list or BOM review is not a one-time event. PCNs issued by GIDEP and other sources should be monitored. When a change occurs, a part that had been determined to be acceptable may be altered in a way that is no longer acceptable. Also, limited life parts and materials application requirements should be assessed to confirm that requirements are met.

When any PM&P selection problems are found, the program office should seek a way to escalate and negotiate their correction. The program office should exert as much influence as possible to make part changes because ultimately, the problem will have to be fixed. Fixing a design issue before fielding is highly preferable.

6.2 Exception Reporting

There may be instances where the program office has a preference for some specific part characteristics (e.g., not a sole source situation) but does not state that preference as a requirement because it may not be possible to achieve everywhere, or it may be too expensive to apply everywhere. In such instances, the program office could require the contractor or GDO to submit "exception reports" describing where the characteristic was not met and why.

[49] *Commercial Off-the-Shelf Critical Electronic Assembly Selection Checklist: Guidance and Background Information*, Revision 1.1, DSPO, October 27, 2021, https://www.dau.edu/commercial-shelf-critical-electronic-assembly-selection-checklist.

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6.2.1 *Information on Associated Contractual Requirements*

Contract requirements (or the equivalent in the case of government design) inform the contractor or GDO of program office interests so the contractor or GDO is better able to satisfy its customer beyond meeting formally stated requirements. For example, the contractor may select a part that is different from the one it would normally select in similar circumstances. Similarly, the contractor may attempt to obtain some data on a PM&P characteristic or lower tier supply chain company that it typically would not consider.

The SD-26 illustrative language includes requirements for exception reports for parts not meeting the following explicit preferences:

- Microelectronics parts procured from an authorized source
- Microelectronics parts meeting the criteria of International Organization for Standardization/ International Electrotechnical Commission (IEC) 20243-1:2018
- Avoiding rebranded parts
- Microelectronic parts with at least one qualified and approved alternate
- PM&P meeting program protection and hardware and software assurance requirements.[50]

Other exceptions may be used at the discretion of the program office. The contract should specify that the program office has the right to review and inspect material needed to review and understand the content of the exception along with the period of performance of the contract when this review and inspection can occur.

6.2.2 *Oversight Considerations*

Exception reports should be used in less critical situations when monitoring may not provide sufficient leverage to force a change, or the program office wants a better understanding of certain risks. These exception reports should include:

- A list of selected parts where the contractor or GDO did not meet the program manager's preference.
- A description of why the part was chosen.
- An explanation of the risk mitigations put into place to address the risks associated with not meeting the preferences.

When exception reports are required, the contractor or GDO should also be required to allow the program office to view and inspect data, information, and reports (including test reports) that support the mitigations shown in the exception reports.

The requirement for exception reports and their supporting data should be used to:

- Inform the contractor or GDO of program office interests without establishing a firm requirement.
- Enable the program office to reject PM&P that do not meet its preferences on a limited basis. The SD-26 illustrative contract requirement gives the program office 30 days from receipt of the exception report to reject the part. The reason for the 30-day time frame is that the program office does not want to put itself on the critical path of system development for a long period of time. After 30 days, the program office may coordinate with the contractor or GDO about further use of the parts according to the SD-26. Parts are automatically accepted if the program office does not respond to the exception report.
- Highlight areas where program office attention may be needed in the context of its monitoring of the contractor's or GDO's PM&P selection processes and its PM&P approval process.

[50] See SD-26 illustrative requirements 30, 31, and 32.

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6.3 PM&P Approval

This guidance does not suggest a return to the pre-acquisition reform parts control environment that imposed cost and schedule burdens on the government, and schedule burdens on the contractor while waiting for program office approval. Adjustments should be made on both where PM&P approval should be applied and how PM&P approval should be conducted, including timelines.

The concept of program office PM&P approval under certain circumstances is consistent with MIL-STD-11991 requirements for government approval. The risk profile provides the program office with information on critical areas where PM&P approval should be focused based on whatever is necessary to reduce risk to an acceptable level. A program office should make a risk-based, subjective judgement on the extent to which PM&P approval applies to the different aggregations of PM&P selection.

Recognize however, that PM&P approval should lead to lower life-cycle cost for the government. If a part on a defense system does not meet its requirements, the situation is usually rectified. Options for fixing the problem range from finding a substitute PM&P to redesign. Some of these options may have high cost; all these options are costlier than selecting the right PM&P during initial design. Therefore, it is cost effective to allocate resources for front-end oversight to avoid expenses associated with solving a problem that could have been avoided. While other forms of oversight will identify some problems, a PM&P approval process will discover issues that other oversight mechanisms miss.

6.3.1 Information on Associated Contractual Requirements

The illustrative language in the SD-26 (see illustrative requirement 32) requires program office PM&P approval in the following circumstances:

- Part use outside vendor specifications other than in accordance with the criteria in Appendix A of MIL-STD-11991.
- Use of any prohibited parts, materials, and processes from MIL-STD-11991, Appendix C.
- Use of any non-conforming parts without government approval.
- Use of application specific integrated circuits system not procured from a Defense Microelectronics Activity accredited trusted supplier in accordance with DoDI 5200.44.[51]

The contract should specify that the program office has the right to review and inspect material needed to approve PM&P. The material to be inspected or reviewed should be specified along with the period of performance of the contract when this review and inspection can occur.

6.3.2 Oversight Considerations

As a function of the risk profile, program offices' PM&P approval needs may vary. Some program offices may have no need to approve any PM&P, while other program offices may need to approve all PM&P or only all critical PM&P for the entire system or for some selected subsystems. For example, in fixed price contracts where the contractor is responsible for performance, program office PM&P approval may be limited. The concept of risk tolerance may also apply. The extent of PM&P approval should be a function of the mission and a determination of the extent of risk acceptability. Program office PM&P approval should be based on an analysis of contractor or GDO provided data that will provide government SMEs with a basis to accept or reject any PM&P. Data needs may also vary as a function of risk tolerance. PM&P approval should be somewhat automatic when there is a program office–approved PPL. Parts on this PPL have been pre-approved for use.

[51] DoDI 5200.44, *Protection of Mission Critical Functions to Achieve Trusted Systems and Networks* (TSN), February 16, 2024, https://www.esd.whs.mil/portals/54/documents/dd/issuances/dodi/520044p.pdf.

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The following lists some further considerations on the extent of PM&P approval:

- Certain program offices have decided on and implemented disciplined PM&P approval processes and procedures along with associated contract language based on the specific applications and missions of their systems. These program offices should continue to follow their established practices. This guidance should not be construed to reduce the oversight being provided today.
- In general, PM&P approval may be limited to those risk-based subsystem aggregations with the highest risk or some subset thereof as determined by program office experts.
- The use of exception reporting may be a way to focus PM&P approval to the most important issues. Exception reporting also provides the program office with valuable information about the PM&P in systems that can be used in formulating future policy and guidance and for improving efficiency and effectiveness.
- The actual combination of PM&P approval actions should be determined by the program office based on its determination of acceptable residual risk and the cost to mitigate risk to that level.
- PM&P approval may be limited to critical PM&P potentially based on risk tolerance for critical missions.
- PM&P approval should focus on the selection of standard parts as compared to non-standard parts. Standard parts meet minimum levels of quality and reliability as established in a reference Military Specifications (MIL-SPECs). PM&P covered by MIL-SPECs typically meet quality and reliability needs for most military applications.
- PM&P approval should focus on the process for the selection of COTS assemblies. COTS assemblies are frequently utilized because they can improve cost and schedule. Unfortunately, the PM&P management community (industry and government) does not use standard procedures to select COTS electronic assemblies for integration into aerospace, defense, high performance, and space applications primarily because of the difficulty in obtaining BOM data on COTS assemblies.

PM&P approval may be formal or informal during design. Both the formal and informal processes may be program office or contractor/GDO led. When the contractor/GDO leads, it is only necessary for the program office to participate with de facto authority to reject any PM&P it considers unacceptable.

- A formal process might have an official parts control board or a configuration management board that approves parts.[52] The contractor or GDO may be required to formally submit parts for approval along with data for the board to consider when making a formal decision. Potential data requirements for non-standard parts include part information (e.g., manufacturer, vendor, part number), prohibitions or restrictions on the use of the part, mitigations performed to mitigate restrictions, information from a destructive physical analysis, DMSMS information, results from tests of interest (e.g., radiation hardening), PCNs, GIDEP alerts, previous applications, justification for selection, expected life, qualification status, etc. [53] Most of these data requirements are not needed for standard parts.[54]
- There are several informal processes that program offices can use to approve PM&P. For example:
    - One type of informal process considers similar information to the formal process. The principal difference is that there is no formal parts control board. PM&P approval is determined through consensus at a working-level meeting, typically chaired by program office and industry quality or product assurance SMEs. The informal PM&P approval process should be documented in the contractor or GDO PM&P Management Plan. Data requirements should be a function of why government approval is required. For example, if

[52] See MIL-STD-1546 for more information about forming a formal PM&P control board.

[53] These potential requirements are a subset of the requirements in MDA policy.

[54] Since the SD-26 has not created a contract data requirement list and data item description for obtaining this data, either they would have to be developed, or the information could potentially be built into the contractor or GDO PM&P Management Plan since MIL-STD-11991 calls for government approval.

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the contractor proposes the use of a prohibited PM&P, the request for approval should explain exactly why it is essential for that PM&P to be used and should provide detailed mitigation plans associated with why the PM&P is prohibited.

- The exception reports discussed in Section 6.2 represent another somewhat informal mechanism for PM&P approval. Program office PM&P approval is assumed unless the part is rejected within 30 days. The SD-26 information requirement for the exception reports is much less detailed than the potential formal potential data requirements above.
- Theoretically, the program office may use the results of a Technology Readiness Review or a Production Readiness Review for PM&P approval.

Data needs differ when approving the use of a COTS assembly. Testing is difficult without field history of the use of the same version of the part. It is important to investigate whether requirements are met by obtaining as much data as possible. Therefore, in addition to the information contained in the COTS checklist, the program office may request a BOM to assess whether the assembly will meet requirements.[55] Furthermore, the program office may request vendor item control drawings or source control drawings, engineering descriptions, acceptance criteria, and suggested sources of supply. Since the program office is unable to control the manufacturing process, a destructive physical analysis may be conducted to assess quality and to check the accuracy of the data.

PM&P approval may take a different form, depending on the phase of the life cycle. For example, during production and sustainment, the program office is likely to have assumed configuration control as defined by the TDP. Contracts may state that the program office must approve the use of any PM&P not included in the TDP or that changes form, fit, or function.

Under certain circumstances, PM&P approval may not be unconditional. There may be usage restrictions driven by security, environmental, or technical performance considerations.

Any PM&P approval process gives the program office more control over final PM&P selection and usage. However, it may be problematic for the program office to exercise that control. PM&P changes lead to rework, and the scope of that rework increases as a function of the time between initial PM&P selection and PM&P change. This often leads to increased government cost and sometimes schedule slippage.

[55] Commercial vendors may refuse to provide a BOM for their COTS items. When that happens, the program office may use other criteria to approve the part.

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Section 7. Record Keeping

This section discusses PM&P management record keeping as highlighted in Figure 10, which extracts the pertinent aspects of Figure 1. Section 9 of this document suggests how analyses of those records might be used to assess and improve the government's PM&P management oversight efficiency and effectiveness.

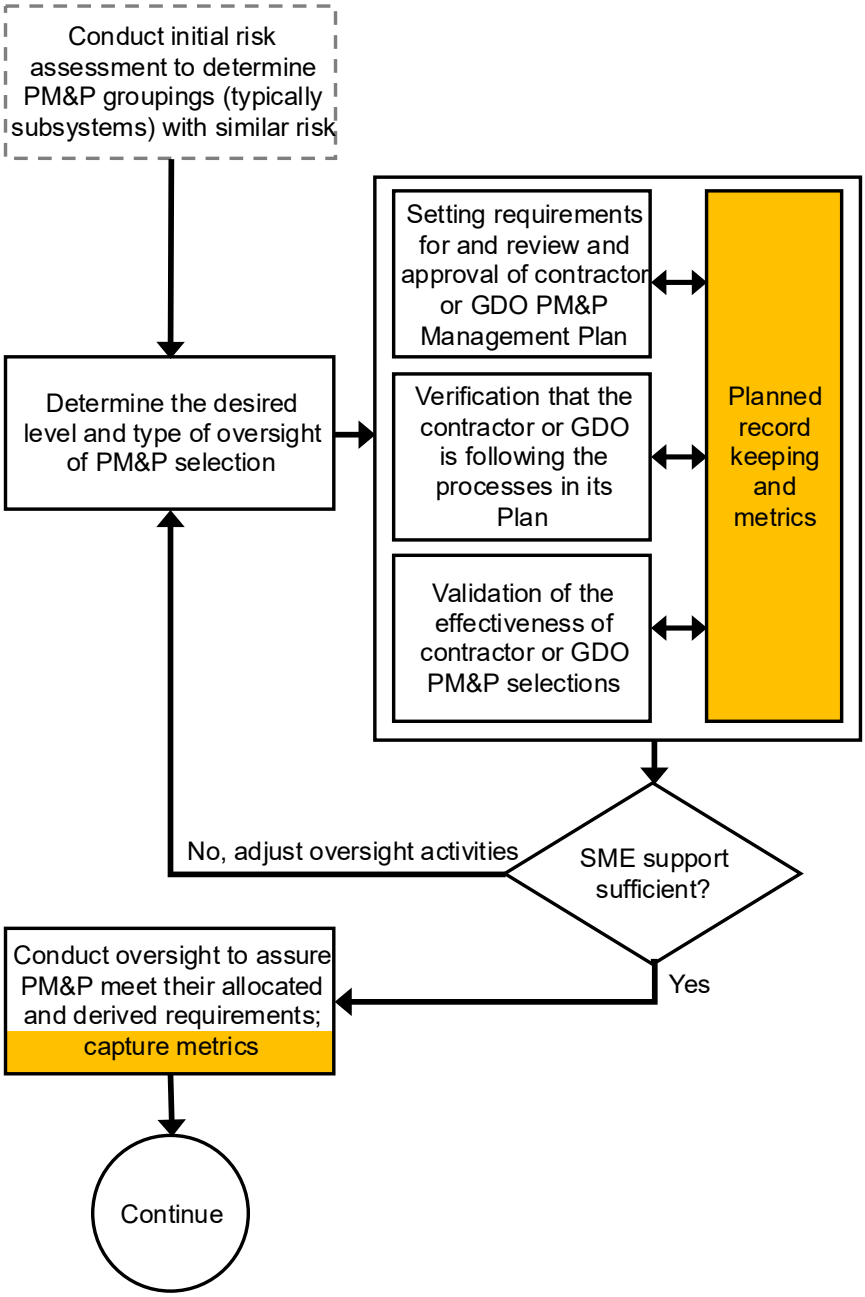


Figure 10. Record Keeping and Capturing Metrics

The guidance contained in this section focuses on (1) records that should be collected from contractors or GDOs regarding PM&P management practices, as well as specific part selections and (2) records the

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program office should collect describing its own oversight functions.[56] Because it's often not known in advance how certain records will be useful, program offices should also consider keeping records that *may* be useful when the overhead cost of collecting them is not high. Examples are provided to aid program offices in considering what records may be useful.

This document is meant to provide initial record keeping guidance for a new PM&P management paradigm. Subsequent versions will incorporate additional material such as lessons learned, use cases of observed benefits of PM&P management record keeping, and standardized data dictionaries to make record keeping more effective.

There are two basic approaches to keeping PM&P management records for the purposes of assessing and improving the government's PM&P management oversight. The first is to start by determining what questions should be answered, and then considering what records are needed to answer those questions. The second is to rely primarily on intuition and experience to determine records that are likely to have future value, possibly without a clear hypothesis in mind. The latter approach is the subject of this subsection; Section 9 addresses the former.

It's often difficult to conceive of every possible question the government may need answered in the future, hence gathering records without a specified use is of value. To aid program offices in determining a set of records to be kept without the context of a specific question, three subcategories of records have been defined: *contract*[57] *and system records, parts records*, and *practices records*.

- *Contract and system records* provide needed context to make other records more useful. Data elements about the contract will include things such as the company name (and subdivision if applicable), location(s), the contract's purpose (such as a development of a new system or component, or a redesign), and timelines. Data elements about the system include items such as the expected users and expected warfare areas.
- *Parts records* describe the parts being selected for a system. This will include information about specific parts, such as whether the part is already used in other DoW systems, whether the part required explicit program office approval, and whether any exception reports were required for the part to be selected. Parts records also include *aggregate* data elements across all parts selected, such as the number of parts selected which are (not) already in the DoW supply system, the number requiring program office approval, and how many parts were included in exception reports.

  While data elements about the specific parts selected are more informative than aggregate data (the latter can always be derived from the former), in situations where collecting part-level records is not practical, aggregates are a valuable alternative. Collecting part-level records may be impractical, for example, when the cost to collect and provide them to the program office is high. Most parts records are more useful if further qualified with reason codes—knowing why a part appeared on an exception report (DMSMS criteria, microelectronics traceability, etc.) is more useful than knowing only that it appeared on one. Parts records can also include part attributes (e.g., the required and selected mean time to failure), which can often be valuable when assessing the effectiveness of PM&P management. This type of data is often gathered by other stakeholders (e.g., engineers); what's important is that someone within the program office gathers it. Lastly, parts records can include administrative aspects of the specific parts selected, such as manufacturer part numbers and commercial and government entity (CAGE) codes.
- *Practice records*, in contrast, describe the PM&P management process as a whole vice specific parts and aggregates. For instance, practice records could include whether there was a requirement to have a contractor or GDO PM&P Management Plan, and whether the Plan was

[56] In general, this record keeping section focuses more on parts than materials and processes since parts are ultimately being selected and the materials and process used to create those parts are inherent in the selection itself. Eventually, record keeping may capture more information on materials and processes as more experience is achieved.

[57] While the term contract will be used throughout this section, in the case of a GDO, it would be the contract-equivalent document the program office uses to provide GDO requirements.

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delivered to the program office. Other examples include whether and how the program office conducted audits on contractors' or GDOs' PM&P management processes, and overall costs related to PM&P management oversight (often measured in manhours). Practice records play a crucial role in assessing the effectiveness of the government's PM&P management efforts.

Sections 7.1, 7.2, and 7.3 list further examples for each category of records. Most of these examples are records with clearly defined values, such as categorical responses (e.g., a contractor or GDO Plan *was* or *was not* required) or quantitative values (e.g., the MTTF was *x*). However, at times the examples suggest a narrative description of an aspect of the contract or system (e.g., description of the need for a system redesign). This is because centralized PM&P management record keeping in the DoW is an evolving field, and standardized, categorical responses for many questions have not yet been established. Later iterations of this document will include a data dictionary to standardize the set of possible responses to non-quantifiable questions, but for the time being, narrative descriptions are valuable because they can be used in conjunction with manual or automated language processing. Lacking a centralized data dictionary, a best practice is for individual program offices to create their own well-documented data dictionary.

These lists are meant to stimulate program offices to consider records that would be valuable to have. Because there are innumerable records which could be of value, program offices should balance the time and effort of gathering records with their potential value. As preliminary guidance, each record in the below lists is marked as either a *basic*, *intermediate*, or *advanced* PM&P management record keeping practice. Regarding record keeping formats, there are not yet standardized reports program offices are required to submit, so program offices should determine the best format to capture records. Program offices are also left to determine the best means of collecting records, such as who collects the records (the prime contractor, a GDO, a subcontractor, or the program office itself) and when.

7.1 Contract and System Records

Contract and system data elements are intended to provide useful context that will make other records more useful. Examples are provided in Table 1.

Table 1. Contract and System Records Examples

Category	Level	Description
Contractor or GDO information	Basic	Contractor's company name (and subdivision if applicable) or GDO name.
	Basic	Contractor's or GDO's location.
	Intermediate	Names, locations, and the tasks assigned to any subcontractors, by tier. Note that subcontracting tiers 1 and 2 generally capture the bulk of significant functions.
	Intermediate	Technical points of contact for the contractor or GDO.
Contract purpose	Basic	Whether the contract is for the development of a new system, a system redesign, or to replace an existing part.
	Basic	In the case of a system redesign, a narrative description of the capability being improved or deficiency being corrected.
	Intermediate	In the case of a contract to replace an existing part or parts, a narrative description of the event that triggered the required replacement.
Contract information	Basic	Timelines for the contractor or GDO to submit its PM&P Management Plan to the government (including initial submittals, final submittals, and updates to parts lists or BOMs).
	Basic	The actual dates the contractor or GDO PM&P Management Plan was submitted to the government (initial, final, and updates to parts lists).
System purpose	Basic	A narrative description of the systems purpose, with details such as the expected warfare areas the system will support (e.g., anti-air warfare, electronic warfare, missile defense).
	Basic	A narrative description of expected users of the system. To ensure records are specific yet general enough to compare across systems, keywords such as the service component

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Category	Level	Description
		(e.g., U.S. Army, U.S. Navy), warfighter type (e.g., infantry, surface fleet, submarine fleet), and platform (e.g., F-35, Arleigh Burke class destroyer) should be included.
System information	Basic	Total number of unique parts to be selected as indicated by the design.
	Intermediate	Target levels of operational availability (Ao) and other performance metrics for the system.

7.2 Parts Records

Parts records describe the parts that were selected for use in a system. This can include attributes particular to PM&P management (e.g., whether an exception report was required, commonality information), as well as more conventional attributes related to the part performance (e.g., MTTF and CFOM). Parts records can consist of part-level data elements describing each part selected on an individual basis, as well as aggregates summarizing all parts selected. While the former is more valuable, in certain circumstances gathering the latter from PM&P management contractors or GDOs may be more practical, so examples of each are given. Section 7.2.1 provides examples of aggregates and 7.2.2 lists part-level examples.

Note that the examples provided need not only apply to the parts that were ultimately selected for use in the system; they can, in general, apply equally well to the parts initially submitted by the contractor or GDO, some of which may have been rejected by the program office. As an example, one aggregate listed in Table 2 is the number of parts already having a national stock number (NSN). As a means of understanding PM&P management's value, it's useful to compare how many parts initially submitted had an NSN to how many parts actually used had an NSN. See example 3 in Section 7.2 for a discussion of this topic.

7.2.1 Aggregate Examples of Parts Records

Table 2 provides examples of parts record aggregates. For concision, many are specified as the "number" of parts meeting certain criteria; this should be interpreted as both the raw number and percentage of parts meeting those criteria.

Table 2. Aggregated Parts Records Examples

Category	Level	Description
Standardization	Basic	Number that are MIL-SPEC.
	Basic	Number of non-standard parts.
	Basic	Number of restricted use parts.
Qualified alternates	Basic	Number that have qualified alternates.
FOCI	Intermediate	Number considered to have elevated FOCI risk. Program offices can define for themselves what's considered an elevated FOCI risk; criteria should be briefly documented within their PM&P management records.
Commonality	Basic	Number used in other government assets, as evidenced by already having an NSN.
Hazardous materials	Basic	Number that are, or contain, hazardous materials.
Government approval	Basic	Number that require government approval before selecting.
	Intermediate	Number that require government approval before selecting, categorized by reason. Program offices may record the amount and percent of parts requiring approval because of, e.g., their use in flight safety systems, hazardous material, costs, and the expected criticality of part failure.
	Intermediate	Number of parts that were rejected during government review. It's also valuable to record the number of approved parts which were proceeded by another part being rejected. The latter distinguishes between cases where, for example, 10 parts were

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Category	Level	Description
		the contractors second choice (10 total rejections against 10 parts), and cases where a single part was the contractor's 11th choice (10 rejections against a single part).
	Intermediate	Number that were rejected after government review, categorized by reason. Reasons for rejection may include pending obsolescence, limited supplier capacity, FOCI, cyber vulnerability, etc.
Exception reports	Basic	Number of parts that were submitted on an exception report and were ultimately approved for use in the system.
	Intermediate	Number of parts that were submitted on an exception report and were ultimately approved for use in the system broken out by category of exception report.
	Basic	Number of parts that were submitted on an exception report and ultimately not allowed to be used in the system.
	Intermediate	Number of parts that were submitted on an exception report and ultimately not allowed to be used in the system broken out by category of exception report.

7.2.2 Part-level Examples of Parts Records

Table 3 provides examples of part-level records. As previously mentioned, in addition to typical part features (such as whether parts were MIL-SPEC, already had an NSN, or contained materials of interest), parts records can also include features regarding part performance typically found in engineering or other disciplines, such as MTTF and CFOM. One entry in Table 3 captures examples of this latter category collectively. These are often gathered by other sections of the program office; the PM&P management practitioner need not record the data already collected elsewhere. PM&P management practitioners are responsible for understanding what data elements are being collected elsewhere, and thus what data elements they will be responsible for collecting.

Table 3. Part-level Parts Records Examples

Category	Level	Description
Category of part	Basic	The parts category, e.g., electrical, electronic, mechanical, material, process, COTS assembly.
Description	Basic	A narrative description of the part.
Standardization	Basic	Whether the part is a standard part, evidenced as being included on a Qualified Products List (QPL), PPL, or similar document.
	Basic	List any strict prohibitions for a part.
	Basic	Whether a part is or is not MIL-SPEC.
Qualified alternates	Basic	Whether a part does or does not have a qualified alternate.
	Advanced	The number of qualified alternates for a part (as opposed to only listing whether there are any qualified alternates).
Commonality	Basic	Whether the part does or does not already have an NSN.
	Intermediate	How many other systems use this part.
	Advanced	Miscellaneous SCRM data for parts already in use by the government, such as average wholesale stock-on-hand, demand rates, procurement lead times, repair turn-around times, and vendor-specific risk assessments.
DMSMS	Advanced	A quantifiable DMSMS risk assessment for the part. Quantifiable terms may include the expected years of availability and expected production capacity over time.
Material composition	Basic	Whether a part contains materials of interest. Particular materials will vary by system and over time, but would include materials such as lead, plastics, or various rare earth minerals.
Hazardous materials	Basic	Whether a part is, or contains, hazardous material or not.

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Category	Level	Description
Government approval	Basic	Whether a part requires government approval before selecting.
	Intermediate	If government approval for a part is required, provide a narrative description explaining why. Keywords may include, for example, use in flight safety systems, hazardous materials, costs, and the expected criticality of part failure.
	Advanced	For parts requiring government approval, the number of parts that were rejected, by reason code, before a part was approved. Reason codes may include pending obsolescence, limited supplier capacity, FOCI, cyber vulnerability, etc.
Criticality	Intermediate	Whether the program office has identified a part as critical.
	Advanced	Measurement of a part's criticality. In the most rigorous case, this would involve assigning a numeric score to each part's criticality. The least rigorous method that still provides more context than a simple "Yes" or "No" response would be a subjective assessment of whether a part has "High," "Medium," or "Low" criticality.
Exception reports	Basic	Whether a part required an exception report to be approved before selection.
	Intermediate	The reason/category for any exception reports this part appeared on. Note that in theory a part may require more than one exception report.
Conventional performance data	Basic	If not recorded by other stakeholders in the program office (e.g., engineers), PM&P management practitioners should consider collecting records such as: operational and non-operational temperature ranges; MTTF and mean time between critical failure; mean time between maintenance actions and mean time to repair; CFOM or other quantifications of cybersecurity; part size; and weight.
Part details	Basic	Manufacturer part numbers for parts selected.
	Basic	Manufacturer CAGE codes for parts selected.
	Basic	Manufacturer name for each part.
	Basic	Quantity used per assembly and system
	Intermediate	Manufacturer technical point of contact for each part.
	Basic	Technical information identifying where a part is installed in the system.

7.3 Practice Records

Practice records describe the quality of contractors' or GDOs' and the program office's overall PM&P management practices. They also include overarching PM&P management oversight costs. Practice records are therefore crucial for evaluating the cost-effectiveness of the program office's PM&P management oversight. Examples are provided in Table 4.

Table 4. Practice Records Examples

Category	Level	Description
Standardization	Basic	Whether or not the contract requires adherence to any existing DoW PM&P management documents. If so, list them, such as MIL-STD 11991 (general requirements) or MIL-STD 11991 (detailed requirements).
	Basic	List any prohibitions that apply to all parts.
PM&P Management Plans	Basic	Whether the program office had a documented PM&P Management Plan prior to the contract's start date.
	Basic	Whether the prime contractor or GDO had a PM&P Management Plan, and if so whether it was required to be reviewed and approved by the program office before the contractor or GDO could begin work.
	Intermediate	How many subcontractors, by tier, had PM&P Management Plans, and whether any were required to be reviewed and approved by the program office. If some but not all subcontractors required a Plan, provide reasons.

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Category	Level	Description
	Intermediate	If any subcontractor PM&P Management Plans required government approval, list any that were initially rejected.
	Advanced	Indicator variables for whether the program office considered various factors as part of its PM&P management efforts. Indicators are useful to distinguish basic program office PM&P Management Plans from more advanced ones. Example indicators might include part reliability (most/all Plans will consider this), part commonality with other DoW systems, and the manufacturer's financial health (less robust Plans may not consider these latter two). Note that in some cases these indicators could also be determined retroactively by reading a program office's program office PM&P Management Plan.
Verification	Basic	Whether verification efforts (e.g., audits, personnel embeds, DCMA surveillance) were performed by the government to confirm contractor or GDO PM&P Management Plan processes were being followed. If so, when, how many, and on which contractors/GDOs.
	Intermediate	If performed, the specific aspects of a contractor or GDO PM&P Management Plan that were verified.
	Basic	Verification methods used. If used, how many, where, and for what reasons.
	Advanced	Narrative descriptions of the outcomes of verification efforts.
Costs	Intermediate	Number of PM&P management FTEs. If applicable, records should specify how FTEs were distributed across different PM&P management tasks. The costs of employing these FTEs should also be specified.
	Advanced	Costs associated with specific PM&P management oversight functions. Costs can generally be measured in manhours. Examples of PM&P management functions would include reviewing a contractor or GDO PM&P Management Plan; performing audits; embedding government personnel in contractor or GDO facilities; and reviewing and approving specific part selections.
Resourcing	Intermediate	Binary indicator for whether the program office PM&P Management Plan was resourced per its requirements (required FTEs, planned audits, embeds, etc.).
	Advanced	The resourcing shortfall for executing the program office PM&P Management Plan. That is, the dollar value of resources received to perform PM&P management oversight, versus the dollar value required to execute the Plan.
	Intermediate	Binary indicators for whether KSAs were acquired in accordance with the program office PM&P Management Plan. Binary indicators should be recorded for each individual KSA identified in the Plan (SCRM, system security, manufacturing, etc.).
Triggering event	Advanced	Triggering events which, after a risk assessment, lead to a reselection of parts. Examples of triggering events may include a new FOCI risk entering the part's supply chain, or even a change in supplier that introduces an increased risk of tampering. If a part has triggering events, identify the part number and CAGE code, and provide the quantitative triggering criteria where possible, or a narrative description when necessary. Quantitative criteria are preferred because they are unambiguous; note quantitative criteria bring an implicit requirement to keep records on the data needed to calculate them.
Exception reports	Basic	Whether a program office PM&P Management Plan requires the use of exception reports.
	Basic	Number of exception reports submitted to the government, broken out by category.
	Intermediate	The dates any exception reports were received and responded to.
	Advanced	The average number of manhours required to process an exception report.

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Section 8. Special Considerations for PM&P Selections by Other Government Organizations

Sections 4 through 7 explained considerations for using a risk profile to determine the level and type of PM&P management oversight associated with:

- A PM&P Management Plan,
- Verification that the processes in that Plan are being followed,
- Validation that the PM&P selections are effective, and
- Keeping records and metrics that can be used to improve the efficiency and effectiveness of PM&P management oversight.

The material in those sections applies to both contractors and GDOs, but in practice, the bulk of PM&P management is performed by contractors. This section does not refer to a specific portion of Figure 1 as was done in previous sections. Instead, this section very briefly covers a few special considerations for situations where government organizations select PM&P. Those situations are when:

- Design activity is performed by a GDO;
- GFP is used in the system; and
- The government makes PM&P changes during maintenance.

8.1 Government Design

The only significant difference between GDO design activity and contractor design activity is that contract language is not applicable for a GDO. A document other than a contract (e.g., a memorandum of agreement, a memorandum of understanding, or a service agreement) is used to secure design services.

Government organizations (typically government laboratories) may design an entire system, a portion of a system, or a prototype that will be used by a contractor to develop a system. In all these situations, government PM&P selections could be maintained into production and sustainment. Program office oversight of government organization PM&P selection should therefore be identical to program office oversight of contractor PM&P selection. The program office should not relax any requirements or constraints when another GDO is performing the work.

Resources necessary to perform the oversight and conduct design activities should be obtained by the program office. The GDO should understand its requirements in advance and the resources for meeting these requirements should be factored into its cost quotation.

8.2 Use of Government Furnished Property

This section is concerned with government-acquired property, not contractor-acquired property. Government property is defined as follows:

> *"Government property means all property owned or leased by the Government. Government property includes both Government-furnished property and contractor-acquired property. Government property includes material, equipment, special tooling,*

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special test equipment, and real property. Government property does not include intellectual property and software."[58]

While there are contract requirements associated with government property, those requirements deal with how a contractor should manage government property and are not discussed in this document.[59]

The use of GFP is included in this document because the contractor cannot apply the same level of PM&P selection discipline and rigor to GFP that the contract requires for contractor-acquired property. Furthermore, the amount of PM&P selection rigor and discipline used in GFP development may be unknown. GFP is selected for use by the program office because it meets the requirements of the system being developed and is suitable for use. GFP cannot be modified without the approval of the contracting officer. If modifications are requested, then a logical conclusion is that either there is a defect or the GFP does not meet the requirements and should not have been selected for the program under development. In addition, the program office would have to pay its contractor to make changes.

From an oversight perspective, the government program office should apply rigor like what is used to approve the use of COTS on the system. Examples include:

- Verifying that contractor test results meet performance requirements of program office using the GFP;
- Verifying contractor processes for protecting critical information when GFP is being used; and
- Verifying contractor processes for avoiding situations conducive to malicious tampering as the GFP is installed and operated.

If the candidate GFP cannot satisfy a similar level of scrutiny, then perhaps it should not be selected, although in some circumstances (e.g., communication equipment) there may be no other acceptable alternative.

8.3 Government PM&P Changes during Maintenance

Analogous to section 8.1, contractual language is not applicable because there is no formal contract with another government organization.

As was the case for instances of government design, GDOs performing maintenance should be treated the same as contractor organizations performing maintenance from a PM&P management oversight perspective, commensurate with the risk and likelihood of PM&P selection. While maintenance does not normally involve many PM&P selections, PM&P changes may occur because of obsolescence, other part availability issues, or small engineering changes. For example, oversight could include:

- Review and approval of a production or sustainment organization's tailored PM&P Management Plan that describes processes for making PM&P changes;
- Program office approval for changes not previously approved in the technical data package; or
- Program office approval for changes to form, fit, or function.

[58] See Federal Acquisition Regulation 45.101, Definitions, https://www.acquisition.gov/far/45.101.

[59] See Federal Acquisition Regulation 52.245-1, Government Property, https://www.acquisition.gov/far/52.245-1.

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Section 9. Evaluating Metrics to Improve the Efficiency and Effectiveness of PM&P Management

Pertinent sections of Figure 1 associated with evaluating metrics are included in Figure 11.

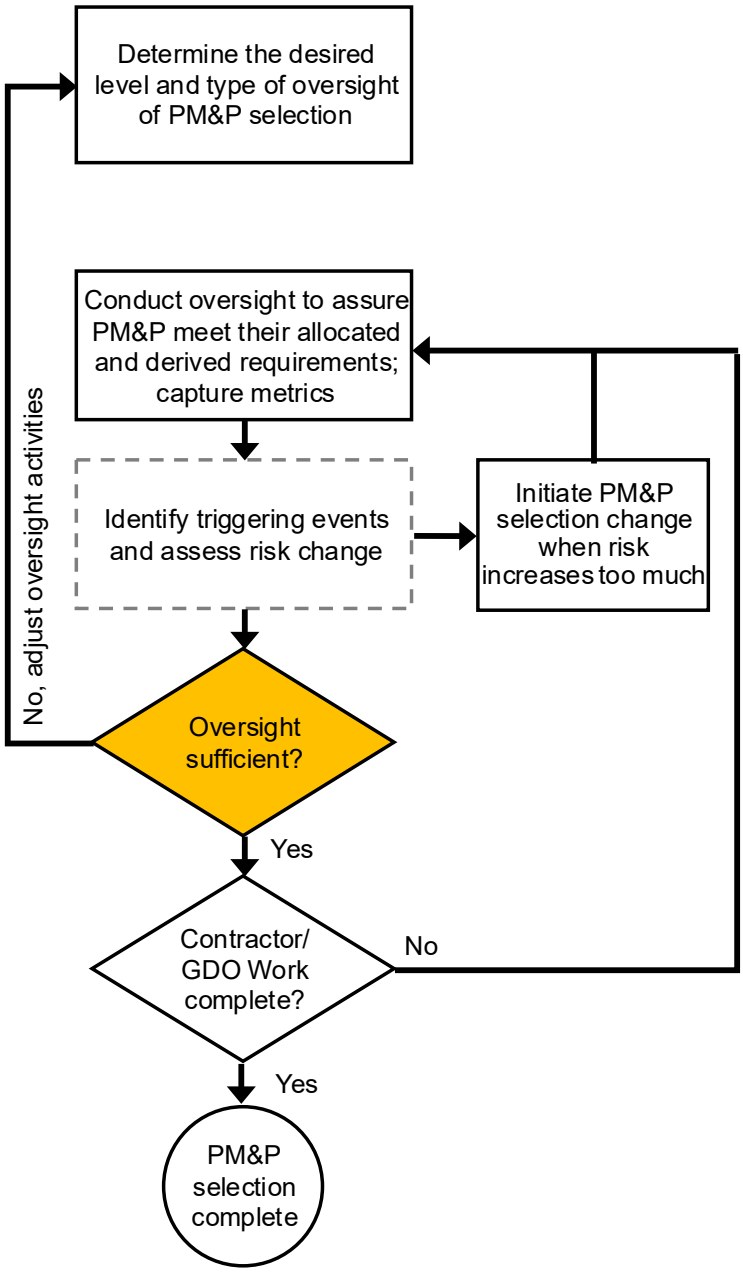


Figure 11. Evaluation of Metrics

Previous sections of this document have described how to determine and conduct the necessary level and type of PM&P management oversight. Record keeping and metrics capture elements of that

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oversight. As illustrated in Figure 11, throughout the oversight process, PM&P management records should be analyzed to assess the efficiency and effectiveness of that oversight and changes should be made to implement improvements as needed.

There are many benefits to analyzing PM&P management records. For instance:

- Identifying PM&P management resource constraints and justifying additional resources. PM&P management records are useful in measuring the scope of the PM&P management problem space and illustrating the potential value of dedicating increased resources to PM&P management. *See example 1.*
- Improved PM&P management efficiency. For a given set of resources, records allow program offices and higher echelon commands to observe where PM&P management resources are being allocated and where the most value is being added and to reallocate to the highest value-added areas. *See examples 1 and 2.*
- Reducing systemic DoW risks. Program offices make many decisions during part selection, some of which introduce risks. While individual risks may seem innocuous, records are useful in cumulative large risks that emerge from the confluence of multiple decisions made across multiple program offices. *See example 6.*
- Measuring the effects of disciplined PM&P management. While it's often claimed that PM&P management has downstream benefits such as improved system reliability and reduced life-cycle costs, records can provide the empirical evidence decision makers need to justify PM&P management resourcing decisions. *See examples 1, 2, 3, 7, and 8.*
- Monitoring trends in PM&P management practices. As the new PM&P management oversight paradigm is established, records will allow higher echelon commands to observe whether sound PM&P management practices are being adopted throughout the DoW, observe barriers to adoption, and develop courses of action to overcome them. *See example 4.*
- Improved government-contractor/GDO relations. Records can be used to produce summary statistics and identify trends in contractors' or GDOs' PM&P selection processes and performance. This will aid the government in strengthening relationships and building trust with contractors or GDOs who select PM&P. It will also provide program offices an opportunity to indicate their PM&P selection preferences to contractors and GDOs.[60] *See example 5.*

This section provides a series of concrete examples of how PM&P management records as described in section 8 can be used in practice. The benefits of PM&P management record keeping are illustrated by tying specific records to specific outcomes. The examples show how some of the records may be used to gain meaningful PM&P management insights, for both government program offices and higher-echelon commands. However, by necessity, the examples are an incomplete list of uses of records, and program offices should carefully consider additional records that may be of value.

PM&P management record keeping has several benefits for program offices and higher echelon commands. As mentioned previously in this document, disciplined PM&P management produces effects such as reduced life-cycle costs, improved system reliability, and improved cyber resiliency. PM&P management record keeping allows these effects to be rigorously documented and quantified. PM&P management records can also highlight areas where further analyses and fact finding should be conducted to determine root causes of potential issues. Both features improve decision-making with respect to PM&P management activities and investment.

The examples progress from simple cases which provide immediate insights for PM&P management practitioners, to more detailed empirical studies explaining causal relationships between PM&P management and warfighting effectiveness. Program offices can use these examples to consider

[60] A shared data repository for PM&P information such as approval dispositions and qualification data provide a method to effectively retain information over long periods.

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additional questions they would like answered, and hence what records should be kept. The examples and their benefits are not intended to apply to a single program office. Behavior identified by one program office may be useful to another, and some of the examples require collecting records from multiple program offices for analysis by a higher echelon command.

Each example is structured in three sections: *Records required*, which lists the records that should be collected during system development; the *Benefits and Metrics* from collecting these records; and a *Narrative* section providing additional context. For more advanced examples, additional records will normally need to be collected after the PM&P selection process, possibly by entities outside the program office. Such post-selection records are not listed under the *Records* section but would be mentioned under *Narrative*.

Example 1: Benefits of, and the capacity for, program office review and approval of parts

Records required

- Number of parts explicitly reviewed and approved by the program office.
- Number of parts selected only after rejecting a contractor's initial selection.
- Total parts selected.

Outcome

Greater understanding of the need to dedicate additional resources to program office approval of parts.

Narrative

By observing the number of parts it explicitly reviewed, as well as the number it rejected during review, a program office can gain an understanding of how frequently contractors are submitting part selections that do not meet the program office's expectations. If the rate of rejection is high, *and* the program office only had resources to review and approve a small fraction of total parts, then the program office can justify additional resources being allocated to part review and approval. This may involve reallocating resources within the program office or seeking more resources from its headquarters.

While a simple example that could be improved by further analysis such as the demonstrated long-term effects of part approval (see example 7), this has the benefit of being intuitive and providing immediate insights. Presumably, a program office would only reject a part if there were an eventual benefit to the warfighter.

Example 2: Benefits of verification efforts on the part selection process

Records required

- How many verification efforts a program office conducted to review a contractor's part selection process, categorized by type (audits, personnel embeds, DCMA surveillance, etc.).
- What percentage of parts reviewed by a program office were rejected.
- If applicable, all the above records should be categorized based on the division of the prime contractor, or subcontractor, for which the verification effort was conducted, or by whom the rejected part was selected.

Outcomes

- Higher echelon commands gain an understanding on whether verification efforts reduce the rate at which contractors select parts that must be rejected by program offices.
- In the situation where a single program office has collected records from multiple contractors, subdivisions, or subcontractors, that program office can likewise learn the effect of its verification efforts on part rejection rates.

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Narrative

Gathering records on verification efforts and part rejection rates will allow the program office or higher echelons to estimate the magnitudes of causal effects for each verification on rejection rates. Presumably, if verification efforts are valuable, then more of them would lead to lower rates of rejection. This is valuable because a lower rate of rejection implies less need to review part selections. However, verification efforts can be expensive if they require significant manhours. By improving understanding of how much value verification efforts add, the government will be better positioned to decide whether to invest more in these functions.

Example 3: Effects of PM&P management on reducing part proliferation

Records required

- Number of parts selected that did not previously have an NSN.
- Number of parts that were rejected by the program office and did not previously have an NSN, and were eventually replaced with a part that did have an NSN.

Outcome

A measurement of how much less part proliferation occurred due to the program office prioritizing commonality of parts.

Narrative

This simple example provides an easy way of measuring a core function of PM&P management, which is the reduction of the proliferation of parts. Comparing the number of new NSNs that would have been needed had the government not rejected any parts shows the effectiveness of PM&P management in reducing proliferation. The analysis can be enhanced using a cost-savings measurement for reduced proliferation, which could be either internally generated or informed by existing publications such as NAS 1524, “Standardization Savings, Identification, and Calculation” (refer to Section 7 for more information on NAS 1524).

More complex extensions can lead to additional insights. For instance, rather than looking only at whether a part had an NSN or not, collecting records on the actual demand for parts selected by other DoW users will provide a fuller picture of the system’s supportability. From a higher echelon perspective, indicator variables specifying whether program offices did or did not consider part commonality will help explain differences across program offices with respect to part proliferation.

Example 4: Changes in the use of PM&P Management Plans over time

Records required

- For each program office, whether they used a PM&P Management Plan to guide their PM&P management efforts.
- For each program office, how many contractors they employed (primes and subcontractors) which did and did not have a contractor PM&P Management Plan.

Outcomes

- Insight into the prevalence of program office and contractor PM&P Management Plans throughout the DoW over time, which is a useful indicator of whether program offices are or are not emphasizing PM&P management.
- Insight into the effect of program office and contractor PM&P Management Plans on key outcomes such as life-cycle costs, system reliability, and part proliferation.

Narrative

While providing useful insight, a time series of PM&P Management Plan usage may help higher echelon commands explain other phenomena being observed, such as changes in life-cycle costs, system reliability, or parts proliferation. For instance, explaining life-cycle costs would require a

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robust empirical study involving several factors, but if records on the use of plans are kept, analysts may find that the use of government and/or contractor PM&P Management Plans has a statistically significant effect on costs. Though not listed as a separate example, higher echelon commands should also consider collecting time series records on PM&P management costs. When combined with analyses of the effects of PM&P Management Plans on outcomes like life-cycle costs and system reliability, a cost-benefit analysis for PM&P management could be conducted.

Example 5: Knowledge sharing between program offices for improved contractor selection

Records required

For each program office and system developed, narrative descriptions of the outcomes of any audits that were performed.

Outcome

A database, that can be shared across program offices, to review lessons learned and experiences stemming from audits of part selection contractors. This will allow program offices developing new systems to make more informed decisions when choosing a contractor for part selection.

Narrative

In contrast to example 2, this example uses the details of particular audits to extract useful information for PM&P management practitioners. Unlike the other examples presented in this section, it relies exclusively on qualitative data, namely the narrative descriptions of program offices' audits. When a program office chooses a contractor to perform part selection for a new system, the choice may rely on prior experience, direct information from the candidate contractors, or a combination of these. By collecting the findings and lessons learned from past audits, program offices will also be able to incorporate knowledge from other program offices to improve their own decision-making.

In time, such a database could become large, making it impractical to read every entry before selecting a contractor. Simple techniques can reduce the size of relevant entries, such as searching for keywords, and more advanced techniques such as natural language processing may also be useful. As PM&P management records keeping matures as a discipline, a higher-echelon command can play a coordinating role in making the database more useful by creating standardized data entry requirements following a PM&P management audit.

Example 6: Monitoring the extent of exception reports

Records required

For each program office, the number of exception reports received by category.

Outcome

Higher echelon commands can identify potential sources of risk stemming from extensive use of exception reports throughout the DoW.

Narrative

Collecting records on the use of exception reports is extremely valuable for higher echelon commands analyzing supply chain risks. For example, a single exception report on the traceability of microelectronics may seem innocuous to a program office; however, when aggregated across the DoW, several such exception reports may indicate a systemic risk to the DoW's microelectronic supply chain. Continuous monitoring of the use of exception reports will allow higher echelons to intervene before risks get unacceptably high.

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Example 7: Effect of program office review and approval on Ao

Records required

- For each program office and system, the percentage of parts that required program office approval.
- For each system, the target level of Ao.

Outcome

An understanding of whether the extent of program office approval of parts leads to a higher chance that a system meets its target level of Ao once deployed.

Narrative

Like example 4, this example highlights a situation where PM&P management records can play a role in a broader empirical analysis. For deployed systems, an empirical assessment of why they have or have not met its targeted Ao levels could be performed using multiple factors. For instance, the realized availability and training of maintenance personnel would be an important factor, as would the funding and availability of spare parts and the environmental conditions in which a system has operated. Another important factor may be the quality of PM&P management during system development, which can be proxied by the percentage of selected parts the government was able to review and approve. If an empirical study were to find that more government review and approval leads to a higher chance of meeting Ao goals, it would provide justification for allocating more resources toward part review.

Other PM&P management records would be useful for this empirical analysis, such as the extent of audits and embeds during part selection, or a simple indicator of whether a program office PM&P Management Plan was used. Prior to performing the study, it's unknown which records will prove explanatory for realized Ao, so collecting more records is generally better. Also note that, like most empirical studies, the one described here may have confounding factors. For instance, it may be the case that program offices with stronger PM&P management are also those who are more diligent, have a better understanding of the systems they develop, and hence are better at estimating deployed Ao. While challenging, analysts are accustomed to dealing with such confounding factors, and having records to perform detailed empirical studies is still valuable.

Example 8: Effect of cybersecurity KSAs on reducing cyber incidents during system deployment

Records required

For all program offices, binary indicators for whether the program office acquired the needed cybersecurity-related KSAs per its PM&P Management Plan.

Outcome

Understanding of how resourcing cybersecurity expertise to support part selection impacts the prevalence of cyber incidents during system deployment.

Narrative

Collecting records across multiple program offices on whether the requirements for cybersecurity KSAs were properly resourced enables higher echelon commands to assess whether properly resourced program offices do, in fact, experience fewer cyber incidents. Records on the number of observed cyber incidents also need to be collected during system deployment. To make the analysis more complete, multiple binary indicators should be collected to identify the presence or absence of specific cyber-related KSAs, such as knowledge of critical hardware and software vulnerabilities, knowledge of cyber-attack and mitigation strategies, and so on.

This is another example where records from different program offices are being compared. To ensure like-systems are being compared to one another, higher echelons can (1) identify control factors that suggest two systems should have a similar amount of cyber incidents and perform a

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rigorous empirical analysis (as in examples 4 and 7) or (2) more informally compare results among a set of systems that are believed to be similarly vulnerable to cyber-attacks *before accounting for KSAs during part selection*.

These examples are inherently an incomplete list, as the types of insights that can be gained from PM&P management records are only limited by the practitioner's ability to ask probing questions and develop a record keeping plan to answer them. The provided examples are meant to fuel discussion within program offices and higher echelons which will result in empirically backed findings that will inform future PM&P management investment decisions. Updates to this document will include examples of record keeping being used in practice as the PM&P management community accumulates real-world findings on PM&P management effectiveness.[61]

[61] Update frequency will be a function of how many program offices are early adopters of this guidance.

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Section 10. Detecting and Evaluating Risk Changes

Figure 12 highlights the elements of Figure 1 concerned with how a program office PM&P Management Program should continuously seek to identify changes to the risk profile, quantify the change in risk, and assess whether the risk change is severe enough to change a PM&P selection. Note that identifying risk changes is mostly performed outside of PM&P management.

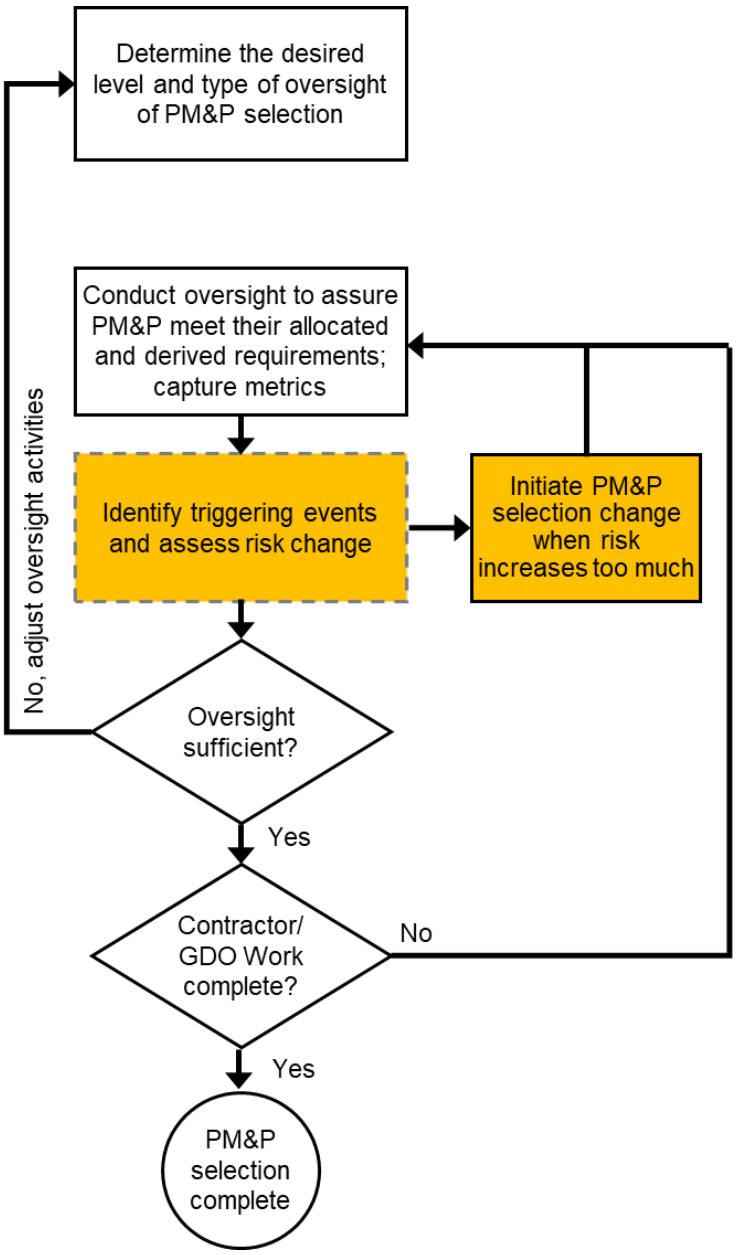


Figure 12. Changing PM&P Selections Based on Increased Risk

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Figure 12 uses the term triggering event. The concept of reassessing supply chain risk changes is derived from the idea of a triggering event as discussed in SAE JA7496 and defined as:

> *An occurrence that results in a risk assessment to identify any changes in the risk profile and the associated impact of those changes on a system and parts within the system. Examples of such occurrences are changes in the existing design or interface; changes in the risk profile of the system and its external supporting processes or procedures; or planned assessments.*[62]

All changes shown in the definition affect PM&P selection risk, even when the changes were made in response to a short-term availability issue.

10.1 Information on Associated Contractual Requirements

The current version of the SD-26 has no contract language associated with this subject. A future update to the SD-26 will include material for some of the monitoring activities listed below that may be performed by a prime contractor.

10.2 Oversight Considerations

If rigorous, risk-based PM&P selection occurred during initial design (and redesign), any part performance, system security, or availability risks should have been identified and mitigated if there was a determination that any of the risks were unacceptably high. However, the risk profile that existed during PM&P selection could have changed because of a triggering event. For example:

- Different maintenance equipment may have been introduced;
- A different maintenance provider may have been selected, or a current maintenance provider may be under new ownership;
- The OEM may have changed one of its (lower tier) suppliers—the OEM part number may or may not have changed;
- There may be changes to configuration items being refurbished or remanufactured;
- Engineering changes may have been made for product improvement purposes or to resolve obsolescence issues;
- New regulations may have driven product changes;
- New cyber weaknesses or vulnerabilities may have been discovered;
- The part's reliability may have decreased; or
- An adversary could have placed its representation on the Board of Directors of a (lower tier) supplier or a foreign government may have made an investment in a supplier.

PM&P management oversight should first assure that processes are in place to monitor proactively for triggering events throughout the life cycle and that data are accumulated on the events themselves.[63] Information about potential triggering events should be derived from multiple sources, including:

- Test failures or information reported through the Product Data Reporting and Evaluation Program;
- The results of a failure modes, effects, and criticality analysis that identifies potential failures in systems and equipment;

[62] Adapted from SAE JA7496, *Cyber-Physical System Security Engineering Plan* (CPSSEP), June 2022.

[63] One impediment to the identification of triggering events is limited supply chain visibility. However, limited visibility may also impact the amount of rigor applied when a PM&P was initially selected. The program office should nevertheless monitor for triggering events as thoroughly as data permits taking criticality into account.

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- Product changes identified through commercial industry issued PCNs, GIDEP reporting, or provided by the prime contractor;
- Lead time changes identified by part procurement organizations;
- Company financial health concerns as detected by the industrial base community;
- Changes to part vulnerabilities applicable to program protection or the protection of CPI or mission critical functions to achieve TSN as identified by system security engineers;
- Yield changes identified by manufacturing organizations;
- Part availability changes observed by commercial or government maintenance providers;
- Supply chain changes identified by logistics centers or SCRM organizations;
- Counterfeit issues detected in part acceptance processes;
- Obsolescence as reported by the organizations performing DMSMS management for the system;
- PPP compliance or supply chain risk identified during supplier assessments;
- Active Committee on Foreign Investment in the United States filings where a foreign entity seeks to acquire control over critical PM&P or components; and
- Open source, intelligence community, or counterintelligence community information that may have an impact on potential foreign influence or other vulnerabilities on a PM&P.

The PM&P management practitioner in the program office should be responsible for ensuring that all appropriate sources are collecting triggering event data and then proactively, during all life cycle phases, providing that data to the organization responsible for:

- Assessing the extent of risk change caused by the triggering event and
- Determining whether that risk change is serious enough to recommend a different part, change a supplier or a support (e.g., maintenance) provider, or different mitigating action (e.g., redesign).

During production, these responsibilities are often performed by the prime contractor with oversight from the program office. During sustainment, the program office either performs the work itself or arranges for contractor support to conduct the analysis. Typically, this is overseen by a SCRM focal point within the logistics function in the program office and carried out using an IPT. The PM&P management practitioner in the program office should be a member of that IPT and not only provide subject matter expertise but also inform the SCRM lead if some important monitoring activities are being overlooked.

Caution should be exercised that changes made after a triggering event are subject to rigorous oversight from the program office. It is important to review and approve:

- The portion of a production or sustainment organization's tailored PM&P Management Plan that describe processes for making PM&P changes and
- Changes not previously included in the technical data package.

Finally, the SCRM IPT in the program (or sustainment) office should provide oversight and ensure that all approved mitigating actions are being implemented. Depending on the scope of the change (e.g., whether there is a change to form, fit, or function), a field logistics organization may perform or oversee the implementation. In some specific circumstances, oversight may be led by a system security organization. The PM&P management practitioner in the program (or sustainment) office should monitor the thoroughness of these activities and point out any PM&P selection concerns.

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Section 11. Program Office PM&P Management Plan

11.1 Best Practices for a Program Office PM&P Management Plan

As previously discussed, the steps in a program office PM&P Management Program are listed below:

1. Assess risk to create a risk profile used to determine the level and type of oversight to be performed by the program office.
2. Set requirements for a contractor PM&P Management Plan that describes processes to assure that PM&P selections meet all allocated and derived requirements.
3. Verify that the contractor is following the processes in its Plan.
4. Validate that the contractor's PM&P selection is effective.
5. Keep records and develop metrics designed to improve PM&P selection.
6. Perform steps 2–5 when PM&P selection performed by GDOs.
7. Evaluate metrics and changing oversight processes where needed.
8. Detect risk changes generated by triggering events and adjust PM&P selections when warranted.

The program office PM&P Management Plan to oversee PM&P selection should be organized around these steps as follows:

- Section 1 of the Plan summarizes objectives, roles and responsibilities, and risk assessment results. It is based on step 1 of the Program supplemented by overarching information about infrastructure, resources, and rationale for performing the Plan.
- Sections 2–4 of the Plan lay out the risk-based oversight activities that the program office will perform associated with steps 2–4 of a program office PM&P Program when PM&P selection is being done by a contractor.
- Section 5 of the Plan describes how the record keeping and metrics will be used to improve PM&P selection. It aligns with step 5 of the Program.
- Section 6 of the Plan covers the risk-based oversight activities that the program office will perform associated with steps 2–5 of a program office PM&P Program when PM&P selection is being done by a GDO. Since oversight of a contractor is the norm in DoW, this section of the Plan can often be eliminated. However, Section 6 of the Plan may be necessary when the term GDO is applied in its broadest possible sense to encompass any government organization that selects PM&P.
- Section 7 of the Plan covers evaluating metrics and changing oversight processes where the evaluation identified inefficiencies per the material in step 7 of the Program. When section 6 of the Plan is eliminated because no GDO is involved, it is normally convenient to combine sections 5 and 7 of the Plan.
- Section 8 of the Plan describes monitoring for events that could trigger changes to PM&P selection and assessments of whether the risks associated with such triggering events warrant changes to be made. This corresponds to step 8 of the Program.

Formulation of the program office PM&P Management Plan should begin early in a system's life cycle. Such a Plan should be developed for *every company or GDO* that is involved in:

- Design activity;
- PM&P selection, replacement, or substitution; or
- The use of prototypes, NDI, or GFP.

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The emphasis on establishing a program office PM&P Management Plan for every company or GDO is crucial because PM&P management oversight of PM&P selection efforts is dependent on the functions performed on the system. For example, oversight of design is different than oversight of maintenance. Plans for different companies or GDOs may be combined into a single document at the discretion of the program office. Plans should be updated as appropriate whenever there is a contract modification or the exercise of a contract option with that company or the equivalent with a GDO.

There also are other reasons for updating the program office Plan. For example, system requirements or missions or operating environment may change, and such changes will impact the risk assessment that determined the appropriate level of program office oversight. Therefore, the program office Plan should be a "living document" that is relatively easy to modify.

A senior manager, e.g., the program manager, chief engineer, or quality lead, should approve the program office PM&P Management Plan. Because the program office PM&P Management Plan describes oversight activities for contractor and GDO systems engineering functions to select PM&P, the Plan should be discussed and referenced in the technical tracking sections of the SEP. The SEP outline calls for documentation of how the program office implements various aspects of systems engineering which encompasses PM&P selection considerations (e.g., corrosion, system security, reliability and maintainability). Implementation includes contract requirements and program office oversight of how the contractor or GDO satisfies those requirements per the program office's PM&P Management Plan.

A program office's Plan should include what it specifically intends to do based on resources available and why; it should not be a tutorial on best PM&P management processes. References to other applicable documents should be made as much as possible to avoid duplication. The program office should provide a copy of its PM&P Management Plan to the contractor or GDO to help guide PM&P selection activities and better align them with the program office's requirements and expectations. If the program office's Plan contains information inappropriate for sharing, the program office should redact such material.

11.2 Program Office PM&P Management Plan Contents

11.2.1 Program Office PM&P Management Plan Introduction

11.2.1.1 Scope and Applicability

The scope and applicability section of the program PM&P Management Plan should answer the following questions:

- **What are the program office's near-term and long-term objectives associated with its PM&P Management Plan?**

  The overall objective of a program office PM&P Management Program is to ensure that the parts selected for the applicable systems meet their allocated and derived requirements to include performance, quality, security, supply chain risk, etc. However, in most cases, a program office's long-term objective of its Plan is to establish the government's PM&P management activities associated with oversight of contracts to manage system security, part availability, and part performance risks. In the short term, however, depending on where the system is in its life cycle, a program office may have one of multiple, possible objectives. For example, one option may be ensuring the adequacy of the processes in the contractor PM&P Management Plan and ensuring that the contractor is following those processes. Another example is deciding the timeline for PM&P acceptance (to be defined by the program office, e.g., within five days) a contractor's approval submission.
- **On what system(s) is PM&P management oversight being conducted?**

  This question identifies the weapon system(s) covered by the program office PM&P Management Plan. While development and production contracts are usually associated with a single weapon system, support contracts may not be. If there will be no PM&P management activity on any of

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the systems under the purview of the program office, indicate which systems are excluded and the rationale for exclusion. PM&P management activities on test and support equipment should be encompassed in the PM&P management activities of the system(s) they support when included in the contract.[64]

- **What is the contractor or government supporting organization scope of work?**

  Indicate whether the contractor or government supporting organization is performing development, production, maintenance, modification, or some combination thereof and describe the extent of PM&P selection activities taking place for each of those functions. For example, maintenance may not involve a great deal of PM&P selection if common multiple-source parts are the only ones being consumed. On the other hand, development could involve significant PM&P selection.

11.2.1.2 Resources

The resources section of the program office PM&P Management Plan should identify the people involved in PM&P management oversight. In general, there will be two sets of skills involved—the general PM&P management skills necessary to perform oversight functions and detailed subject matter expertise needed under specific, technical circumstances.[65]

While it is not unusual for SMEs to be part of multiple program office organizations (or even external to the program office, since their expertise will not be needed all the time), the program office may choose whether to centralize general PM&P management oversight. A decentralized organization could place the PM&P management practitioners in various subsystem-related integrated product teams (IPTs) based on how the program office is organized.

Program office leadership should appoint a single, overall PM&P management lead early in the design process.[66] This PM&P management lead should coordinate the objectives of the PM&P Management Program and organize a PM&P management IPT (or equivalent). The IPT is generally responsible for implementing the program office PM&P Management Program, which includes interfacing with other program office IPTs. Contractors and GDOs that select parts should be represented on the IPT.

The resources section of the program office PM&P Management Plan should identify PM&P management practitioners needed to perform oversight, their roles and responsibilities, their areas of expertise, and the (program office or other) organization where they are assigned. A matrix as shown in Table 5 would suffice.

The Plan should also show the interactions needed for PM&P management practitioners to be able to perform their functions effectively. For example, the Plan could list the other communities/teams in the government and at the contractor with whom PM&P management practitioners should network (e.g., SCRM, design, system security engineering, cyber) and then describe the purpose and frequency of that networking.

Table 5. Example Matrix

Name	Roles and responsibilities	Areas of expertise	Organization
xxxx	PM&P management lead		
yyyy	PM&P management IPT member		

[64] A contract associated with a single system and its test and support equipment, would identify only that single system regardless of whether the support equipment is applicable to other systems. A contract only associated with test and support equipment for multiple systems should identify the test and support equipment itself as the system involved.

[65] A single individual may provide both types of skills.

[66] The PM&P management lead may not be a full-time function.

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Finally, the program office Plan should also indicate the total FTEs needed and available. The program office Plan should be executable within available resources. If the available FTEs are less than what was needed, the Plan should highlight the changes made to accommodate the shortfall.

11.2.2 Program Office PM&P Management Plan Oversight Description

Sections 2–4 of the Plan lay out the risk-based oversight activities that the program office will perform. These sections summarize a program office's oversight decisions based on the risk profile and the associated determination of the type and level of oversight that will be conducted.

- **Section 2 of the Plan covers the contractor or GDO PM&P Management Plan**. The program office Plan should state whether and where contractor or GDO PM&P Management Plans are required. When not required, an explanation of the exclusion rationale should be provided. In addition, the risks incurred from excluding a requirement for a contractor or GDO Plan should be identified in the program office Plan along with an explanation of the acceptability of those risks.

  When contractor or GDO Plans are required, the program office Plan should explain where and why the requirements of MIL-STD-11991 (or whatever document was used to establish contractor or GDO requirements) were tailored. When the requirements are stated in another way, the program office Plan should explain the basis for those requirements. The program office Plan should describe how it will examine contractor or GDO PM&P Management Plans and subcontractor PM&P Management Plans if applicable.[67] The Plan should also identify the contractor or GDO processes that will receive the greatest scrutiny along with the associated rationale. The program office Plan should additionally discuss review and approval procedures. Review procedures should include a list of reviewers and the elements of the contractor or GDO Plans where each reviewer will focus his or her efforts.
- **Section 3 of the Plan covers verification that contractors or GDOs are following the processes in their Plans**. The contents of this section should indicate when and how verification activities may differ as a function of the risk associated with subsystems (or other part groupings) as established in the risk profile. The program office Plan should discuss where there will be verification, how such oversight will be conducted, and the associated rationale. When verification is excluded, the program office Plan should explain why the risks were acceptable. Verification options, as discussed earlier in this document, include:
  - Official program office audit,
  - DCMA surveillance,
  - Embedding engineers with contractors or GDOs,
  - Formal program office reviews, and
  - Informal site visits and other interactions.
- **Section 4 of the Plan concerns validation of the effectiveness of contractor or GDO PM&P selection processes**. Based on the risk profile, the program office PM&P Management Plan should list the aspects of the system (subsystems) where the program office will apply additional PM&P selection oversight along with the rationale for focusing on those aspects of the system. This may be for all PM&P, only critical PM&P, PM&P associated with PPP identified subsystems, subsystems with similar likelihood and severity of risk, or something else depending on the nature of the risk assessment. This document described three validation oversight activities:
  - Monitoring of contractor's PM&P selections;
  - Assessing the viability of PM&P selected that do not meet government preferences (exception reporting); and
  - Program office approval of PM&P.

[67] Part of the review should be at the contractor's facility.

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Finally, the program office's PM&P Management Plan should briefly identify the combination of these three validation activities it will perform for each of those system aspects, as well as the rationale for performing those validation activities.

For monitoring, this section of the program office Plan should discuss whether there will be monitoring and include a risk-based explanation of why monitoring was excluded if that is the case. When monitoring is included, the program office Plan should describe the monitoring activities to be performed and how they will be carried out. In addition, the Plan should describe the areas that will be emphasized.

For exception reporting, the program office PM&P Management Plan should include a breakdown by the risk-differentiated (and therefore oversight differentiated) subsystem aggregations, as appropriate, of the following:

- The program office preferences to be put on contract.
- How the preferences are expressed for exception reporting. This may include a measure of criticality.
- Information to be reviewed that supports a decision to reject the part, coordinate with the contractor or GDO about further use of the parts listed in the exception report after 30 days, or to take no action which is equivalent to approving the use of the part.
- A description of the process for rejecting PM&P.
- A description of the process for coordinating further use of the PM&P and a plan for phasing out PM&P usage.

For program office approval of PM&P selections themselves, the Plan should discuss:

- A breakdown by the risk-differentiated (and therefore oversight differentiated) subsystem aggregations, as appropriate, of the PM&P requiring program office approval;
- The process for receiving approval requests;
- The initial data necessary to make an approval decision (e.g., evidence that the part meets its allocated and derived requirements) and how that data will be obtained;
- The decision-making processes and rationale for rejections;
- The process for resolving disagreements within the program office; and
- Decision record keeping and tracking.

Section 5 of the Plan establishes the record keeping and metrics that will be used to improve PM&P selection. It should identify the data elements the program office will collect, where the data will be maintained, how and how often the data will be collected, and the data sources.

Section 6 of the Plan is only applicable when a GDO selects PM&P. It should cover two additional subjects:

- The program office PM&P Management Plan should describe any measures taken to avoid the use of GFP because of PM&P selection risk concerns.
- The program office PM&P Management Plan should call for the maintenance performing government organizations to submit their own tailored PM&P Management Plans. The program office Plan should either include a provision to approve the use of any parts that have not been previously approved in drawings or similar documents or direct program office part selection when a part is unavailable. If there are no engineering changes, the maintenance organization's PM&P Management Plan could be limited to such an approval process.

Section 7 of the Plan discusses evaluating the data elements being collected per section 5 of the Plan. It should identify analyses to be conducted and where analyses results will be reported. When Section 6 of the Plan is not applicable, Sections 5 and 7 of the Plan should be combined.

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Section 8 of the Plan deals with triggering events. It should identify the processes used to:

- Proactively and methodically keep informed of triggering events identified by external organizations, internal elements of the program office, and contractors.
- Obtain triggering event information and provide that information to the organization that will assess risk changes and recommend PM&P selection mitigation when the risk is too high; and
- Oversee PM&P changes by production and sustainment organizations.

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Section 12. PM&P Management Support

DoW acquisition activities and contractors have several resources available to aid in effective PM&P management. The following sections detail resources and services, tools, and available training. More information on some of this material may be found on the Defense Standardization Program website.[68]

12.1 Resources, Tools, and Services

12.1.1 Item Reduction Program

The DoW Item Reduction (IR) Program consolidates items in the Federal Logistics Information System (FLIS) with generally similar form, fit, and function. The program reviews the items in FLIS, which affords visibility of items across program offices, military services, and federal agencies. Successful IR actions facilitate parts selection by achieving the following:

- Increased standardization of items in FLIS. Eliminating redundant, superseded, out-of-date, or no longer required items reduces the number of parts that can be selected.
- Economies of scale in acquisition. Increased demand for a standardized item of supply increases competition from suppliers, lowers costs, and shortens acquisition lead times.
- Efficiencies in storage and distribution. Reduction in items of supply from standardization decreases storage and distribution requirements at all levels, from shipboard to major depot, shrinking the required logistic footprint.
- Increased reliability. Increased item standardization results in more extensive performance history for individual parts, which leads to more precise performance estimates.

The IR Program operates independently, and no actions by PM&P management practitioners are needed to obtain these benefits.

12.1.2 Government-Industry Data Exchange Program

The GIDEP is a Defense Standardization Program (DSP)–sponsored activity that brings together government and industry participants with the aim of sharing, and consolidating technical information essential during the research, design, development, production, and operational phases of the life cycle of systems, facilities, and equipment. GIDEP-consolidated data materially improves decision-making during the acquisition and logistics phases of the life cycle and reduces costs in the development and manufacture of complex systems and equipment. The byproduct of this is improved quality and reliability of systems and components.

From a parts selection perspective, GIDEP provides information that part selecting organizations should consider and that program office PM&P management practitioners should use when conducting oversight. Most importantly in that regard, GIDEP contains reports identifying some potential counterfeit risks and disseminating part discontinuation notices. Another important feature is that GIDEP publicizes PCNs that may lead to the need to replace parts that have already been selected.

GIDEP's web address is www.gidep.org.

[68] See https://www.dsp.dla.mil/Programs/Parts-Management.

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12.1.3 DMSMS and PM&P Management Working Groups

The DMSMS and PM&P Management Working Groups offer a forum to promote, shape, and coordinate government and industry approaches to effective PM&P and DMSMS management procedures in support of the warfighter. Organized by the Office of the Secretary of War for Research and Engineering, the working groups provide organizations with several benefits, such as networking opportunities with multiple levels of government and industry, a forum to share innovative tools and practices, international partner collaborations, pertinent news and policies related to PM&P and DMSMS management, increased knowledge of resources, and opportunities to contribute to the strategic objectives of both communities.

12.1.4 ASSIST

ASSIST, formerly known as Acquisition Streamlining and Standardization Information System, is a web-based application that serves as the DoW's official source for standardization documents developed, maintained, and used by DoW. ASSIST is a critical DoW application that houses requirements documents cited in defense acquisitions for the development, acquisition, and logistic support of systems and equipment used by the DoW. Through ASSIST, the DoW maintains centralized control over the indexing, cataloging, management, and distribution of standardization documents and related information in accordance with DoDM 4120.24. ASSIST is located at https://assist.dla.mil.

12.1.5 Defense Logistics Agency Land and Maritime Document Standardization Unit

The Defense Logistics Agency (DLA) Land and Maritime Document Standardization Unit is the preparing activity for thousands of standardization documents covering a wide variety of electronic components and other items. The DLA Land and Maritime website's MIL-SPEC page has search tools to aid in the identification and selection of high-quality, reliable standard electronic components. The Document Standardization Unit coordinates and prepares technical documents in 70 Federal Supply Classes and offers engineering support to DoW customers using these documents: https://landandmaritimeapps.dla.mil/offices/Doc_Control.

12.1.6 Pin Point

Pin Point is a government-only query engine for researching parts in the federal supply chain. Pin Point provides users with rich technical and logistics information about NSNs and tens of millions of commercial parts. Defense and industry part managers use the system to reduce part duplication; manage risks such as long lead times, obsolescence, and the presence of restricted materials; evaluate part similarity; and conduct market and price research.

Pin Point aggregates data from multiple sources, including databases and legacy systems internal to the DoW enterprise, as well as external web-based sources, such as manufacturer and supplier websites. Pin Point uses artificial intelligence to extract and infer part properties from narrative text, specifications, part numbers, and product descriptions, and provides parametric searching on physical and logistic part properties. Users have access to complete, continually updated technical, management, and reference data about NSNs and commercial parts. Pin Point is restricted to DoW common access card (CAC) holders and can be accessed through ASSIST at https://assist.dla.mil or https://pinpoint.xsb.com. Direct questions can be sent to pinpoint-admin@exiger.com.

12.1.7 Weapon System Impact Tool

The Weapon System Impact Tool (WSIT) provides access to information about weapon systems and specifications associated with NSNs. The tool helps to group parts influenced by a specification and helps program offices evaluate the effect of specification changes on weapon systems. WSIT's query feature

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provides a list of weapon systems in which the NSN associated with the specification is used. In addition to the raw number of weapon systems using the affected part, WSIT provides a breakdown of the part's use by military service component.

WSIT allows users to search by military specification, federal specification, commercial item description, or non-government standard. It can supply a list of the NSNs and controlling part numbers referenced in that document. WSIT is restricted to CAC holders and can be accessed through ASSIST at https://assist.dla.mil or at https://wsit.xsb.com.

12.2 Training

12.2.1 WarU Courses Related to PM&P and DMSMS Management

- LOG 0630 Introduction to PM&P Management (revision planned 2026)
- LOG 0640 DMSMS: What the Program Manager Needs to Do and Why
- LOG 0650 DMSMS Fundamentals
- LOG 0660 DMSMS Executive Overview
- LOG 0670 DMSMS Research Essentials

12.2.2 WarU Parts and Material Life Cycle Management Credential

- LOG 0380 Provisioning and Cataloging
- LOG 0390 Additive Manufacturing Overview
- LOG 0470 Sustaining Engineering
- LOG 0510 System Retirement, Disposition, Reclamation, Demilitarization, and Disposal
- LOG 0621 Counterfeit Risk Management and Mitigation
- LOG 0630 Introduction to PM&P Management (revision planned 2026)
- LOG 0640 DMSMS: What the Program Manager Needs to Do and Why
- LOG 0650 DMSMS Fundamentals
- LOG 0660 DMSMS Executive Overview
- LOG 0670 DMSMS Research Essentials
- CON 0040 Market Research
- CLE 019 Modular Open Systems Approach
- CLE 026 Trade Studies

The Parts and Material Life Cycle Management Credential (CLCL 014A) is available to workforce members from any defense acquisition workforce functional community, including life cycle logistics, systems engineering, production, quality and manufacturing, and program management. It includes a scenario-based end-of-credential assessment. The total time to complete the 14 online-training modules is approximately 40 hours, excluding the time to complete the end-of-credential assessment. Additional information on the WarU credentialing program is available at https://www.dau.edu/credentials.

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Appendix A. PM&P Management Policy and Guidance

As described in Section 1.3, PM&P management reflects two broad types of policy and guidance—overarching and that which is pertinent to engineering design considerations. This appendix provides more detailed information on pertinent policies and guidance related to the acquisition pathways, systems engineering, system security engineering, and SCRM. The tables in this appendix often use the term "parts management" instead of "PM&P management." That is because the source documents themselves have not been updated to use the most recent terminology.

A.1 Pertinent Guidance Related to the Acquisition Pathways

Table 6. PM&P Management Guidance Content by Acquisition Pathway

Acquisition Pathway	Summary of PM&P Management–Related Guidance Content
MCA	PM&P management content appears in the roles and responsibilities, as well as activities with the acquisition phases. During the TMRR and EMD phases, the systems engineer is responsible for developing or updating a plan for the management and mitigation of PM&P issues, which includes the identification of the technical data needs to support mitigations. During the production and deployment phase, the systems engineer is further responsible for the identification of long-lead items and critical parts and materials and the mitigation of impacts on production and sustainment. During the operations and support (O&S) phase, the systems engineer is responsible for performing parts and DMSMS risk analysis to identify, prioritize, and mitigate any issues. In addition, the MCA chapter lists parts management as one of several specialty engineering disciplines. Specifically, parts management is described as a fundamental process that should be tailored to meet the needs of a program office, starting during early design to ensure that systems engineering and manufacturing objectives are met. The PM&P management process continues to be important through the O&S phase to reduce costs, enhance logistics readiness and interoperability, reduce acquisition lead-time, increase supportability and safety of systems, and enhance reliability and maintainability engineering.
MTA	There are parts management sections under specialty engineering, corresponding to rapid prototyping and rapid fielding. Program managers should implement parts management, as appropriate, during parts selection, parts redesign, and further address parts management during design reviews
UCA	There is a parts management section under specialty engineering. Program managers should implement parts management, as required, when parts require modification or the system will enter longer term sustainment.

A.2 Pertinent Guidance Related to the System Engineering Guidebook

Table 7. Location of PM&P Management Content in Systems Engineering Guidebook

Location	Summary of PM&P Management–Related Guidance Content
Section 2	• Under "System-Level Considerations," systems and systems of systems are described as consisting "of parts, relationships, and a whole that is greater than the sum of its parts." Systems engineers oversee the implementation of disciplined systems engineering processes and "the selection of qualified, trusted vendors for parts, materiel, and processes (for hardware and software)." • Under "Systems Engineering Role in Contracting," a contractor PM&P Management Plan and associated deliverables is listed as one type of typical content to appear in "Section H: Special Contract Requirements" for a request for proposal.

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Location	Summary of PM&P Management–Related Guidance Content
Section 3	As part of a PDR, the evaluation of parts lists for compliance with the PM&P Management Plan is listed as one type of criteria for consideration as part of the technical baseline documentation/digital artifacts (allocated).
Section 5	Under the discussion of the DMSMS design consideration, one of the practices listed to promote DMSMS resilience is the proactive assessment of parts obsolescence risk when selecting parts. The verification of the plans for the selection and application of parts and materials to minimize reliability risks is listed among reliability and maintainability activities during preliminary design. Furthermore, during detailed design, and production, reliability and maintainability activities include the verification that parts, materials, and processes meet system requirements. The latter activity for during design includes a reference to MIL-STD-1546, *Parts, Materials, and Processes Control Program for Space and Launch Vehicles*; MIL-STD-1547, *Electronic Parts, Materials, and Processes for Space and Launch Vehicles*; and MIL-STD-11991, *General Standard for Parts, Materials, and Processes*. Parts management and selection are also discussed for the purposes of standardization.

A.3 Pertinent Policy and Guidance Related to Systems Engineering

Table 8. Policy and Guidance Pertinent to Design Considerations for Systems Engineering

Design Consideration	Policy or Guidance	Document
Overarching	Policy	DoDI 5000.88, *Engineering of Defense Systems*, November 18, 2020.
	Guidance	*Engineering of Defense Systems Guidebook*, February 2022.
		Systems Engineering Guidebook, December 2021.
		Department of Defense Systems Engineering Plan Outline, Version 4.0, May 2023.
Manufacturing and Quality	Policy	DoDI 5000.93, *Use of Additive Manufacturing in DoD*, July 6, 2021.
	Guidance	*Early Engineering and Quality Engineering Guide*, July 2022.
		DoD Manufacturing and Quality Body of Knowledge (M&Q BoK), January 2021.
Reliability and Maintainability	Guidance	*DoD Reliability and Maintainability Engineering Management Body of Knowledge*, August 2018.
Standardization	Policy	DoDM 4120.24, *Defense Standardization Program (DSP) Procedures*, October 15, 2018.
Supportability	Guidance	*Product Support Manager Guidebook*, May 2022.
		Life Cycle Sustainment Plan: Sample Outline, Version 3.0, October 2022.
		Integrated Product Support Element Guidebook, January 2024.

A.4 Pertinent Policy and Guidance Related to System Security Engineering

Table 9. Policy and Guidance Pertinent to Design Considerations for System Security Engineering

Design Consideration	Policy or Guidance	Document
Assurance	Policy	DoDI 5200.50, *Access to Assured Microelectronics*, June 27, 2025.
	Guidance	(Forthcoming)
Program Protection	Policy	DoDI 5200.39, *Critical Program Information (CPI) Identification and Protection Within Research, Development, Test, and Evaluation (RDT&E),* October 1, 2020.
		DoDI 5200.44, *Protection of Mission Critical Functions to Achieve Trusted Systems and Networks*, February 16, 2024.

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Design Consideration	Policy or Guidance	Document
		DoDI 5000.83, *Technology and Program Protection to Maintain Technological Advantage*, May 21, 2021.
	Guidance	*Program Protection Plan Outline & Guidance*, Version 1.0, July 2011.
		Technology and Program Protection Guidebook, July 2022.
		Trusted Systems and Networks (TSN) Analysis, June 2014.
		Program Protection Plan Evaluation Criteria, Version 1.1, February 2014.
Anti-Tamper	Policy	DoD Directive 5200.47E, *Anti-Tamper*, December 22, 2020.
		DoDI 5200.39* and DoDI 5000.83.* [Full citations provided under Program Protection]
Cybersecurity	Policy	DoDI 5000.90, *Cybersecurity for Acquisition Decision Authorities and Program Managers*, December 31, 2020.
		DoDI 8500.01, *Cybersecurity*, October 7, 2019.
		DoDI 8510.01, *Risk Management Framework for DoD Systems*, July 19, 2022.
		System Protection—DoDI 5200.39.* and DoDI 5200.44.* [Full citations provided under Program Protection]
	Guidance	*DoD Program Manager's Guidebook for Integrating the Cybersecurity Risk Management Framework (RMF) into the System Acquisition Lifecycle*, Version 1, September 2015.

A.5 Pertinent Policy and Guidance Related to Supply Chain Risk Management

Table 10. Policy and Guidance Pertinent to Design Considerations for Supply Chain Risk Management

Design Consideration	Policy or Guidance	Document
Classical SCRM	Policy	DoD Directive 3000.16, *Vendor Threat Mitigation*, July 6, 2022.
		DoDI 4140.01, *DoD Supply Chain Materiel Management Policy*, September 2, 2025.
		DoDI 4140.67, *DoD Counterfeit Policy*, February 2, 2024.
		DoDI 4245.15, *Diminishing Manufacturing Sources and Material Shortages*, September 2, 2025.
		DoDM 4245.15, *Diminishing Manufacturing Sources and Material Shortages*, October 26, 2022.
		DoDI 5000.91, *Product Support Management for the Adaptive Acquisition Framework*, November 4, 2021.
		DoDM 4140.01, *DoD Supply Chain Materiel Management Procedures*: Vol. 3, Materiel Sourcing, August 26, 2022. Vol. 4, Make and Maintain Materiel, November 1, 2019.
	Guidance	*DoD Supply Chain Risk Management Guidebook*, June 2025.
		[See Systems Engineering—Supportability guidance]

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Appendix B. PM&P Management Questions for System Engineering Technical Reviews, Independent Logistics Assessments, and Sustainment Reviews

Table 11 contains examples of questions that may be asked during the system engineering technical reviews (SETRs), independent logistics assessments (ILAs), and sustainment reviews throughout the acquisition life cycle. The questions correspond to plans or requirements associated with PM&P management processes. Answers to all applicable questions should be yes to move to the next phase of the life cycle.

The first five columns of Table 11 correspond to five of the SETRs as follows:

- System Requirements Review (SRR) where top-level requirements are understood.
- System Functional Review (SFR) where the functional baseline is established.
- PDR where the baseline is shown to meet the requirements and development is ready to begin detailed design.
- CDR where the design is stable and fabricating, integrating, and higher-level testing is about to start.
- Production Readiness Review (PRR) where the system is ready to begin low rate production.

ILAs occur at the following points in the life cycle of a system as shown in the next to last column of Table 11:

- Milestone (MS) B where a decision is made to begin detailed design.
- MS C where low rate production is approved.
- Full rate production (FRP) where an FRP decision is made.

Sustainment reviews should occur at initial operating capability (IOC), where sufficient assets have been fielded to operate the system within a unit, and at least every five years after IOC.

Table 11 is written to reflect only the first time a question is posed. It is implied that program offices must continuously thereafter ensure the requirement is being adhered to.

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Table 11. Examples of Questions to Be Addressed at Different Stages of the Acquisition Life Cycle

	SRR	SFR	PDR	CDR	PRR	MS B, MS C, and FRP	IOC & After
Government risk assessment							
	Has the program office begun a risk assessment to determine the appropriate level of PM&P management oversight?	Has the level and type of program office PM&P selection oversight been determined?					
Resourcing government oversight							
		Have the required skillsets and numbers of FTEs to perform oversight been identified?	Have the required resources to perform PM&P management oversight been obtained?				
Contract or GDO requirements establishment							
		Have PM&P management contract or GDO requirements been drafted to correspond to the necessary level of oversight?	Have all PM&P management requirements been put on contract or an equivalent document for a GDO?		Are PM&P management contract or GDO requirements being updated as needed?		
Program office PM&P Management Plan							
		Has a preliminary program office PM&P Management Plan been developed?	Has the program office's PM&P Management Plan been approved?		Is the program office PM&P Management Plan being updated as needed?		
Contractor or GDO PM&P Management Plan							
		Is a contractor or GDO PM&P Management Plan required in the contract or elsewhere for a GDO?	Has the program office approved the contractor or GDO PM&P Management Plan?		Is the contractor or GDO PM&P Management Plan being updated as needed?		
Lead free risk mitigation plan							
		Is a lead-free control requirement established in the contract?	Has a lead-free control plan been developed, implemented, and approved by the program office?				

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	SRR	SFR	PDR	CDR	PRR	MS B, MS C, and FRP	IOC & After
Counterfeit risk mitigation plan							
		Is a counterfeit control requirement established in the contract?	Has a counterfeit control plan been developed, implemented, and approved by the program office?				
COTS management plan							
		Is a COTS management requirement established in the contract?	Has a COTS management plan been developed, implemented, and approved by the program office?				
PM&P selection and approval							
		Has the program office determined how it will verify that the contractor or GDO is following the processes in its approved Plan?	Has the program office verified that the contractor or GDO is following the processes in its Plan?				
			Has the program office verified that all design considerations are being taken into account during contractor or GDO PM&P selection?				
Subcontractor management							
		Have contractor flow down requirements been established?	Are requirements being flowed down correctly?		Are requirements being flowed down to new subcontractors?		
		Has the program office's role in overseeing subcontractors been established?	Is the program office carrying out its role?			Is the program office role being updated?	
Monitoring and record keeping							
		Does the contract specify that BOMs must be provided to the program office?	Is the program office reviewing BOMs (and updates to BOMs) to validate that parts meet their allocated and derived requirements?		Are BOM updates being received?		

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	SRR	SFR	PDR	CDR	PRR	MS B, MS C, and FRP	IOC & After
		Does the program office PM&P Management Plan include participation in program reviews and other on-site oversight?	Are program office PM&P management practitioners participating in program reviews and conducting other on-site analyses?				
		Have record keeping requirements for PM&P management processes been established (what must be collected, and when)?	Are the appropriate data being collected, maintained, and analyzed?				
Exception reports							
			Have procedures for submitting and reviewing parts listed on exception reports been documented?	Are exception reports being analyzed?			
PM&P approval							
		Has responsibility been established for PM&P approval?					
		Have data requirements been established for PM&P approval?	Are PM&P approval processes in place and being followed?		Are PM&P approval processes being updated as needed?		
Triggering events							
			Are triggering events being identified, analyses being conducted to determine additional risk being imposed by triggering events, and changes being implemented to reduce that risk where needed?				

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Appendix C. Description of a Formalized Approach for a Risk Assessment

C.1 Risk Categories and Types

Evaluating the risks associated with inadequate PM&P selection begins with the identification of potential risks. The following is a representative list of risk categories and associated risks that may be used.[69] There may be overlaps between the categories and among the risks within and across the categories. The risk categories themselves may be defined differently in other DoW policy and guidance; however, these differences do not impact the PM&P management function, which relies primarily on the existence of the risks. Depending on the circumstances of the specific system involved, additional PM&P management risks from the program office's risk register should be added to the following.

Part performance is the first risk category, and it is also a traditional focus of PM&P management oversight. Other disciplines are concerned with these risks as well. Part performance risks include:

- Inadequate reliability or maintainability
- Inability to meet requirements in all pertinent operating environments
- Lack of supplier quality
- Inadequate design
- Inadequate qualification and testing
- Inadequate manufacturing processes
- Improper application/use of soldering processes and materials
- Improper parts derating
- Use of prohibited PM&P
- Use of life-limited parts
- Adverse impact to operating environment (e.g., humidity causing corrosion).

The second category is system security risk. Such risks have been given high attention in recent years, but PM&P management practitioners traditionally limit their efforts to trying to ensure that system security or cyber engineers consider such risks. Program offices should prevent situations where the PM&P management and/or the PM&P approval function incorrectly assumes that system security risks have been examined by SMEs. Consideration of all PM&P selection risks should be a PM&P management oversight responsibility. System security risks include:

- Known weaknesses and vulnerabilities
- Unsecure networks and systems
- Inadequate hardware and software assurance
- Inadequate traceability
- Unprotected information
- Tampering[70]

[69] Other categories may be used.

[70] Exploitation of CPI in U.S. defense systems in domestic and export configurations to impede countermeasure development, unintended technology transfer, or alteration of a system due to reverse engineering.

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- Malicious intrusions
- Unsecure design.

The third risk category is part availability. PM&P management practitioners have traditionally considered these risks when oversight is performed. Some of these risks are considered and overseen by other disciplines. Part availability risks include:

- DMSMS and their root causes (e.g., financial viability of the suppliers, market for the part, technological advances, …)
- Hard to find parts
- Long lead times
- Counterfeits
- Cost
- Use of hazardous material
- FOCI
- Sole sources
- Lack of standardization
- Increased range of parts to be made available in the wholesale supply system.

Determination of risk should not be limited to the weapon system itself. Associated test and support equipment should also be considered when they are part of the same contract. PM&P used on such equipment can impact the larger system mission, life-cycle cost, and readiness. For example, poor PM&P selection for test equipment may generate invalid test results which in turn may impact reliability and performance in an actual mission.

C.2 Evaluating Risk Likelihood

Likelihood criteria are often portrayed on a five-level as shown in Table 12, which was extracted from the *Risk, Issue, and Opportunity Guidebook*.[71]

Table 12. Typical Likelihood Criteria

Level	Likelihood	Probability of Occurrence
5	Near Certainty	> 80% to ≤ 99%
4	Highly Likely	> 60% to ≤ 80%
3	Likely	> 40% to ≤ 60%
2	Low Likelihood	> 20% to ≤ 40%
1	Not Likely	> 1% to ≤ 20%

The following bullets describe overlapping risk drivers that a program office should consider when determining the likelihood of applicable risks at any point in the life cycle.

- ***Extent of design (and therefore PM&P selection and procurement) activities.*** If there is absolutely no PM&P selection, due to no design or limited design (e.g., a repair contract where the contractor has no authority to change parts), then PM&P management contract requirements

[71] Office of the Executive Director for Systems Engineering and Architecture, *Department of Defense Risk, Issue, and Opportunity Guidebook*, September 2023, incorporating change 2.2 as of December 2023, section 2.3.2, https://www.cto.mil/wp-content/uploads/2024/05/RIO-2023-2-2.pdf.

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and associated oversight can be minimal. However, it is important to recognize that system designs and their component parts typically change over time.

- ***New or different PM&P may be introduced to a system design.*** As more PM&P are selected or changed, there is an increased likelihood that the consequences of changed parts not meeting their requirements may be significant. There are numerous reasons for part changes: (1) A part manufacturer may replace one part with a newer version and may or may not change the part number.[72] (2) There are situations where the process to make the part changes or the suppliers change. (3) Parts may no longer be available and an equivalent part that may or may not be mentioned on the drawing may be used.
- ***Possibility of major or minor configuration/ECPs being made.*** This driver is closely associated with the extent of design activities anticipated during production and sustainment. If there are only minor (alias class II) engineering changes, then a lower level of oversight may be appropriate because the likelihood of PM&P changes (and their associated risks) would be lower. For major (alias class I) changes, more oversight is desirable, since the risk likelihood should be greater when the ECP scope increases. As shown in the box below, major changes will impact the system's capability. The parts selected for such major changes are also more likely to be critical and have higher consequences if they do not meet their allocated and derived requirements.

Definition of Major (alias class I) ECP

An ECP proposing a change to approved configuration documentation for which the government is the current documentation change authority (CDCA) or that has been included in the contract or statement of work by the tasking activity, and:

(1) Affects any physical or functional requirement in approved functional or allocated configuration documentation.

(2) Affects any approved functional, allocated, or product configuration documentation and cost, warranties or contract milestones, or affects approved product configuration documentation.

Source: Military Handbook (MIL-HDBK)-61B, *Configuration Management Guidance*, April 7, 2020.

- ***Life cycle phase and acquisition pathway.*** As indicated above, all contracts with design activities (whether they be initial design, redesign, or smaller ECPs) involve part selection, which may impact risk likelihood.
  - PM&P selection is clearly of interest during the development phase.
  - PM&P also change during production and, possibly to a lesser extent, during sustainment, when obsolescence can be a principal driver for the selection of new parts to replace those parts that have gone out of production or are no longer available.
  - Spares procurement contracts, on the other hand, generally have fewer part changes.

  The acquisition pathway also can impact the amount of PM&P selection required.
  - Service contracts have very little PM&P selection, unless they are associated with providing system maintenance.
  - The amount of PM&P selection through the software acquisition pathway is similarly small.
  - Very little PM&P selection takes place for UCA pathway programs, because the development effort is minimal and deployment ends in less than two years from the start of development.

[72] The document uses the term part number exclusively and does not refer to item numbers. There are two types of part numbers. The original component manufacturer (OCM) part number is the number assigned by the manufacturer of the part. The original equipment manufacturer (OEM) part number is assigned by the OEM of the parent assembly of the OCM part.

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- The MTA pathway could have significant development activity during rapid prototyping unless the system transitions immediately to rapid fielding. In other situations, the MTA pathway can transition to the MCA pathway where PM&P selection is substantial. In general, greater PM&P selection implies greater risk likelihood and the need for more oversight.

- ***Threats to and opportunities for system compromise.*** In addition to a part not meeting quality, performance, and supportability specifications, a part may not meet requirements due to *external* factors that can compromise a part. External factors include adversary exploitation of vulnerabilities and weakness, malicious tampering or counterfeits, protections designed into the system, and the likelihood that the system will be operating in an environment where there are threats to these external factors.
- ***Design control.*** When a contractor has design control, a program office may not be able to influence PM&P selection; however, even in such situations, this does not mean that the government will never assume design control in the future. Therefore, there may be situations where the contractor has full control of a system's design, but the program office retains some level of oversight on PM&P selection as a future contingency to minimize design control risk. When that is the case, all other risk likelihood principles apply.
- ***Use of commercial-off-the-shelf (COTS) assemblies or non-developmental items (NDIs).*** Similar to "design control," the use of a COTS assembly or NDI may have an impact on risk likelihood. Risks associated with parts internal to COTS/NDI may not be known or may not be easily attainable. Furthermore, it may be very difficult or expensive to address such risks when they are found. Depending on the criticality, sensitivity, or vulnerability to compromise, different oversight mechanisms (e.g., obtaining and assessing the risks associated with the COTS/NDI bill of material [BOM]) may be employed.
- ***Use of GFP or government developed prototypes.*** Risk is a function of the extent of PM&P management discipline that the government imposed during the development of the GFP or prototypes. While all program offices should apply disciplined PM&P management, a new development program may use GFP or prototypes that were not subject to the same level of rigor. If the PM&P management discipline was inadequate, risk will also be affected by the extent to which the receiving program office can and does take corrective action where needed after the fact.

C.3 Evaluating Risk Severity

The Risk, Issue, and Opportunity Guidebook also contains a matrix with consequences associated with cost, schedule, and performance.[73] While that breakdown is appropriate for PM&P management, the criteria for determining a risk level may not be fully applicable. Consequence levels in that matrix are minimal impact, minor impact, moderate impact, significant impact, and critical impact. A program office should define its own criteria based on its specific situation.

The mission of a system along with negative consequences to that mission are important aspects of risk severity. In addition, life-cycle cost or readiness impacts can result from a PM&P being compromised, a part being unavailable, or a failure of the part to perform adequately or reliably. There are several interrelated considerations that provide subjective data for evaluating the severity of the risks from PM&P selection. These considerations and those discussed for risk likelihood drive a program office's initial decision on the extent of PM&P management oversight it should perform.

- ***Reliability requirements.*** This consideration, especially when the performance requirements are demanding, indicates a situation where the system missions and operating environments drive the need for high reliability parts. When high reliability is critical to mission success, the consequences of a part failure are high and therefore the extent of oversight should be greater. However, such performance requirements do not always drive the inclusion of system reliability

[73] Ibid, Section 2.3.1.

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requirements in the contract. Unreliable parts (which often are less costly parts) may meet performance requirements, but rigorous oversight should minimize their use.

- ***Complex, critical, or sensitive functions.*** PM&P selection associated with this consideration indicates circumstances where the risk severity might be much higher, and therefore, PM&P management oversight should also be more rigorous. One such circumstance is a critical (either mission critical or safety critical) function. Rigorous PM&P management is more important when mission assurance is critical. The same is true for sensitivity, which refers to security classification or communications. A part not meeting its requirements could compromise information that leads to significant repercussions. Technically challenging functions, including operations in harsh environments, are often related to critical or sensitive operations.
- ***Long-life systems with minimal planned changes.*** The extent of PM&P selection for systems with this characteristic impacts the supportability of the system. Selection of high quality and reliable parts, not near obsolescence, generally lowers life-cycle cost and tends to increase readiness. Therefore, there may be greater severity of poor PM&P selection in these somewhat stable situations and the level of oversight (and associated contract requirements) should reflect those consequences.
- ***Life-cycle cost.*** In addition to complex, critical, or sensitive functions, systems with unnecessarily high life-cycle cost because of inadequate PM&P management imply that PM&P selection oversight should be more rigorous. High cost could be a function of poor reliability, inadequate quality, or design characteristics driven by PM&P selection.
- ***Use of COTS assemblies or NDIs.*** This consideration was also listed as a driver of risk likelihood. In one sense, a COTS assembly or an NDI must meet its allocated and derived requirements just like any part. If requirements are not met, then the consequences of a failure must be identified. Factors such as criticality, sensitivity, and vulnerability to compromise should be considered.

C.4 Aggregating Risk by Subsystem

Working with a long list of risk likelihoods and severities may impede decision-making about the appropriate level of oversight. It may be useful to identify subsystems (or some other grouping of work breakdown structure) where those lists can be aggregated into a single, overall likelihood and severity.[74,75]

The aggregation may begin with an evaluation of the likelihood and severity of each individual risk followed by a combining of all the risks to the subsystem level. For example, there may be a sufficient portion of the individual risks with high likelihood or severity to assign an aggregated likelihood or severity to the subsystem. Something similar could be done for a combination of low likelihood or severity individual risks. If there were no clear way to combine the individual risks, then the subsystem aggregation could be medium.[76] These examples assume that all individual risks are weighted equally. Alternatively, a weight could be assigned to each individual risk when aggregating to the subsystem level.

Aggregation facilitates a determination of the level of program office oversight to be applied to contractor PM&P selection. The term *subsystem* is used somewhat generically. While subsystem is a convenient way to aggregate, more fundamentally the PM&P needing special attention are identified to enable focus on a smaller subset of items where more than normal oversight is needed.

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[74] MIL-STD-881F, *Work Breakdown Structures for Defense Materiel Items, Standard Practice*, May 13, 2022, https://www.secnav.navy.mil/rda/OneSource/Documents/New%20MIL-STD-881C%20Work%20Breakdown%20Structures%20for%20Defense%20Materiel%20Items.pdf.

[75] PM&P themselves have not yet been selected at this stage of the risk analysis.

[76] The approach uses a three-level scale for likelihood and severity as compared to the five-level scales mentioned earlier. One way of reducing the five-level likelihood scale to three levels is to classify not likely and low likelihood as low, likely as medium, and highly likely and near certainty as high. For the three-level severity scale, high could represent critical or significant impact, medium could represent moderate impact, and low could represent minor or minimal impact.

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Appendix D. Considerations for Specifying PM&P Requirements on Contract

Detailed PM&P selection requirements typically limit the selection of parts to a specific type, grade, or class. This is commonly accomplished by putting a document on a contract that specifies selection requirements and describing the processes for doing this in the contractor PM&P Management Plan. There are multiple mechanisms for specifying PM&P requirements and prohibitions.

D.1 Standards and Specifications

MIL-STDs are a common way to express PM&P requirements. A MIL-STD is used to satisfy military-unique applications. There are five types of DoW standards: interface standards, design criteria standards, manufacturing process standards, standard practices, and test method standards.[77] Other types of documents serving similar functions include:

- MIL-HDBKs provide standard procedural, technical, engineering, or design information about the materiel, processes, practices, and methods covered by the DSP.[78]
- MIL-SPECs describe the essential technical requirements for military-unique materiel or substantially modified commercial items.[79]
- MIL-PRFs are performance specifications that state requirements in terms of the required results with criteria for verifying compliance, but without stating the methods for achieving the required results. A performance specification defines the functional requirements for the item, the operational environment, and interface and interchangeability characteristics.[80] Military Details are detailed specifications that identify design requirements, such as materials to be used, how a requirement is to be achieved, or how an item is to be fabricated or constructed. A specification that contains both performance and detail requirements is still considered a detail specification.[81]

The most used MIL-STDs and similar documents for specifying detailed PM&P selection requirements are listed below.

- MIL-STD-11991
- MDA-QS-003-PMAP-Rev C for missile defense[82]
- MIL-STD 882E for system safety[83]

[77] MIL-STD-962D, *Defense Standards Format and Content*, May 6, 2025, https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=36064.

[78] Ibid.

[79] MIL-STD-961E, *Defense and Program-Unique Specifications Format and Content*, July, 16, 2020, https://quicksearch.dla.mil/qsDocDetails.aspx?ident_number=36063.

[80] Ibid.

[81] Ibid.

[82] MDA-QS-003-PMAP-Rev C, *Missile Defense Agency: Parts, Materials, and Processes Mission Assurance Plan*, n.d.

[83] MIL-STD-882E, *System Safety*, May 11, 2012, https://www.nde-ed.org/NDEEngineering/SafeDesign/MIL-STD-882E.pdf.

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- SMC-S-009 and SMC-S-010 for space and missile systems[84]
- NASA/TP—2003--212242.[85]

Industry standards may also establish detailed PM&P requirements. Many industry standards are referenced in the above list of documents. Commonly used industry standards include the following:

- EIA-STD-4899 for component management[86]
- IEC TS 62239-1 for component management[87]
- AIAA R-100A for component management[88]
- SAE AS5553 for counterfeit prevention[89]
- SAE AS6174 for counterfeit prevention[90]
- SAE AS12500 for corrosion prevention[91]
- SAE EIA933 for COTS management[92]
- SAE GEIA-STD-0005-1 for lead free control management[93]
- SAE GEIA-STD-0005-2 for tin whisker risk management[94]
- SAE AS8030 for electronic materials and processes[95]
- IPC J-STD-001 for electronic assembly[96]
- IPC-A-620 for cable and wire harnesses[97]
- SAE AMS7000 series for additive manufacturing.[98]

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[84] Space and Missile Systems Center Standard (SMC-S)-009, *Parts, Materials, and Processes Control Program for Space Vehicles*, April 12, 2013, https://apps.dtic.mil/sti/pdfs/ADA619373.pdf and SMC-S-010, *Technical Requirements for Electronic Parts, Materials and Processes Used in Space Vehicles*, August 14, 2020, https://apps.dtic.mil/sti/pdfs/AD1112325.pdf.

[85] National Aeronautics and Space Administration, Goddard Space Flight Center, NASA/TP—2002—212242, EEE-INST-002: Instructions for EEE Parts Selection, Screening, Qualification, and Derating, April 2008, incorporated addendum 1, https://nepp.nasa.gov/docuploads/FFB52B88-36AE-4378-A05B2C084B5EE2CC/EEE-INST-002_add1.pdf.

[86] EIA-STD-4899C, *Requirements for an Electronic Components Management Plan*, May 9, 2017.

[87] IEC 62239-1:2018, *Process management for avionics—Management plan—Part 1: Preparation and maintenance of an electronic components management plan*, September 24, 2018.

[88] AIAA R-100A:2018, *Recommended Practice: Recommended Practice for Parts Management*.

[89] SAE AS5553D, *Electronical, Electronic, and Electromechanical (EEE) Parts: Avoidance, Detection, Mitigation, and Disposition*, April 14, 2022.

[90] SAE AS6174A, *Counterfeit Materiel; Assuring Acquisition of Authentic and Conforming Materiel*, July 29, 2014.

[91] SAE AS12500, *Corrosion Prevention and Deterioration Control in Electronic Components and Assemblies*, April 24, 2018.

[92] SAE EIA933C, *Requirements for a COTS Assembly Management Plan*, August 3, 2020.

[93] SAE GEIA-STD-0005-1A, *Performance Standard for Aerospace and High Performance Electronic Systems Containing Lead-free Solder*, March 2012.

[94] SAE GEIA-STD-0005-2A, *Standard for Mitigating the Effects of Tin Whiskers in Aerospace and High Performance Electronic Systems*, May 2012.

[95] SAE AS8030, *Requirements for an Electronic Materials and Processes Management Plan*, August 16, 2018.

[96] IPC J-STD-001H, *Requirements for Soldered Electrical and Electronic Assemblies*, September 2020.

[97] IPC/WHMA-A-620E, *Requirements and Acceptance for Cable and Wire Harness Assemblies*, October 2022.

[98] SAE ASM7000A, *Laser-Powder Bed Fusion (L-PBF) Produced Parts, Nickel Alloy, Corrosion- and Heat-Resistant, 62Ni - 21.5Cr - 9.0Mo - 3.65Nb Stress Relieved, Hot Isostatic Pressed and Solution Annealed*, May 16, 2022.

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D.2 Qualified Manufacturers List and Qualified Products List

Another way to specify part requirements is through the mandated use of products found on a Qualified Manufacturers List (QML) or a QPL. The parts listed in either a QML or a QPL are ones where there is a MIL-SPEC that invokes a QPL or QML requirement.

QMLs and QPLs provide lists of parts that have successfully met the qualification and testing requirements in the associated specification and are approved for use in DoW systems. More formally, a QML is a list of "qualified processes and materials at each facility that have been successfully subjected to a defined set of qualification and periodic tests using processes and worst case designs or materials to verify the end product's design, performance, quality, and reliability meet all the applicable specification requirements."[99] A QPL is a list of "products or families of products that have successfully completed the formal qualification process (including all specified periodic tests) that examines, tests, and verifies that a specific product design meets all the applicable specification requirements."[100] The DSPO is responsible for QML and QPL policy and procedures.

"Qualification is a process in advance of, and independent of, an acquisition by which a manufacturer's capabilities or a manufacturer's or distributor's products are examined, tested, and approved to be in conformance with specification requirements, …."[101] The preparing activity for a MIL-SPEC is responsible for qualification.

DSPO is responsible for the policy and procedures, but the preparing/qualifying activity is responsible for qualification of the items under their purview.

There are three ways to mandate the use of QML or QPL parts:

1. Specify that a part shall meet the requirements of a MIL-SPEC invoking a QPL or QML.
2. Specify a mandatory order of precedence for selecting parts in the contract and have all MIL-SPEC qualified parts as the highest preference. When this is done, a QML or QPL part that meets the requirements should be selected. The selection is not guaranteed however because there may be a justification for selecting another part, e.g., the form or fit of the MIL-SPEC part may not be acceptable. This type of contract requirement is not discussed in the SD-26.
3. Require an order of precedence in the processes described in the contractor PM&P Management Plan. In this situation, the Plan should not be approved by the government without an acceptable order of precedence.

D.3 Preferred Parts List

Although the same part can be on a QPL and a PPL, they are entirely different. There are two types of PPLs. The first is a program office–approved PPL. Inclusion of a part on this PPL typically implies that the program office has agreed in advance that parts selected from the list will automatically be approved by any program office part selection approval process established in the contract or in the contractor PM&P Management Plan. Typically, parts included in a program office–approved PPL will have been successfully used in similar applications. The program office should also approve any changes to the PPL. The process to make changes should be in the contractor PM&P Management Plan.

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[99] SD-6, *Provisions Governing Qualification: Qualified Products Lists and Qualified Manufacturers Lists*, DSPO, August 2019, https://www.navsea.navy.mil/Portals/103/Documents/QPL/13.%20SD-6%20Provisions%20Governing%20Qualification.pdf.

[100] Ibid.

[101] Ibid.

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The second type of PPL is a contractor-chosen PPL. That is a commercial company artifact that identifies the parts to be given the highest selection priority, typically through a computer-aided design tool when developing a product. Parts are chosen for a PPL based on factors such as usage, sources, community acceptance, performance, availability, technology, cost, and lead time. There are many reasons for a program office's part selection value proposition. To satisfy the requirements of a specific acquisition, it may be desirable to tailor the selection of parts from the PPL. This can be accomplished by limiting the selection of parts to a specific type, grade, or class as discussed above.

D.4 PM&P Prohibitions

Just as certain PM&P requirements may be specified, PM&P prohibitions may also be established. Appendix C of MIL-STD 11991 contains prohibited parts, materials, and processes. The listed PM&P "are considered to be reliability suspect, have limited application, involve restricted or special controls, or are otherwise unacceptable for use in aerospace and defense electronic hardware and associated mechanical hardware."[102] These PM&P should not be used without advanced and explicit government approval. The Parts, Materials, and Processes Mission Assurance Plan (PMAP) similarly includes PM&P prohibitions throughout its requirements section. PMAP Appendix E contains restricted PM&P requiring special consideration before use.

[102] See MIL-STD-11991B, *General Standard for Parts, Materials, and Processes*, October 3, 2023, https://www.dau.edu/sites/default/files/2024-01/MIL-STD%2011991B.pdf.

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Appendix F. Abbreviations

AAF	Adaptive Acquisition Framework
ACAT	Acquisition Category
Ao	Operational Availability
BOM	Bill of Material
CAC	Common Access Card
CAGE	Commercial and Government Entity
CDCA	Current Document Control Authority
CDR	Critical Design Review
CFOM	Cyber Figure of Merit
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
CPSSEP	Cyber-Physical System Security Engineering Plan
DCMA	Defense Contract Management Agency
DLA	Defense Logistics Agency
DMSMS	Diminishing Manufacturing Sources and Material Shortages
DoD	Department of Defense
DoDI	Department of Defense Instruction
DoDM	Department of Defense Manual
DoW	Department of War
DSP	Defense Standardization Program
DSPO	Defense Standardization Program Office
ECP	Engineering Change Proposal
EEE	Electronical, Electronic, and Electromechanical
EMD	Engineering and Manufacturing Development
FLIS	Federal Logistics Information System
FOCI	Foreign Ownership, Control, or Influence
FRP	Full Rate Production
FTE	Full-Time Equivalent
GDO	Government Design Organization
GFP	Government Furnished Property
GIDEP	Government-Industry Data Exchange Program
ICT	Information Communication Technology
IEC	International Electrotechnical Commission
ILA	Independent Logistics Assessment
IOC	Initial Operating Capability
IPC	Initial Production Check
IPT	Integrated Product Team

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IR	Item Reduction
KSA	Knowledge, Skill, and Ability
LCSP	Life Cycle Sustainment Plan
L-PBF	Laser-Powder Bed Fusion
M&Q BoK	Manufacturing and Quality Body of Knowledge
MCA	Major Capability Acquisition
MDA	Missile Defense Agency
MIL-HDBK	Military Handbook
MIL-PRF	Military Performance Specification
MIL-SPEC	Military Specification
MIL-STD	Military Standard
MS	Milestone
MTA	Middle Tier Acquisition
MTTF	Mean Time to Failure
NAS	National Air Standard
NDI	Non-Developmental Item
NIST	National Institute of Standards and Technology
NSN	National Stock Number
O&S	Operations and Support
OCM	Original Component Manufacturer
OEM	Original Equipment Manufacturer
PCN	Product Change Notice
PDR	Preliminary Design Review
PM&P	Parts, Materials, and Processes
PMAP	Parts, Materials, and Processes Mission Assurance Plan
PPL	Preferred Parts List
PPP	Program Protection Plan
PRR	Production Readiness Review
QML	Qualified Manufacturers List
QPL	Qualified Products List
RDT&E	Research, Development, Test, and Evaluation
RMF	Risk Management Framework
SCRM	Supply Chain Risk Management
SD	Standardization Document
SEP	Systems Engineering Plan
SETA	Systems Engineering and Technical Assistance
SETR	Systems Engineering Technical Review
SFR	System Functional Review
SMC-S	Space and Missile Systems Center Standard

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SME	Subject Matter Expert
SRR	System Requirements Review
SSS	Systems Security Symposium
TDP	Technical Data Package
TMRR	Technology Maturation and Risk Reduction
TSN	Trusted Systems and Networks
UCA	Urgent Capability Acquisition
WarU	Warfighting Acquisition University
WSIT	Weapon System Impact Tool

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